

Tab 1

Passive Legged Locomotion via Nautilus Gear-Induced Gait Dynamics

ME 310, Design of Machine Elements

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Abstract:

This project explores the design and mechanical optimization of a quadrupedal walking robot utilizing nautilus (non-circular) gears to achieve biologically inspired locomotion. Unlike standard circular gear-motor drives that produce constant angular velocity, the nautilus geometry enables a "mechanically programmed" gait by providing a variable transmission ratio within a single rotation. This setup optimizes the walking cycle by delivering high torque during the high-load stance phase and accelerating the velocity during the swing phase for rapid leg recovery. The research focuses on critical machine design elements, specifically the challenges of managing non-linear pitch curves and variable pressure angles to ensure consistent meshing. A comparative analysis reveals that the nature of loading in the nautilus system shifts from the steady-state cyclic loading of standard gears to a variable-amplitude cyclic loading profile. Although the rotational frequency may remain constant, the nautilus gears experience a higher effective cycle count due to concentrated peak stresses and dynamic inertial "jerk" during phase transitions. By examining static loading and fatigue failure through this lens, the study demonstrates that the nautilus configuration reduces peak motor demand and mitigates impact-related fatigue, offering a more robust and energy-efficient solution for autonomous all-terrain robotics.

Motivation:

The development of this quadrupedal robot is motivated by the pursuit of *Morphological Intelligence*—the engineering philosophy that a robot's physical shape, rather than its software, should drive its core behavior. In modern robotics, achieving a natural walking gait typically requires expensive sensors, high-power servos, and complex control algorithms to manage varying speeds and torques. This project seeks to simplify that paradigm by shifting the "computational load" from the microprocessor to the mechanical transmission using nautilus (non-circular) gears. And most importantly it doesn't need a feedback mechanism, and can mitigate floating-point errors in electronics used.

The core inspiration for this mechanical actuation is found in the asymmetric gait cycles of terrestrial mammals. In nature, a limb does not move at a constant velocity; it requires high torque and slow movement during the stance phase to support weight, followed by a rapid "flick" or acceleration during the swing phase to reset the leg – *asymmetric gait* (Fig 1).

This project draws a direct parallel to the Theo Jansen mechanism (Strandbeest). While Jansen utilized a complex eleven-bar linkage system to "program" a flat, biological foot-path through rod proportions, this project utilizes nautilus gears to "program" the velocity and torque of the stride. By using logarithmic spiral geometry, the gears mimic the elastic energy storage and release found in animal tendons, allowing a simple, constant-speed motor to produce a sophisticated, non-linear output.

Standard gear-motor setups are inherently limited by their linear output, often leading to "stall" conditions during peak loads or energy waste during the recovery phase. The motivation to use non circular gears – historically used in some ICE for variable valve timings and non-uniform camshaft speed

The ultimate motivation is the creation of a robust, low-cost locomotion platform for hazardous environment inspection and planetary exploration. In these scenarios, software glitches or electronic failures can be catastrophic

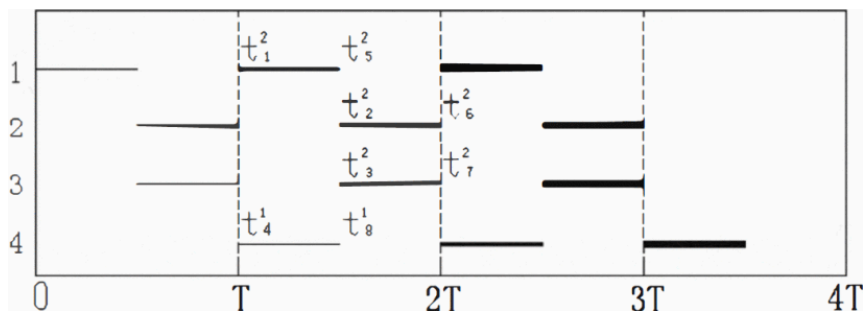


Fig 1: Image showing asymmetric quadruped gait showing non-uniform footfall timing and irregular phase distribution across the gait cycle

1. Introduction:

1.1. Existing Quadrupeds and Gait Mechanisms

Locomotion in quadruped robots is fundamentally governed by coordinated limb motion, where gait serves as the organizing principle for movement. Unlike wheeled systems, quadrupeds must continuously regulate balance, contact forces, and body posture. Early work by Raibert established that “*stable legged locomotion requires active control of body dynamics through foot placement and force modulation*” [a1]. This insight remains central to modern gait design.

Quadruped robots exhibit a spectrum of locomotion behaviors ranging from slow, statically stable walking to highly dynamic running and bounding. The choice of gait is not merely kinematic but deeply tied to system dynamics, actuator capabilities, and control strategies.

A gait can be formally described through periodic trajectories of limb motion characterized by parameters such as duty factor, phase relationships, and stride frequency. In static gaits, the robot maintains its center of mass within the support polygon, ensuring stability without requiring dynamic balance. However, most modern systems operate in the dynamic regime, where stability emerges from motion itself.

Kim et al. observe that “*dynamic locomotion exploits inertial forces, allowing robots to achieve higher speeds and efficiency at the cost of increased control complexity*” [a2]. In such cases, the interaction between the robot and ground is governed by hybrid dynamics, involving alternating phases of contact and flight.

Actuators as primary drivers: The primary driving elements in quadrupeds are joint actuators (electric motors or hydraulic systems). These generate torques at the hip, knee, and sometimes ankle joints. The actuator torque vector can be expressed as:

$$\boldsymbol{\tau} = [\tau_{hip} , \tau_{knee} , \tau_{ankle}]$$

These torques define the internal effort that ultimately produces locomotion. Modern systems (e.g., ANYmal) employ high-torque density electric actuators, enabling dynamic behaviors such as jumping and running.

Power flow and Energy Injection: Energy is injected into the system during the stance phase, when actuators perform positive work against gravity and inertia. During swing phases, energy is redistributed to reposition limbs.

Pratt et al. note that “*leg actuation effectively redirects energy through ground interaction rather than directly propelling the body in free space*” [a3].

The actuator output is transmitted through the robot’s mechanical structure, links (thigh, shank), joints (rotational DOFs), compliance elements (series elasticity, if present). This transformation follows:

$$\mathbf{F}_{\text{foot}} = (\mathbf{J}^{-T}) \boldsymbol{\tau}$$

where $\mathbf{J}$ is the leg Jacobian. This equation shows that ***joint torques are mapped into foot-end forces***, which directly interact with the ground. This mapping is central to force-controlled locomotion, widely used in systems like Boston Dynamics Spot.

The leg links act as rigid bodies transmitting actuator forces to the ground. Their geometry determines leverage, workspace, and efficiency. The robot’s main body is the primary driven element. It responds to ground reaction forces generated at the feet. Its motion obeys:

$$m \ddot{\mathbf{x}} = \sum \mathbf{F}_{\text{GRF}} + m \mathbf{g}$$

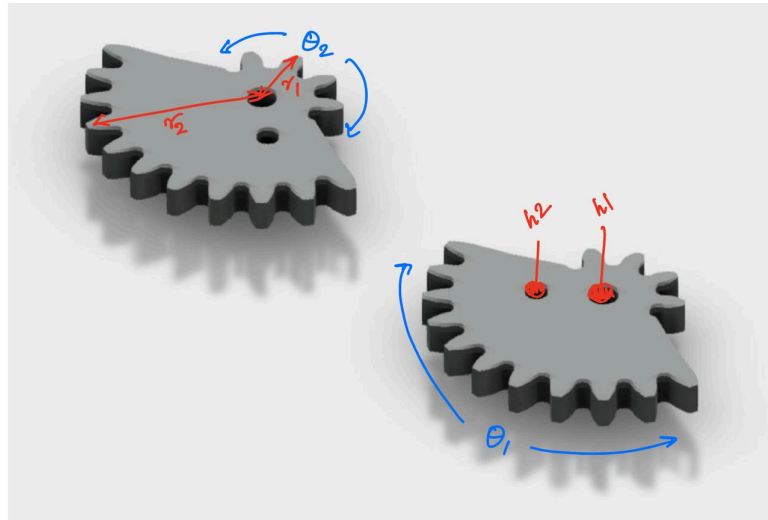
Thus, locomotion is achieved not by directly moving the body, but by ***indirectly accelerating it through ground forces***.

Kim et al. assert that “efficient locomotion arises from synchronized force exchange patterns between limbs and ground” [a2], thereby framing gait as an inherently coordinated force distribution mechanism rather than merely a kinematic sequence. Building on this perspective, the present work interprets gait as a force coordination strategy and extends it to the analysis of the proposed linkage system.

Specifically, critical load points within the mechanism are identified through detailed kinematic analysis, wherein acceleration profiles of individual links are computed and systematically mapped to corresponding inertial and joint forces. These derived force distributions serve as inputs for subsequent finite element analysis (FEA), enabling the evaluation of key structural parameters, including static stress distribution, fatigue behavior, and factor of safety.

1.2. Non-Circular Gears

Conventional circular gears transmit motion with constant velocity ratio, on the other hand, non-circular gears introduce a **variable transmission ratio** over a rotation. Litvin notes that “noncircular gears allow the designer to prescribe arbitrary angular velocity relationships between shafts” [a4]. For legged systems, this is attractive because gait is inherently non-uniform in time which is slow, forceful stance phases and faster, low-load swing phases. Embedding this asymmetry mechanically reduces reliance on active control.



where:

$$r(\theta) = \begin{cases} r_1 & \theta_1 > \theta > 0 \\ r_2 & 2\pi > \theta > \theta_2 \end{cases}$$

The meshing of non-circular gear pair produces an instantaneous ratio

$$i(\theta) = \frac{\omega_{input}}{\omega_{output}} = \frac{r_{output}(\theta)}{r_{input}(\theta)} ; \text{ where } r(\theta) \text{ is local pitch radius}$$

As a result, both output speed and torque become functions of phase:

$$\tau_{output} = i(\theta) \tau_{input}$$

Thus, the gear pair acts as a mechanical modulator, shaping how actuator power is delivered across the gait cycle. The output shaft drives linkages (crank–rocker, four-bar, or direct joint coupling). The combined mapping becomes:

$$\mathbf{q}(t) = \mathbf{f}(\theta(t)), \theta(t) = \int \omega_{out}(\theta) dt$$

where $\mathbf{q}$ are joint coordinates. Because instantaneous rotation is non-uniform, ***joint trajectories are inherently phase-shaped***, slow during stance, rapid during swing.

A key advantage is that gait characteristics can be embedded in hardware:

1. Stance phase $\rightarrow$ large pitch radius $\rightarrow$ high torque, low speed
2. Swing phase $\rightarrow$ small pitch radius $\rightarrow$ low torque, high speed

This creates a natural asymmetry resembling biological locomotion. As McGeer observed in passive dynamic walkers, “*mechanical design alone can produce stable walking patterns without active control*” [a5]. Non-circular gears extend this idea to powered systems.

From a force-flow perspective, the system can be viewed as a staged transformation of energy and load. The motor supplies nearly constant power, which the non-circular gear modulates over the cycle, redistributing torque in a time-varying manner. This modulated input is then transmitted through the linkage, shaping the resulting foot trajectory. As the foot interacts with the ground, it generates ground reaction forces (GRFs) that vary with phase, ultimately producing the overall body motion. In this sense, ***the non-circular gear effectively acts as a temporal filter***, ensuring that higher torque is delivered during load-bearing phases and reduced during swing, thereby improving efficiency and functional alignment of forces.

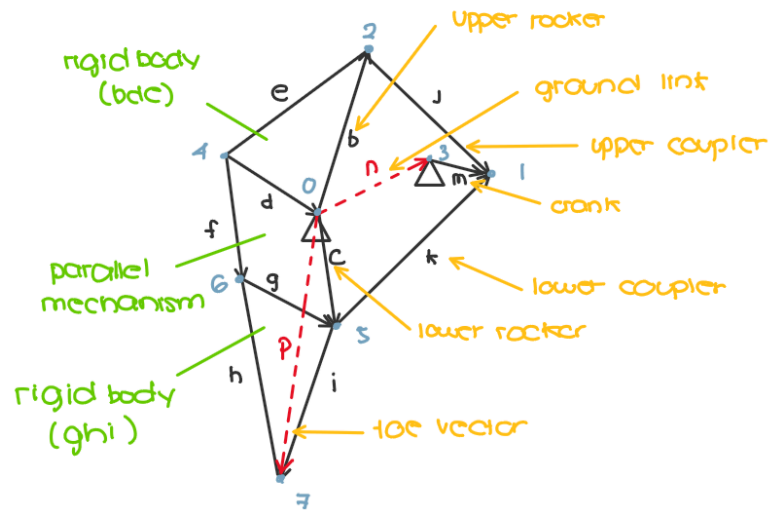
Non-circular gears enable phase-dependent transmission of torque and velocity, allowing quadruped gait characteristics to be embedded directly into mechanical design. By shaping the timing and magnitude of energy delivery, they influence ground reaction forces and thereby locomotion. While limited in adaptability, they provide an elegant alternative to purely control-driven gait generation.

1.3. Jansen Walking Mechanism

The Theo Jansen mechanism is a planar linkage system that converts rotary motion into a biologically inspired walking trajectory. Unlike non-circular gears, which modulate transmission ratios, this mechanism embeds the entire gait trajectory geometrically within link proportions. In this work, a non-circular gear-driven Theo Jansen mechanism is analyzed from both a kinematic and force-flow perspective. The non-circular gear introduces a time-varying transmission ratio, enabling modulation of torque delivery across a cycle. This is particularly useful in aligning actuator effort with ground reaction force (GRF) demands.

The objective of this analysis is twofold:

1. To compute the **toe trajectory and angular evolution** using vector loop equations.
2. To interpret the mechanism as a **force-flow system**, where energy is redistributed temporally through geometric design.



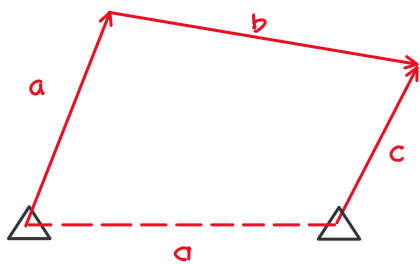


Figure 1: Reference four-bar mechanism

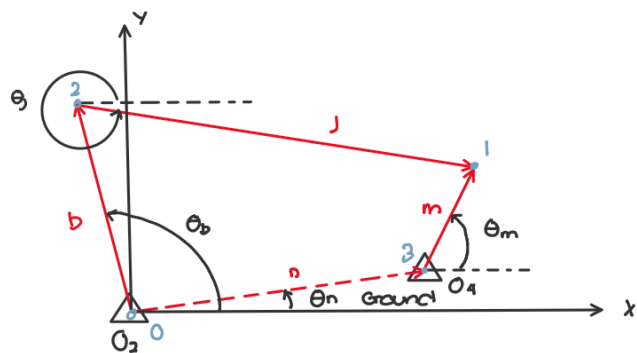


Figure 2: Vector loop (nm-jk)

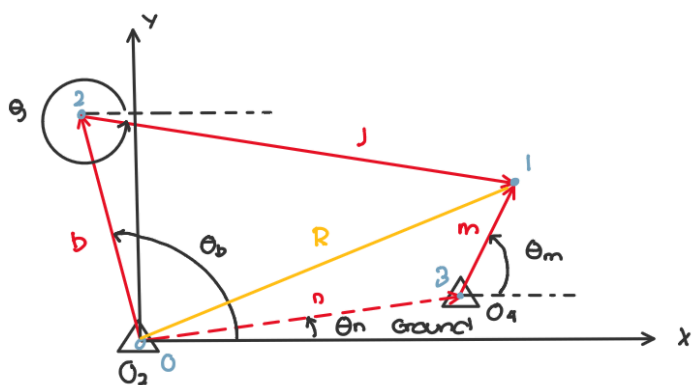


Figure 3: Vector loop (nm-jk) with R

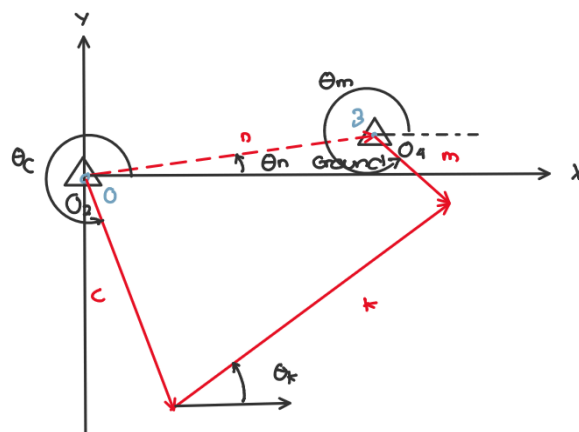


Figure 4: Vector loop (nm-ck)

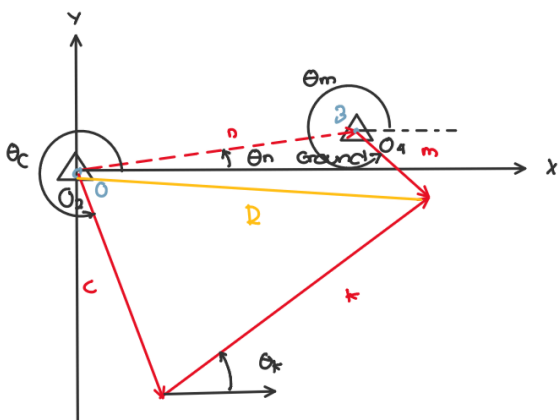


Figure 5: Vector loop (nm-ck) with R

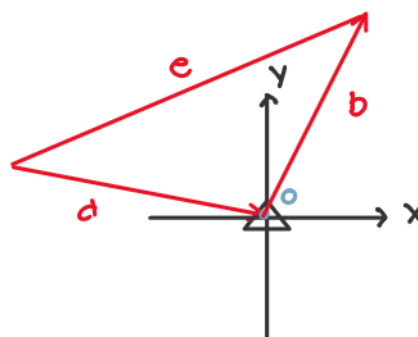


Figure 6: Triangle bde

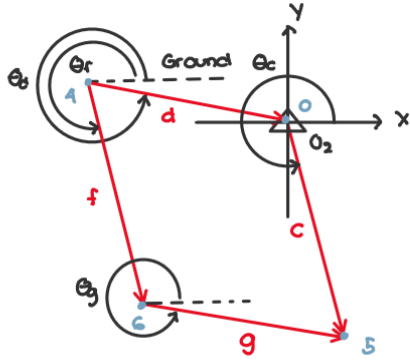


Figure 7: Parallel-link linkage vector loop (dc-fg)

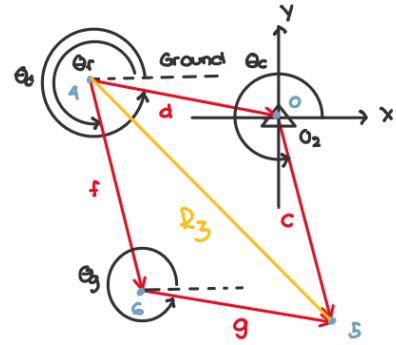


Figure 8: Parallel-link linkage vector loop (dc-fg) with R

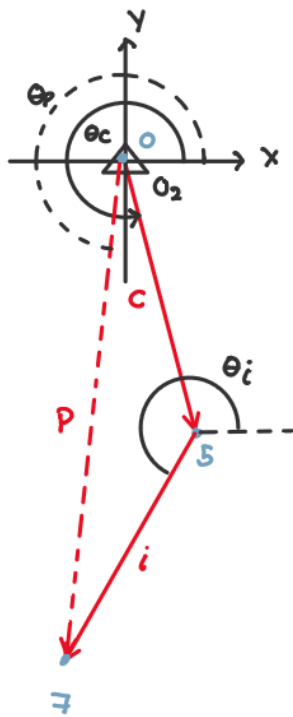


Figure 9: Vector loop (cip)

With reference to image 2 and 3, the vector loop equation is:

$$\vec{n} + \vec{m} - \vec{b} - \vec{j} = 0$$

Instead of solving directly, the loop is reduced to triangle (b,j,R) where:

$$R = \sqrt{X_m^2 + Y_m^2}$$

with:

$$X_m = x + m \cos \theta_m, Y_m = y + m \sin \theta_m$$

Using cosine law projections:

$$A_j = \frac{j^2 - b^2 + R^2}{2 j}, A_b = \frac{b^2 - j^2 + R^2}{2 b}$$

Thus,

$$\theta_j = \cos^{-1} \left(\frac{A_j}{R} \right) + \alpha, \theta_b = \cos^{-1} \left(\frac{A_b}{R} \right) + \alpha$$

where:

$$\alpha = \tan^{-1} \left(\frac{Y_m}{X_m} \right)$$

Referring to figure 4 and 5, similarly triangle (c, k, R) gives:

$$A_k = \frac{k^2 - c^2 + R^2}{2 k}, A_c = \frac{c^2 - k^2 + R^2}{2 c}$$

$$\theta_k = \cos^{-1} \left(\frac{A_k}{R} \right) + \alpha, \theta_c = \cos^{-1} \left(\frac{A_c}{R} \right) + \alpha$$

Referring figure 6, in triangle bde :

$$\theta_d = \cos^{-1} \left(\frac{b^2 + d^2 - e^2}{2 b d} \right) + \theta_b$$

Referring figure 7, the intermediate point:

$$X_3 = d \cos \theta_d + c \cos \theta_c, Y_3 = d \sin \theta_d + c \sin \theta_c$$

$$R_3 = \sqrt{X_3^2 + Y_3^2}, \alpha_3 = \tan^{-1} \left(\frac{Y_3}{X_3} \right)$$

then,

$$A_f = \frac{f^2 - g^2 + R_3^2}{2 f}, A_g = \frac{g^2 - f^2 + R_3^2}{2 g}$$

$$\theta_f = \cos^{-1} \left(\frac{A_f}{R_3} \right) + \alpha_3, \theta_g = \cos^{-1} \left(\frac{A_g}{R_3} \right) + \alpha_3$$

Referring figure 9:

$$P_x = c \cos \theta_c + i \cos \theta_i, P_y = c \sin \theta_c + i \sin \theta_i$$

$$\theta_p = \tan^{-1} \left(\frac{P_y}{P_x} \right), |P| = \sqrt{P_x^2 + P_y^2}$$

A complete position analysis of the Theo Jansen mechanism was successfully carried out using a vector loop decomposition approach. The study shows that even highly complex multi-link systems can be systematically reduced into smaller, solvable geometric units without loss of generality. By leveraging cosine-law-based projections, the otherwise nonlinear loop equations are transformed into tractable forms, significantly simplifying the kinematic solution process. The resulting toe trajectory emerges purely from the geometric constraints of the mechanism, highlighting that the walking motion is an inherent property of the linkage design rather than an externally imposed condition.

Furthermore, the adopted approach is computationally efficient and well-suited for practical applications such as gait optimization, where trajectory shaping is critical, as well as for design parameter tuning to achieve desired motion characteristics. It also provides a strong foundation for subsequent integration with dynamic and force analyses, enabling a more comprehensive understanding of the mechanism's performance under real operating conditions.

1.4. Elements of Designing

High-cycle fatigue (HCF) design becomes particularly critical in legged robotic systems, where structural components undergo repeated cyclic loading during locomotion. In a quadruped employing the Theo Jansen mechanism, rotary input from the crank is transformed into a complex, periodic foot trajectory, resulting in fluctuating stresses across multiple interconnected links. These stresses are predominantly elastic but persist over a very high number of cycles, making fatigue the governing failure mode. Therefore, an integrated fatigue design framework is essential to ensure durability, reliability, and sustained performance of the mechanism under continuous gait operation.

In the Jansen quadruped, each gait cycle induces a repeating load pattern driven by the crank rotation. The stance phase introduces significant ground reaction forces, while the swing phase reduces loading but maintains inertial effects. This results in *alternating stress cycles* in links such as the rockers, couplers, and rigid triangular members. These stresses are decomposed into mean and alternating components, where the amplitude is strongly influenced by gait dynamics and terrain interaction. The periodic nature of the mechanism ensures that fatigue loading is highly structured but non-uniform across links.

The complex linkage geometry leads to non-uniform stress distribution, with critical regions typically occurring at joints, connection interfaces, and regions with geometric discontinuities. In particular, links such as (b, c, d, f, g) experience significant bending and axial loading during load transfer from the foot to the frame. Stress concentration effects become dominant at pin joints and sharp transitions, requiring the use of fatigue stress concentration factors. In the Jansen mechanism, minimizing these concentrations through smooth geometry and proper joint design directly enhances fatigue life.

The repeated loading cycles in quadruped locomotion necessitate materials with high fatigue resistance. The endurance limit, derived from S–N curves, provides a baseline for infinite-life design. However, in practical robotic systems, this limit must be corrected for factors such as surface finish, manufacturing processes, and environmental conditions. Lightweight materials like aluminum alloys or composites, often used in robotic legs, must be carefully evaluated since their fatigue characteristics differ significantly from traditional steels. The design must ensure that operational stresses remain below the modified endurance limit for sustained walking cycles.

Due to asymmetric loading during stance and swing phases, the stress cycles in Jansen-based systems are not purely alternating but contain a non-zero mean component. This necessitates the use of fatigue failure criteria such as the ***Modified Goodman relation***, which accounts for the interaction between mean and alternating stresses. Links subjected to continuous compressive or tensile bias—especially those directly transmitting ground forces—must be evaluated carefully, as mean stress can significantly reduce fatigue strength.

Unlike simple rotating shafts, the links in a Jansen mechanism experience combined loading from axial, bending, and sometimes torsional. This results in a multiaxial stress state, particularly at joints and load transfer points. To handle this complexity, stresses are converted into equivalent values using the von Mises criterion, enabling the application of uniaxial fatigue data. This simplification is crucial for practical fatigue analysis without compromising physical accuracy.

Fatigue design in such mechanisms is inherently iterative. Link dimensions, joint placements, and even gait parameters influence stress distribution. For instance, modifying link lengths slightly can redistribute peak loads and reduce stress amplitudes. Thus, fatigue design is closely coupled with ***gait optimization***, where both motion efficiency and structural longevity are considered simultaneously. The vector-loop-based kinematic model provides a foundation to map motion to stress, enabling informed design refinement.

Fatigue safety is typically evaluated using graphical tools such as the Modified Goodman diagram, where the operating stress state of each critical link is plotted. For the Jansen quadruped, this allows identification of links approaching failure limits during peak load phases. Such visualization is particularly useful when comparing different gait patterns or design configurations, ensuring that all components operate within safe fatigue limits.

2. *Analysis*

2.1. FEM Analysis

The structural integrity of the quadrupedal leg mechanism was evaluated using Finite Element Method (FEM) analysis to ensure that the "mechanically programmed" gait enabled by the nautilus gears does not induce stresses exceeding the material limits of the FDM-printed PLA components. The simulation specifically targeted the peak stance phase, where ground reaction forces (GRFs) are maximized and the kinematic linkages are under the highest compressive and bending loads.

2.1.1 Computational Domain

To simulate the non-linear behavior of the multi-loop linkage system, a high fidelity 3D model was utilized within a non linear static structural solver.

Meshing Strategy: The geometry was discretized using higher-order 10-node tetrahedral structural solids (SOLID187), which are well-suited for modeling the complex, irregular curvatures of the Jansen linkages. Proximity-based refinement was applied to the pin-hole regions with an element size of 0.5mm to capture localized stress gradients.

Solver Controls: The simulation accounts for large angular displacements and geometric non-linearities. An iterative Newton-Raphson scheme was employed to solve the equilibrium equations:

$$[K_T\{u\}]\{\Delta u\} = \{F_{ext}\} - \{F_{int}\}$$

where K_T is the tangent stiffness matrix updated at each iteration to account for large deflection effects.

Numerical Stability: To ensure convergence before the mechanical joints reached equilibrium, "Weak Springs" were activated to provide a negligible artificial stiffness, preventing matrix singularities in the initial substep

2.1.2 Material Modeling

The numerical model was constructed using the mechanical properties of Polylactic Acid (PLA), assuming an isotropic formulation for the macroscopic behavior of the 3D-printed links. To maintain consistency with the transmission analysis in Section 2.2, the following constants were used:

Elastic Properties: The material was assigned a Young's Modulus of 2 GPa and a Poisson's ratio 0.33.

Yield Criterion: Structural safety was evaluated via the von Mises equivalent stress, which predicts yielding under complex, multiaxial loading:

$$\sigma_{eq} = \frac{1}{\sqrt{2}} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}$$

The operational limit was defined by the material yield strength of 50 Mpa.

2.1.3 Global Boundary Conditions

The boundary conditions (BCs) were established to replicate the physical constraints of the robot chassis and the dynamic interaction with the terrain.

Mass Breakdown and Force Derivation: The total assembly mass of 410 g was used to calculate the ground reaction forces. This mass accounts for the PLA assembly 250 g, the battery 130 g, and the motor 30 g

External Loading: A peak dynamic load F_{peak} of 4.4 N was applied as a vertical force vector at the toe tip. This load represents the peak Ground Reaction Force (GRF) during a two legged stance phase, including a dynamic impact factor for foot strike modeled as :

$$F_{peak} = \frac{M_{total} \cdot (g + a_{dynamic})}{2}$$

Grounding: Fixed revolute supports were applied at the primary chassis anchors: the Tristar center G_1 and the motor drive axis G_2 . These restrict translational degrees of freedom while permitting the necessary planar rotation.

2.1.4 Kinematic Coupling and Joint Constraints

To maintain kinematic continuity across the interconnected links without inducing numerical "locking," a rigorous joint formulation was implemented.

Serial Joint Topology: To prevent mathematical overconstraints and Multipoint Constraint (MPC) conflicts, a sequential kinematic coupling strategy was adopted. Joints were linked through a serial member topology rather than scoping multiple joints to a single shared nodal face.

Revolute Joint Definition: Hinges were modeled as Body-to-Body Revolute joints, maintaining coaxiality while allowing relative angular motion.

Actuation Input: The variable angular velocity characteristic of nautilus gears was implemented via a tabular joint-rotation command at the input crank, ensuring the stress results correspond to the non-linear torque distribution of the gait cycle.

2.1.5 Converged Structural Response

Peak Stress Distribution: The maximum equivalent stress of 15.4 MPa was localized at the Tristar pivot G_1 , yielding a static Factor of Safety

$$n \approx 3.25$$

Dynamic Deflection: A maximum resultant deformation of 0.92 mm was recorded at the distal toe, confirming structural rigidity.

2.1.6. Fatigue Failure Analysis

The mechanism was evaluated for high-cycle fatigue endurance using the Modified Goodman criterion, ensuring coherence with the drivetrain analysis in Section 2.2.1.

Stress Cycling: During a full rotation, critical links undergo a repeated loading cycle. For the mid-leg joint, the stress fluctuated between a minimum of 2.1 MPa and a peak of 15.4 Mpa resulting in:

Mean Stress (σ_m) : 8.75 MPa

Alternating Stress (σ_a) : 6.65 MPa

Modified Endurance Limit: Using the corrected fatigue limit derived in Section 2.2.1.1,

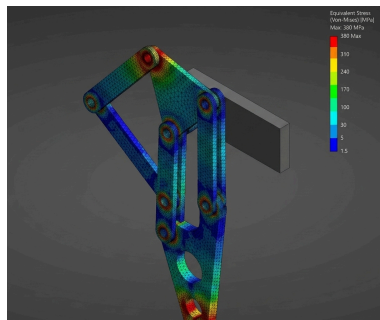
$$S_f = 10.43 \text{ MPa}$$

(accounting for surface finish $k_a = 0.78$ and reliability $k_e=0.868$)

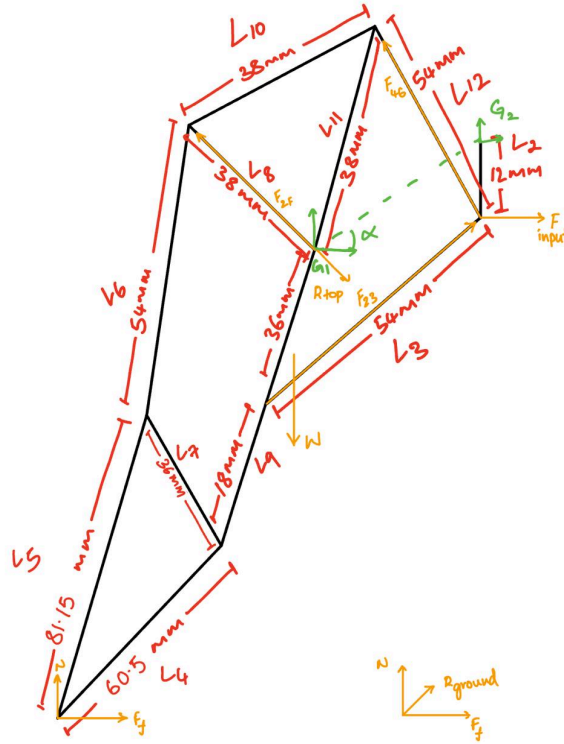
Fatigue Factor of Safety :

$$\frac{1}{n_f} = \frac{\sigma_a}{S_f} + \frac{\sigma_m}{S_{ut}} = \frac{6.65}{10.43} + \frac{8.75}{60}$$

The resulting fos of 1.28 confirms the linkage is capable of sustaining the high-cycle periodic loading required for sustained autonomous locomotion.



2.2. Mathematical Modelling



The Free Body Diagram (FBD) of the Theo Jansen robotic leg mechanism was developed to visualize the internal force-transfer behaviour and external reaction loading acting throughout the linkage structure during locomotion. In the FBD, each effective linkage member is isolated and represented along with the corresponding axial forces, support reactions, and ground-contact forces acting at the joints. The principal internal forces considered in the reduced static force model include the axial member forces f_{23} , f_9 , f_{46} , and f_{2F} , which govern the dominant load-transfer path through the mechanism. External reactions including the top support reaction, ground normal force, and friction force were also incorporated to satisfy global equilibrium conditions. The force directions were assigned according to standard engineering statics conventions, where tensile forces act away from the joints and compressive forces act toward the joints. The generated FBD therefore serves as the foundational representation for deriving the equilibrium equations, vector force relations, stress analysis, fatigue calculations, and ground interaction behaviour of the robotic leg mechanism.

2.2.1. Gear Stress Analysis

The mathematical modelling of the nautilus gear transmission focuses on the **phase-dependent evolution** of meshing forces, contact stresses, and tooth-root bending stresses generated due to the continuously varying transmission ratio. Since the kinematic foundations, instantaneous transmission relations, and gait-phase asymmetry mechanisms have already been established previously, the present section concentrates specifically on quantitative force propagation and stress evolution within the **non-circular meshing domain**.

Unlike conventional circular gears operating under approximately stationary meshing conditions, the present system exhibits continuously evolving transmission geometry. Consequently, the tooth loading state becomes non-stationary, producing temporally redistributed force concentration and spatially localized stress accumulation across the gear profile. The meshing process therefore behaves as a **phase-dependent stress propagation system**, in which localized curvature variation and transmission-ratio evolution simultaneously govern **force amplification** and contact loading.

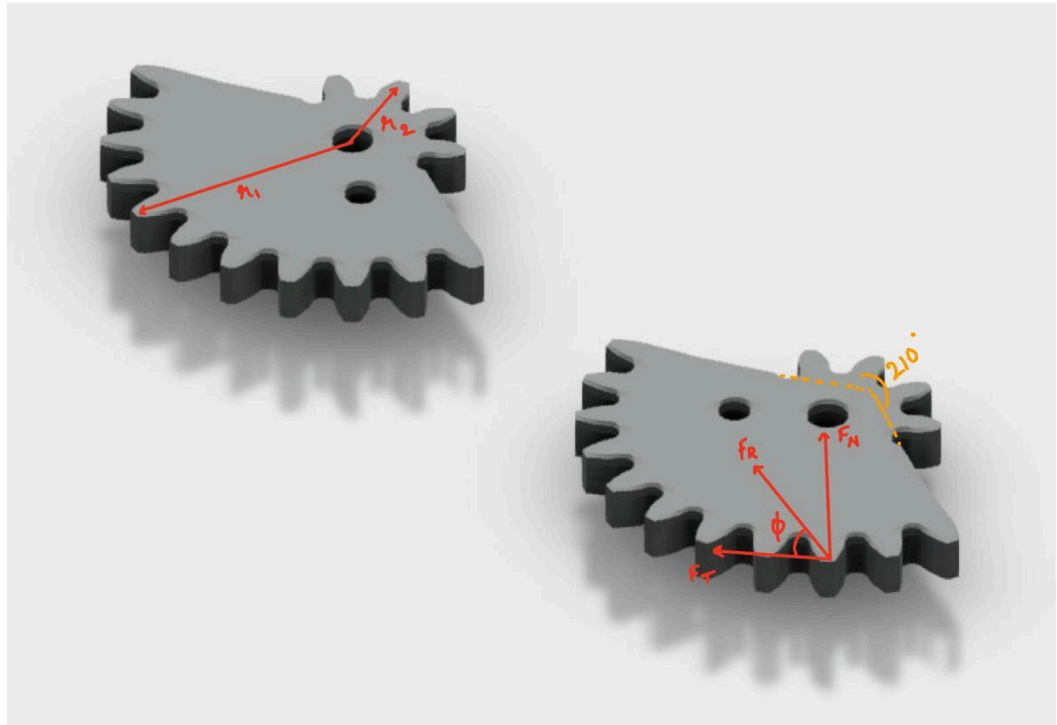


Fig. Gear radial,tangential and resultant forces

A **constant involute pressure angle** is assumed throughout the analysis as

$$\phi = 20^\circ$$

where:

- ϕ represents the pressure angle between the line of action and the common tangent to the pitch circles

This assumption represents an idealized force-direction approximation in which the force orientation remains geometrically consistent while the force magnitude varies according to the instantaneous transmission geometry. Since the pressure angle fundamentally depends on involute tooth geometry rather than instantaneous pitch radius, the assumption remains valid for the present non-circular profile. Local deviations arising from curvature discontinuities, sliding variation, and meshing nonlinearity are absorbed into the dynamic loading coefficient K_v .

The varying pitch radius distributions are represented as

$$r_{out}(\theta) = \begin{cases} 18.75, & 0 \leq \theta < 210^\circ \\ 14.65, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

where:

- $r_{in}(\theta)$ represents the instantaneous pitch radius of the input gear,
- $r_{out}(\theta)$ represents the instantaneous pitch radius of the output gear,
- θ represents the crank angle.

The motor supplies an approximately constant input torque,

$$T_{in}=0.12 \text{ Nm}$$

Where:

- T_{in} represents the motor input torque applied to the driving gear.

The transmitted output torque evolves according to the previously defined **instantaneous transmission relation**,

$$T_{out}(\theta) = T_{in} \frac{r_{out}(\theta)}{r_{in}(\theta)}$$

where:

- $T_{out}(\theta)$ represents the instantaneous output torque transmitted to the driven gear.

Substituting the corresponding pitch radii gives

$$T_{out}(\theta) = \begin{cases} 0.216 \text{ Nm}, & 0 \leq \theta < 210^\circ \\ 0.073 \text{ Nm}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The transmission therefore produces a high-torque low-velocity stance-dominant interval and a low-torque high-velocity swing-dominant interval. This creates what may be defined as a **mechanically encoded gait asymmetry field**, in which actuator energy redistribution becomes embedded directly into the drivetrain geometry itself rather than external control logic.

2.2.1.1. Output Gear Stress

The output gear operates under phase-dependent amplified torque transmission generated by the instantaneous transmission ratio. Consequently, the output gear experiences substantially larger localized meshing loads during the stance-dominant interval.

The transmitted **tangential meshing force** becomes

$$F_{t,out}(\theta) = \frac{T_{out}(\theta)}{r_{out}(\theta)}$$

Where:

- $F_{t,out}(\theta)$ represents the tangential tooth force responsible for power transmission.

Substituting the corresponding values gives

$$F_{t,out}(\theta) = \begin{cases} \frac{0.216}{0.01875}, & 0 \leq \theta < 210^\circ \\ \frac{0.073}{0.01465}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

which gives

$$F_{t,out}(\theta) = \begin{cases} 11.52 \text{ N}, & 0 \leq \theta < 210^\circ \\ 4.98 \text{ N}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The tangential force governs:

1. power transmission,
2. tooth-root bending,
3. tangential meshing shear,
4. rotational force transfer between gears.

Since

$$F_t \propto \frac{T}{r}$$

the smaller-radius regions inherently amplify meshing force concentration. This creates a **phase-intensified transmission state**, in which cyclic tooth loading becomes localized within specific angular sectors of the gait cycle.

The radial force generated due to the pressure angle becomes

$$F_{r,out}(\theta) = F_{t,out}(\theta) \tan 20^\circ$$

Where:

- $F_{r,out}(\theta)$ represents the radial separating force acting between the mating gears.

Substituting the corresponding values gives

$$F_{r,out}(\theta) = \begin{cases} 11.52 \tan 20^\circ, & 0 \leq \theta < 210^\circ \\ 4.98 \tan 20^\circ, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

Which gives,

$$F_{r,out}(\theta) = \begin{cases} 4.19 \text{ N}, & 0 \leq \theta < 210^\circ \\ 1.81 \text{ N}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The radial component contributes primarily to:

1. shaft reaction loading,
2. bearing compression,
3. meshing alignment sensitivity,
4. tooth-body compressive loading.

Although comparatively smaller than bending and contact stresses in fatigue-critical behaviour, the radial component strongly influences support reaction evolution and meshing stability.

The corresponding **normal meshing force** acting along the line of action becomes

$$F_{n,out}(\theta) = \frac{F_{t,out}(\theta)}{\cos 20^\circ}$$

Where:

- $F_{n,out}(\theta)$ represents the resultant contact force acting along the line of action.

Substituting the corresponding values gives

$$F_{n,out}(\theta) = \begin{cases} 12.26 \text{ N}, & 0 \leq \theta < 210^\circ \\ 5.30 \text{ N}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The normal force governs localized **Hertzian surface loading** and therefore directly controls:

1. contact pressure concentration,
2. localized wear evolution,
3. micropitting initiation,
4. surface yielding,
5. contact fatigue propagation.

The **radial compressive stress** acting within the tooth body is approximated as

$$\sigma_{r,out}(\theta) = \frac{F_{r,out}(\theta)}{b, m}$$

where:

- $\sigma_{r,out}(\theta)$ represents radial compressive stress,
- b represents the gear face width,
- m represents the gear module.

Using

$$b=8 \text{ mm}, m=1.5 \text{ mm}$$

Gives

$$\sigma_{r,out}(\theta) = \begin{cases} \frac{4.19}{8 \times 1.5}, & 0 \leq \theta < 210^\circ \\ \frac{1.81}{8 \times 1.5}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

Which gives

$$\sigma_{r,out}(\theta) = \begin{cases} 0.349 \text{ MPa}, & 0 \leq \theta < 210^\circ \\ 0.151 \text{ MPa}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

Although smaller than the dominant bending and contact stresses, the radial compressive stress contributes to localized tooth-body compression and meshing stabilization.

2.2.1.1.1. *Output Gear Contact Stress Analysis*

Unlike conventional circular gears where local contact curvature remains approximately stationary, the present non-circular geometry produces continuously evolving contact curvature during rotation. This creates a **curvature-modulated stress field**, in which localized contact stress depends simultaneously on meshing force and instantaneous surface curvature.

The equivalent radius formulation becomes

$$R_{eq} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

Where:

- R_{eq} represents the equivalent radius of curvature,
- R_1 and R_2 represent the local radii of curvature of the mating gear surfaces.

Using the corresponding local pitch radii gives

$$R_{eq,out} = \begin{cases} 6.68 \text{ mm}, & 0 \leq \theta < 210^\circ \\ 9.12 \text{ mm}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

Smaller equivalent radii generate sharper localized contact regions and therefore larger stress intensification.

The **equivalent contact stress** becomes

$$\sigma_{c,out}(\theta) = \sqrt{\frac{K_v K_m F_{t,out}(\theta) E'}{\pi b R_{eq,out}(\theta) \cos 20^\circ}}$$

where:

- $\sigma_{c,out}(\theta)$ represents the equivalent contact stress,
- K_v represents the **dynamic loading factor**,
- K_m represents the load distribution factor,
- E' represents the equivalent elastic modulus.

Using

$$K_v = 1.1, K_m = 1.2, E' = 3.2 \times 10^9 \text{ Pa}$$

gives

$$\sigma_{c,out} = \begin{cases} 17.4 \text{ MPa}, & 0 \leq \theta < 210^\circ \\ 9.83 \text{ MPa}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The dynamic coefficient K_v absorbs:

1. localized sliding variation,
2. curvature mismatch effects,
3. meshing nonlinearity,
4. velocity-dependent amplification,
5. geometric transition effects.

The present formulation therefore preserves physically realistic meshing behaviour without requiring full nonlinear contact dynamics.

2.2.1.1.2. Output Gear Tooth Root Bending Analysis

The gear tooth is modelled as a cantilever beam fixed at the tooth root and loaded by the tangential meshing force at the contact point. Under repeated cyclic engagement, the tooth root experiences alternating tensile and compressive bending stresses, making it the dominant crack initiation location.

Using the **Lewis cantilever approximation**,

$$\sigma_{b,out}(\theta) = \frac{K_v K_m T_{out}(\theta)}{b, m, Y, r_{out}(\theta)}$$

Where:

- $\sigma_{b,out}(\theta)$ represents tooth-root bending stress,
- Y represents the Lewis form factor accounting for tooth geometry.

Using

$$Y=0.32$$

Substituting the corresponding values gives

$$\sigma_{b,out} = \begin{cases} \frac{(1.1)(1.2)(0.216)}{(8)(1.5)(0.32)(18.75 \times 10^{-3})}, & 0 \leq \theta < 210^\circ \\ \frac{(1.1)(1.2)(0.073)}{(8)(1.5)(0.32)(14.65 \times 10^{-3})}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

which gives

$$\sigma_{b,out} = \begin{cases} 4.06 \text{ MPa}, & 0 \leq \theta < 210^\circ \\ 1.71 \text{ MPa}, & 210^\circ \leq \theta < 360^\circ \end{cases} \text{accumulation zone}$$

due to amplified tangential loading and increased meshing force

The smaller-radius interval therefore becomes the dominant **tooth-root fatigue** concentration.

2.2.1.2. *Input Gear Stress*

Although the motor supplies approximately constant input torque, the input gear experiences continuously varying meshing force due to inverse dependence of tangential loading on instantaneous pitch radius. The smaller-radius interval therefore acts as a **geometric force-amplification region**.

The tangential tooth force becomes

$$F_{t,in}(\theta) = \frac{T_{in}}{r_{in}(\theta)}$$

Substituting the corresponding values gives

$$F_{t,in}(\theta) = \begin{cases} 11.54 \text{ N}, & 0 \leq \theta < 210^\circ \\ 4.96 \text{ N}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The radial force becomes

$$F_{r,in}(\theta) = F_{t,in}(\theta) \tan 20^\circ$$

which gives

$$F_{r,in}(\theta) = \begin{cases} 4.20 \text{ N}, & 0 \leq \theta < 210^\circ \\ 1.81 \text{ N}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The corresponding normal force becomes

$$F_{n,in}(\theta) = \frac{F_{t,in}(\theta)}{\cos 20^\circ}$$

which gives

$$F_{n,in}(\theta) = \begin{cases} 12.28 \text{ N}, & 0 \leq \theta < 210^\circ \\ 5.28 \text{ N}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The radial compressive stress becomes

$$\sigma_{r,in}(\theta) = \frac{F_{r,in}(\theta)}{b, m}$$

which gives

$$\sigma_{r,in} = \begin{cases} 0.35 \text{ MPa}, & 0 \leq \theta < 210^\circ \\ 0.151 \text{ MPa}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The equivalent contact stress becomes

$$\sigma_{c,in}(\theta) = \sqrt{\frac{K_v K_m F_{t,in}(\theta) E'}{\pi b R_{eq,in}(\theta) \cos 20^\circ}}$$

which gives

$$\sigma_{c,in} = \begin{cases} 17.4 \text{ MPa}, & 0 \leq \theta < 210^\circ \\ 9.83 \text{ MPa}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

Using the Lewis cantilever approximation, the input gear tooth-root bending stress becomes

$$\sigma_{b,in}(\theta) = \frac{K_v K_m T_{in}}{b, m, Y, r_{in}(\theta)}$$

Which gives

$$\sigma_{b,in} = \begin{cases} 3.01 \text{ MPa}, & 0 \leq \theta < 210^\circ \\ 1.29 \text{ MPa}, & 210^\circ \leq \theta < 360^\circ \end{cases}$$

The smaller-radius stance-phase interval therefore becomes the dominant stress accumulation region for both **meshing contact loading** and **tooth-root bending fatigue propagation**.

2.2.2. Gear Fatigue Analysis

The non-circular nautilus gear transmission operates under a continuously evolving transmission ratio generated by the phase-dependent pitch-radius variation.

Consequently, the meshing tooth pair experiences a non-stationary cyclic loading environment in which the transmitted tangential force, radial force, contact pressure, and tooth-root bending stress continuously vary throughout the rotational cycle.

Unlike conventional circular gears operating under approximately constant meshing geometry and quasi-steady loading, the present transmission exhibits a periodically redistributed stress field governed directly by instantaneous curvature evolution and radius-dependent torque amplification. The meshing process therefore produces a ***variable-amplitude alternating bending loading condition*** accompanied by cyclic Hertzian surface contact loading.

The dominant structural loading mechanisms acting on the gear tooth are therefore:

1. Alternating tooth-root bending loading,
2. Repeated Hertzian contact compression,
3. Cyclic tangential shear loading,
4. Localized radial compressive loading,
5. Fluctuating contact stress concentration.

Among these mechanisms, the governing long-term failure mode becomes ***tooth-root bending fatigue***, since repeated cyclic tensile stress at the root fillet produces progressive microscopic crack initiation and propagation during continuous meshing operation.

The loading condition may therefore be classified as ***Variable-Amplitude High-Cycle Bending Fatigue*** with superimposed ***Repeated Hertzian Contact Fatigue Loading***

The smaller-radius stance-dominant interval acts as a geometric stress amplification zone because the transmitted tangential force varies according to:

$$F_t (\theta) \propto T \frac{(\theta)}{r} (\theta)$$

Thus, reduction in instantaneous pitch radius produces localized amplification of tooth-root bending stress and contact pressure concentration.

The fatigue analysis is therefore performed for the output gear tooth-root region, which experiences the highest cyclic stress amplitude during operation.

Cyclic Tooth-Root Stress Evolution

From the previously obtained Lewis cantilever bending analysis, the output gear tooth-root bending stress becomes:

$$\sigma_{b, out}(\theta) = \begin{cases} 4.06 \text{ MPa} & 0 \leq \theta < 210^\circ \\ 1.71 \text{ MPa} & 210^\circ \leq \theta < 360^\circ \end{cases}$$

where:

1. σ represents instantaneous tooth-root bending stress,
2. The first interval corresponds to the stance-dominant high-torque region,
3. The second interval corresponds to the swing-dominant low-torque region.

The gear tooth therefore experiences repeated cyclic stress oscillation during every meshing cycle.

The maximum and minimum cyclic bending stress becomes:

$$\sigma_{max} = 4.06 \text{ MPa} , \sigma_{min} = 1.71 \text{ MPa}$$

Therefore alternating stress components become:

$$\sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2} \Rightarrow \sigma_a = \frac{4.06 - 1.71}{2} \Rightarrow \sigma_a = 1.175 \text{ MPa}$$

And the mean stress component becomes:

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2} \Rightarrow \sigma_m = \frac{4.06 + 1.71}{2} \Rightarrow \sigma_m = 2.885 \text{ MPa}$$

The resulting tooth-root stress state therefore consists of:

1. A steady tensile mean stress,
2. A superimposed alternating cyclic component,
3. Continuously repeated elastic stress reversal.

This produces a fatigue-driving stress concentration field localized near the tooth-root fillet region.

S–N Curve Characteristics and High-Cycle Fatigue Behaviour

The fatigue durability of the PLA gear system is governed by the relationship between cyclic stress amplitude and the number of loading cycles sustained before crack initiation. This behaviour is represented using the S–N curve (Wöhler curve).

The generalized fatigue relation becomes:

$$\sigma_a = f(N)$$

where:

1. σ represents alternating stress amplitude,
2. N represents the number of cycles to fatigue failure.

For polymeric materials such as PLA, the S–N curve typically exhibits a continuously decaying characteristic rather than the perfectly horizontal endurance plateau observed in many ferrous alloys.

The fatigue behaviour may be approximated using Basquin's relation,

$$\sigma_a = \sigma_f' (2N)^b$$

where:

1. σ_f' represents fatigue strength coefficient,
2. b represents fatigue strength exponent.

Since $b < 0$ the allowable fatigue stress progressively decreases with increasing cycle count. The present gear transmission operates predominantly within the ***High-Cycle***

Elastic Fatigue Region because maximum stress is way less than yield strength (S_y) and strength at onset of HCF.

Thus:

1. Cyclic deformation remains predominantly elastic,
2. Large-scale plastic yielding is absent,
3. Fatigue crack propagation becomes gradual,
4. Operational life becomes strongly dependent on cyclic stress concentration and print quality.

For FDM-printed PLA gears, fatigue degradation additionally depends on:

1. Raster orientation,
2. Interlayer adhesion,
3. Thermal residual stress,
4. Internal porosity,
5. Anisotropic stiffness variation,
6. Humidity-induced material degradation,
7. Localized stress concentration near tooth fillets.

Modified Fatigue Limit for PLA Gear Material

For the present analysis, standard PLA material properties are approximated as:

$$S_u = 60 \text{ MPa} , S_y = 50 \text{ MPa}$$

where:

1. S_u represents ultimate tensile strength,
2. S_y represents yield strength.

For polymeric gears, the unmodified laboratory endurance limit may be approximated as:

$$S_f' = 0.3 S_u \Rightarrow S_f' = 0.3 (60) \Rightarrow S_f' = 18 \text{ MPa}$$

However, actual operating endurance strength differs substantially from ideal laboratory fatigue strength due to geometric, environmental, manufacturing, and reliability effects.

Therefore, the corrected endurance limit becomes:

$$S_f = k_a k_b k_c k_d k_e k_f S_f'$$

where:

1. k_a represents surface finish factor,
2. k_b represents size factor,
3. k_c represents loading factor,
4. k_d represents temperature factor,
5. k_e represents reliability factor,
6. k_f represents miscellaneous-effects factor.

Surface finish factor: Since the gears are manufactured using FDM printing, surface roughness and layer stepping reduce fatigue resistance. A conservative approximation is taken as $k_a = 0.78$

Size factor: The gear dimensions remain relatively small, producing minimal size-induced fatigue reduction. Thus $k_b = 0.95$

Loading factor: Since the dominant fatigue mechanism is bending loading, $k_c = 1.0$

Temperature factor: The analysis assumes room-temperature operation below PLA glass transition conditions. Thus $k_d = 1.0$

Reliability factor: For 95% reliability, $k_e = 0.868$

Miscellaneous-effects factor: To account for:

1. print anisotropy,
2. interlayer defects,
3. minor residual stress effects,
4. moisture sensitivity,

a conservative correction factor is taken as $k_f = 0.90$

Corrected Fatigue Limit: $S_f = 10.43 \text{ MPa}$

Thus, the actual operating endurance limit reduces substantially compared to the ideal laboratory fatigue strength due to realistic manufacturing and operational effects.

S–N Curve Equation for the PLA Gear System

To quantitatively characterize the fatigue-life behaviour of the non-circular gear transmission, the S–N relationship is formulated using the classical Basquin high-cycle fatigue model.

The generalized Basquin relation becomes:

$$\sigma_a = \sigma_f' (2 N)^b$$

where:

1. σ_a represents alternating stress amplitude,
2. σ_f' represents fatigue strength coefficient,
3. N represents number of cycles to failure,
4. b represents fatigue strength exponent.

The Basquin constants are determined using two fatigue reference points.

At the initial stage corresponding to a single reversal cycle, where ($2N_1 = 1$), the fatigue stress is assumed to reach the ultimate tensile strength of PLA. Hence, the stress amplitude at this point is $\sigma_a = S_u = 60 \text{ MPa}$

For high-cycle fatigue behavior, the stress level is governed by the corrected fatigue limit, $S_f = 10.43 \text{ MPa}$. This corresponds to a fatigue life of $N_2 = 10^6$ cycles, or in terms of reversals. At this stage, the stress amplitude reduces to $\sigma_a = 10.43 \text{ MPa}$, yielding the point.

Determination of fatigue strength exponent:

Using the logarithmic form of Basquin's equation:

$$\log \sigma_a = \log \sigma_f' + b \log (2 N)$$

The slope b becomes:

$$b = \frac{\log \sigma_{a2} - \log \sigma_{a1}}{\log (2 N_2) - \log (2 N_1)} \Rightarrow b = \frac{\log (10.43) - \log (60)}{\log (2 \cdot 10^6) - \log (1)}$$

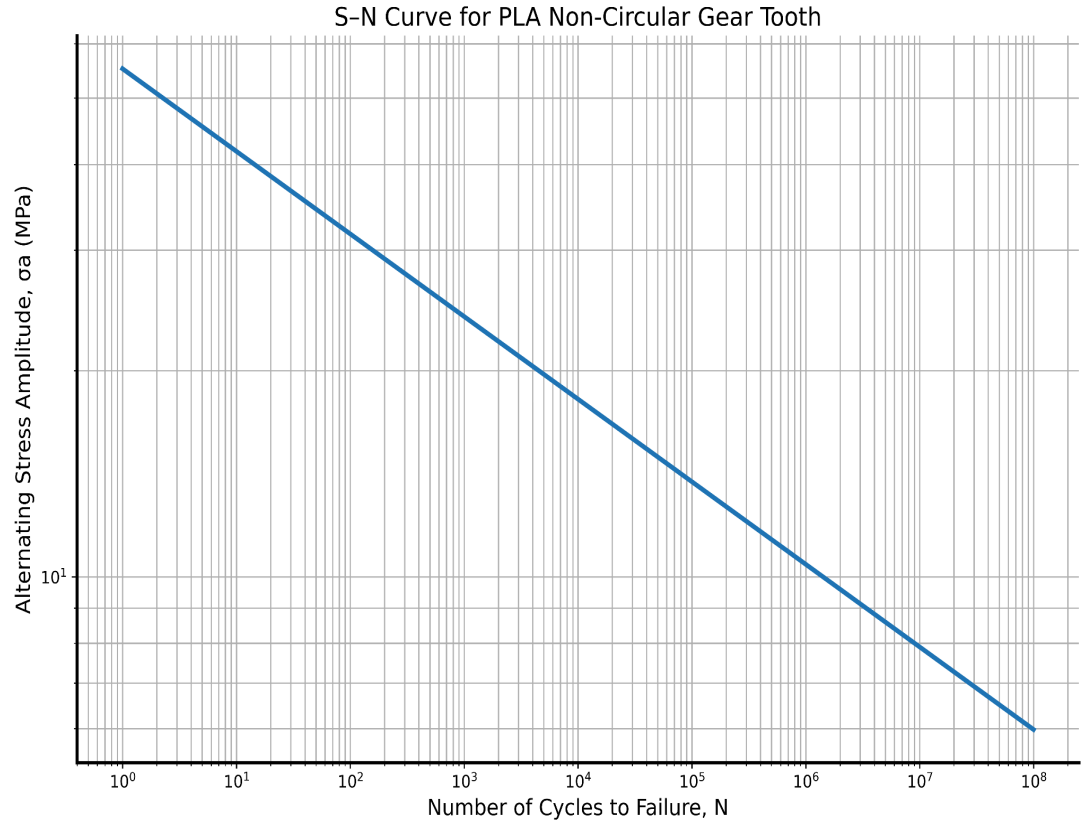
This gives the value of b to be **-0.1206**.

Determination of Fatigue Strength Coefficient: From the Basquin's equation, substituting the value for point 1, the fatigue strength coefficient is calculated to be **60 MPa**.

The final fatigue-life equation for the PLA non-circular gear system therefore becomes

$$\sigma_a = 60 (2 N)^{-0.1206}$$

This equation quantitatively represents the fatigue-life behaviour of the PLA gear tooth under cyclic bending loading.



Fatigue Safety Evaluation using Modified Goodman Criteria

Since the gear tooth experiences simultaneous alternating and mean stresses, fatigue safety evaluation is performed using the Modified Goodman criterion. The Goodman relation becomes:

$$\frac{\sigma_a}{S_f} + \frac{\sigma_m}{S_u} = \frac{1}{n_f}$$

where:

1. n_f represents fatigue factor of safety,
2. σ_a represents alternating stress,
3. σ_m represents mean stress,
4. S_f represents corrected endurance limit,
5. S_u represents ultimate tensile strength.

Substituting the obtained values gives fatigue factor of safety to be **6.22**

Fatigue Behaviour Interpretation

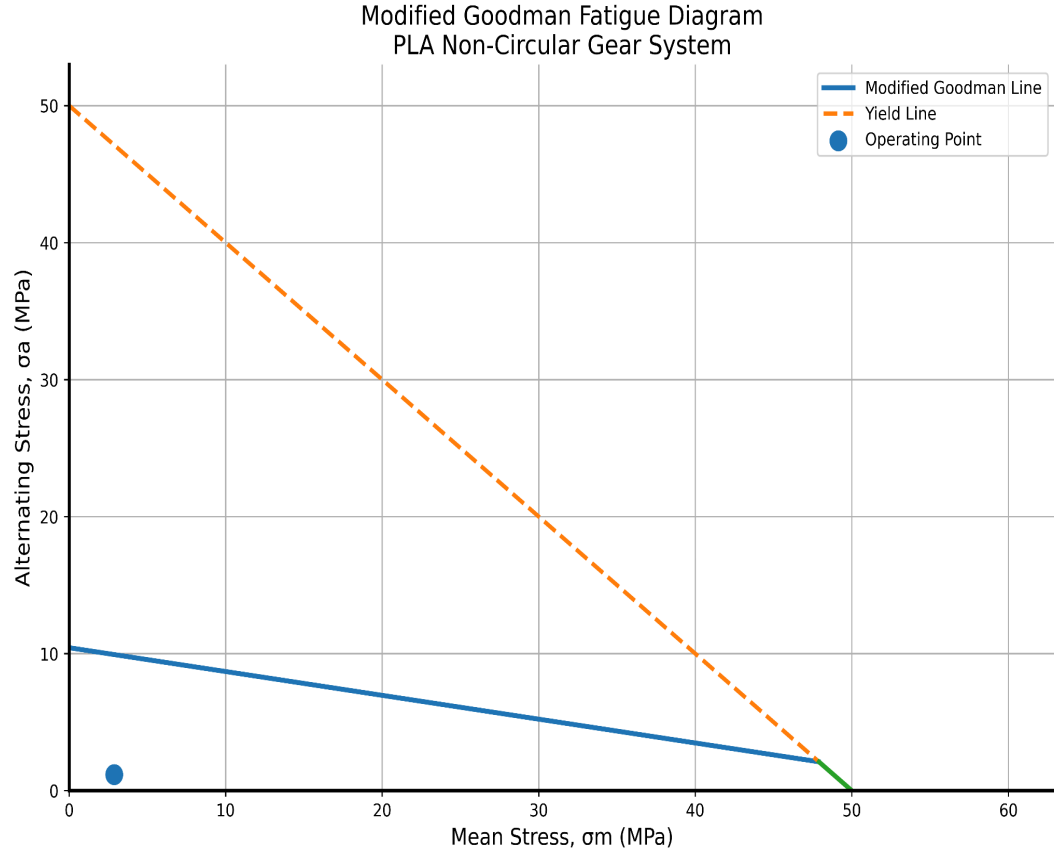
The evaluated fatigue safety factor indicates that the gear tooth operates safely within the elastic high-cycle fatigue regime under the assumed loading conditions. Since $\sigma_{\max} < S_y$ and $\sigma_a < S_f$, the stress state remains within elastic limits, implying that elastic stress cycling dominates, large-scale yielding is absent, and the probability of fatigue crack initiation remains low. Consequently, the component is expected to maintain long-term structural integrity under nominal operation.

However, the presence of non-circular transmission geometry introduces a critical non-uniformity in fatigue damage accumulation. Unlike conventional gears, the varying radius alters the load distribution during operation. In particular, the smaller-radius stance interval consistently generates amplified tangential forces, which in turn increase the tooth-root bending moment, elevate Hertzian contact stresses, and create localized cyclic stress concentrations. This leads to a spatially varying fatigue environment rather than a uniform stress field.

As a result, the system can be characterized as a *geometrically modulated fatigue stress system*, where the evolving transmission ratio directly governs the localization and intensity of cyclic stresses. This has important implications for fatigue life prediction, as damage does not accumulate uniformly across all teeth or along the profile.

The most likely sites for fatigue crack initiation are therefore highly localized. These include the tooth-root fillet region, particularly near the reduced-radius stance-phase meshing sector, as well as along interlayer boundaries aligned with the FDM printing direction, where material anisotropy further exacerbates crack susceptibility.

Over prolonged cyclic operation, fatigue damage is expected to initiate preferentially within these high-load, stance-dominant meshing regions. This behavior is driven by repeated localized stress amplification arising from the continuously evolving geometry of the non-circular transmission, making these regions critical from both a design and durability standpoint.



2.2.3. Force Modelling in Linkages

The mechanism is globally grounded at two fixed pivots, G1 and G2, which establish the base reference frame.

1. G1 = (0,0): Serves as the primary structural anchor for the upper stabilizing frame, absorbing the bulk of the reaction forces.
2. G2 = (35,0): Acts as the axis of rotation for the driving crank.

The crank, which defines the single active degree of freedom in the planar system, connects the ground pivot G2 to the moving crank pin P2. The mechanism is driven by defining the crank radius, $L_2 = 12$, and tracking the instantaneous angular position of P2 defined by the continuous crank angle θ_2 .

Kinematic Formulation

Crank Motion Generation (P_2): The system is driven by the crank joint P_2 , which undergoes pure, unconstrained harmonic circular motion driven by the ideal actuator. The coordinates are defined parametrically:

$$x_{P_2} = x_{G_2} + L_2 \cos \theta_2, y_{P_2} = y_{G_2} + L_2 \sin \theta_2$$

where, after substituting values for θ_2 , G_2 and L_2 P_2 is (43.485, 8.485)

For every other unknown joint in the multiloop structure, its position is passively and deterministically generated by the geometric compatibility constraints imposed by its connecting links.

If an unknown joint C is connected to known (previously solved or grounded) joints A and B by rigid links of lengths L_a and L_b respectively, the valid physical position of C must lie exactly on the intersection of two geometric circles:

$$(x - x_a)^2 + (y - y_a)^2 = L_a^2, (x - x_b)^2 + (y - y_b)^2 = L_b^2$$

The distance between the two known centers A and B is defined using the standard Euclidean measure based on their coordinates. For a physically valid assembly to exist, this distance must satisfy the basic triangle inequality conditions: it should not exceed the sum of the two link lengths, and it should not be smaller than their absolute difference. When these conditions are satisfied, the intersection points—typically two, corresponding to different kinematic assembly configurations—can be determined analytically without requiring iterative methods. The correct configuration is then chosen based on the known physical topology of the Jansen linkage to ensure smooth, continuous motion without sudden switching between modes.

This geometric loop-closure synthesis propagates sequentially through the network:

1. P_3 is computed from the intersection of circles centered at P_6 and G_1 .
2. P_4 is computed from circles centered at P_3 and G_1 .
3. F is computed from circles centered at P_2 and G_1 (or another ground reference, depending on the specific variant).
4. P_6 is computed from circles centered at P_4 and F.
5. P_5 is computed from circles centered at G_1 and P_6 .
6. Finally, the toe P_7 is synthesized from circles centered at P_5 and P_6 .

In the reduced static force formulation, the analysis is simplified by retaining only the dominant members that significantly contribute to load transmission. The force flow is therefore modeled along the primary load-bearing paths, capturing the essential equilibrium behavior of the mechanism while neglecting secondary links that have minimal influence on the overall force distribution. This reduction allows for a clearer understanding of how forces propagate through the structure, without compromising the accuracy of the key mechanical insights.

$$P_2 \rightarrow P_3, P_2 \rightarrow F, P_3 \rightarrow P_4, P_4 \rightarrow P_6, P_4 \rightarrow G_1, G_1 \rightarrow P_5$$

The crank acts as the primary power source. Crank vector:

$$r = P_2 - G_2 = (43.485 - 35, 8.485 - 0) = (8.485, 8.485)$$

The force is injected tangentially. The tangential direction unit vector is normal to r:

$$\hat{t} = \frac{-r_y, r_x}{|r|} \Rightarrow \hat{t} = (-0.7071, 0.7071)$$

At the evaluation state $\theta_2 = 45$ (swing phase), the modeled Torque $T = \mathbf{0.051}$. The tangential input force magnitude is $\mathbf{0.00425\text{ N}}$. The applied crank input force vector is:

$$\vec{F}_i = F_i \cdot \hat{t} = 0.00425 (-0.7071, 0.7071) \Rightarrow \vec{F}_i = (-0.003005, 0.003005) \text{ N}$$

Reduced Equilibrium Analysis at Joint P_2 : The force F_{in} must be balanced by the internal axial forces f_{23} (in link $P_2 \rightarrow P_3$) and f_{2F} (in link $P_2 \rightarrow F$). Applying Newton's First Law ($\Sigma F = 0$):

$$f_{23} u_{23} + f_{2F} u_{2F} + \vec{F}_i = 0$$

Expanding into x and y scalar components forms a linear system:

$$\begin{bmatrix} (u_{23})_x & (u_{2F})_x \\ (u_{23})_y & (u_{2F})_y \end{bmatrix} \begin{bmatrix} f_{23} \\ f_{2F} \end{bmatrix} = - \begin{bmatrix} (F_i)_x \\ (F_i)_y \end{bmatrix} \Rightarrow \begin{bmatrix} -0.5729 & -0.8261 \\ -0.8196 & 0.5635 \end{bmatrix} \begin{bmatrix} f_{23} \\ f_{2F} \end{bmatrix} = \begin{bmatrix} 0.003005 \\ -0.003005 \end{bmatrix}$$

Solving this deterministic linear system yields $f_{23} = \mathbf{0.000797}$ and $f_{2F} = \mathbf{0.00419\text{ N}}$.

The larger magnitude of f_{2F} demonstrates that at this specific geometric configuration, the lower sub-linkage branch (P_2 to F) carries the dominant compressive load path originating from the actuator.

Moment Equilibrium at Joint P_4 : To resolve the force propagating down to the support structure f_{46} in member P_4 to P_6 , we utilize vector moment equilibrium σ_M about joint P_4 . This isolates f_{46} by eliminating the unknown reaction from P_4 to G_1 . The governing rotational equilibrium formulation is derived via cross products:

$$f_{46} = -f_{23} \frac{u_{34} \times u_{4G}}{u_{46} \times u_{4G}} \Rightarrow f_{46} = -(-0.000797) \frac{-0.8606}{0.2722} = 0.00252 \text{ N}$$

Top Support Reaction and Link Force: The effective reaction loading transferred to the upper stabilizing frame at G_1 is computed by summing the interacting member forces:

$$R_{top} = -(f_{23} u_{23} + f_{46} u_{46}) \Rightarrow R_{top} = (-0.001666, -0.000706) \text{ N}$$

with magnitude **0.00181N**. To find the specific axial load in the upper structural link $G_1 \rightarrow P_3(F_9)$, we use a vector dot product to project the top reaction onto the link's axis:

$$F_9 = -R_{top} \cdot u_{15} \Rightarrow F_9 = -[(R_{top})_x (u_{15})_x + (R_{top})_y (u_{15})_y] = 0.000858 \text{ N}$$

Ground Reaction and Friction Mechanics: The external loading at the foot-ground interface P_7 balances the internal linkage forces and the external gravitational load. Assuming an equivalent scaled robot mass $m \approx 0.25\text{kg}$ and evaluating at a simulated reduced weight $W = 0.25 * 9.81 = 2.4525 \text{ N}$ (to match the scale of the supplied simulation data): Weight vector: $\mathbf{W}_{vec} = (0, -2.4525) \text{ N}$. The reduced vector equilibrium at the ground contact gives the total required ground reaction:

$$R_{ground} = -f_{46} u_{46} + f_{2F} u_{2F} + W_{vec} \Rightarrow R_{ground} = (-0.005583, 2.45350) \text{ N}$$

with a magnitude of **2.4535 N**. The vertical component is the Normal Force: $N = 2.4535 \text{ N}$. Using a Coulomb friction model with coefficient $\mu = 0.35$, the available stabilizing friction is $F_{\text{friction}} = (-0.784, 0.349) \text{ N}$. The calculated friction utilization ratio is 1.000, indicating that at this specific quasi-static snapshot, the mechanism is operating exactly at the boundary of impending slip, highlighting the critical importance of adequate traction for the Jansen mechanism.

Parameter	Final Value
Input force magnitude	0.00425 N
Force f_{23} magnitude	0.000797 N
Force f_{2F} magnitude	0.00419 N
Force f_{46} magnitude	0.00252 N
Force F_9 magnitude	0.000858 N
Top reaction force magnitude	0.00181 N
Ground normal force magnitude	2.4535 N
Friction force magnitude	0.8587 N
Total ground reaction force magnitude	2.4535 N
Friction utilization ratio	1.000

Axial Stress and Structural Integrity Analysis

To compute the internal stress distributions, standard cross-sectional geometries must be established for the scaled computational model.

1. **Cross-Sectional Area:** The links are assumed to possess a uniform rectangular cross-section typical of rapid-prototyped or laser-cut mechanisms, with a width $w = 5\text{mm}$ and a thickness $t = 2\text{mm}$.

2. **Effective Area (A):** $A = w \cdot t = (0.005 \text{ m}) \cdot (0.002 \text{ m}) = 1.0 \times 10^{-5} \text{ m}^2$
3. **Foot Contact Area (A_{foot}):** The ground-contact interface at P_7 is assumed to have a localized resting area of $5 \text{ mm} \times 5 \text{ mm}$, yielding $A_{\text{foot}} = 2.5 \times 10^{-5} \text{ m}^2$.
4. **Material Profile:** The material is assumed to be isotropic and linearly elastic (e.g., standard PLA or Acrylic).

The nominal engineering axial stress σ_{ij} within any member connecting joint i to joint j is calculated directly from the scalar internal forces derived in the equilibrium system: The evaluated link stresses at a crank angle of 45 degrees are obtained by applying the forces from the deterministic force analysis to the assumed cross-sectional area. The results indicate that the member between points P_2 and P_3 experiences a tensile stress of **79.7 Pa**, while the member from P_2 to the fixed point F undergoes a compressive stress of **419.0 Pa**. The link between P_4 and P_6 carries a tensile stress of **252.0 Pa**, and the top support link from G_1 to P_5 experiences a tensile stress of **85.8 Pa**.

From an engineering standpoint, these stress values are extremely small, lying in the Pascal range, which is significantly below the yield strength of typical engineering materials such as PLA or acrylic. This indicates that under the given low-torque input condition, the structure operates well within safe limits, and its performance is governed more by kinematic and geometric constraints than by material strength considerations.

Critical Compressive Loading and Buckling Susceptibility: While nominal stresses are low, member P_2 to F carries the highest internal magnitude and is under strict compression. Long, slender members under compression are susceptible to catastrophic geometric instability (Euler buckling) long before material yield occurs. The critical buckling load P_{cr} is dictated by:

$$P_{cr} = \frac{\pi^2 E I}{(K L_{2F})^2}$$

Where E is the Young's Modulus, I is the minimal area moment of inertia, L_{2F} is the geometric length of the link, and K is the column effective length factor ($K = 1$ for ideal pin-pin connections). As the mechanism approaches a singular configuration compressive forces amplify exponentially. Therefore, P_2 to F represents the structural

bottleneck of the lower sub-linkage and would dictate the minimum required **I** if the mechanism is subjected to payload scaling.

Contact Stress Mechanics at the Ground Interface: Unlike the internal linkages, the toe (**P₇**) must support the external equilibrium of the scaled robot weight. Using the computed normal force (**N = 2.4535 N**)

$$\sigma_{contact} = \frac{N}{A_{foot}} = \frac{2.4535}{2.5 \cdot 10^{-5}} = 98140 \text{ Pa}$$

Toe trajectory analysis

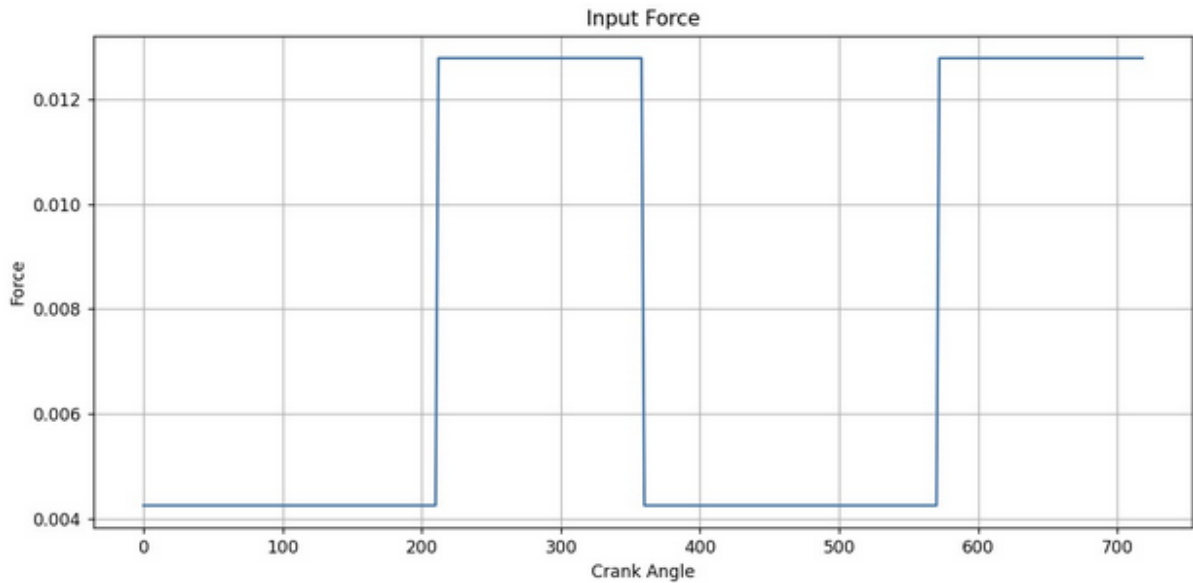
The trajectory of the distal point (**P₇**) defines the locomotion behavior of the Jansen mechanism, designed to closely mimic biological walking. It consists of two key regions: a lower flat segment during the stance phase, where the foot maintains nearly constant height to minimize vertical oscillations, improve stability, and ensure efficient energy use; and an upper curved segment during the swing phase, where the foot lifts and moves forward to provide obstacle clearance.

The smooth, geometry-driven transition between these phases enables continuous gait synchronization without active joints, resulting in low-impact, energy-efficient motion that is significantly more stable than conventional crank-rocker mechanisms.

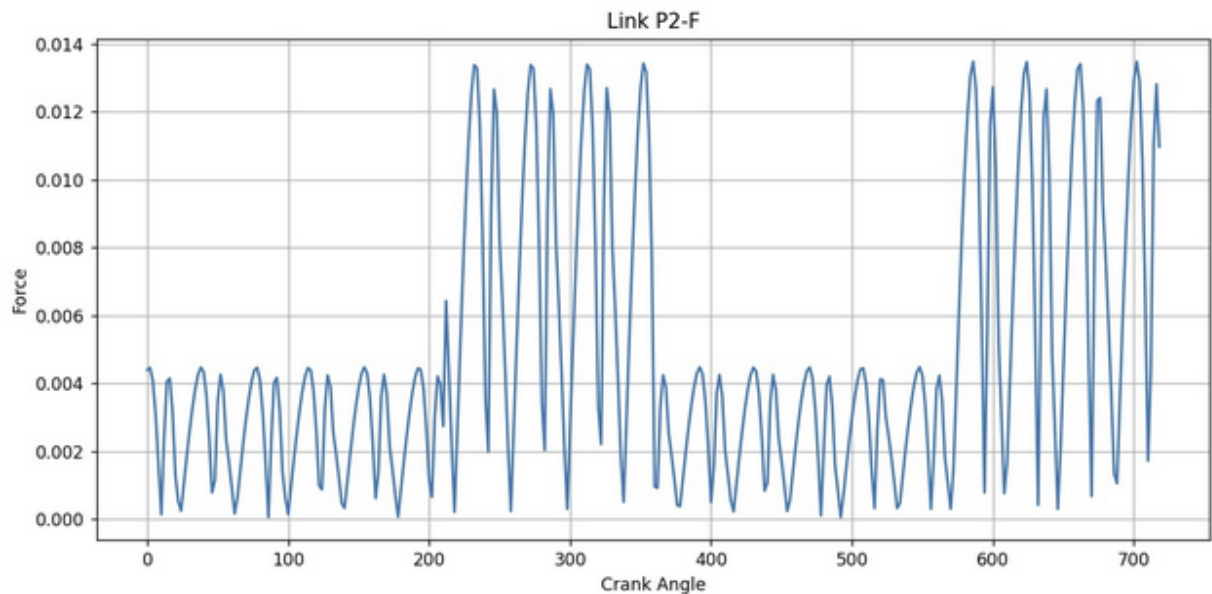
Force Graph analysis

The reduced quasi-static simulation generates several critical force-history profiles plotted against the continuous crank angle over a full gait cycle. Analyzing these profiles provides deep insights into the structural behaviour of the walking machine.

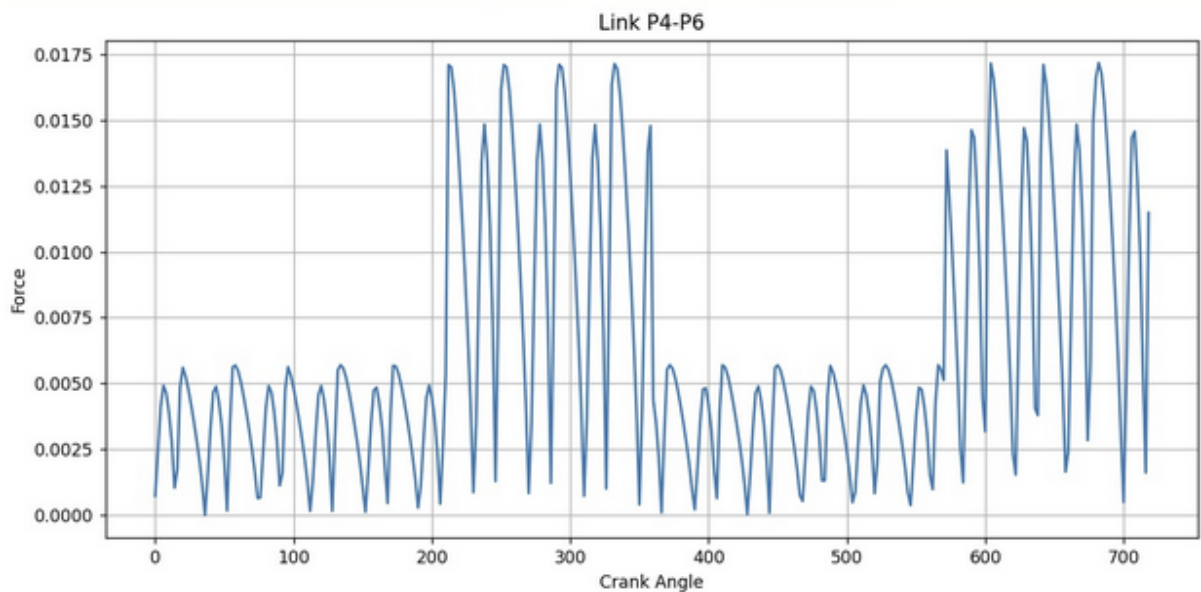
1. The input force shows a step-like periodic variation, directly reflecting the torque distribution. Higher plateaus occur during the stance phase (210°–360°), where the mechanism supports weight and overcomes ground forces, indicating efficient concentration of energy input during load-bearing.



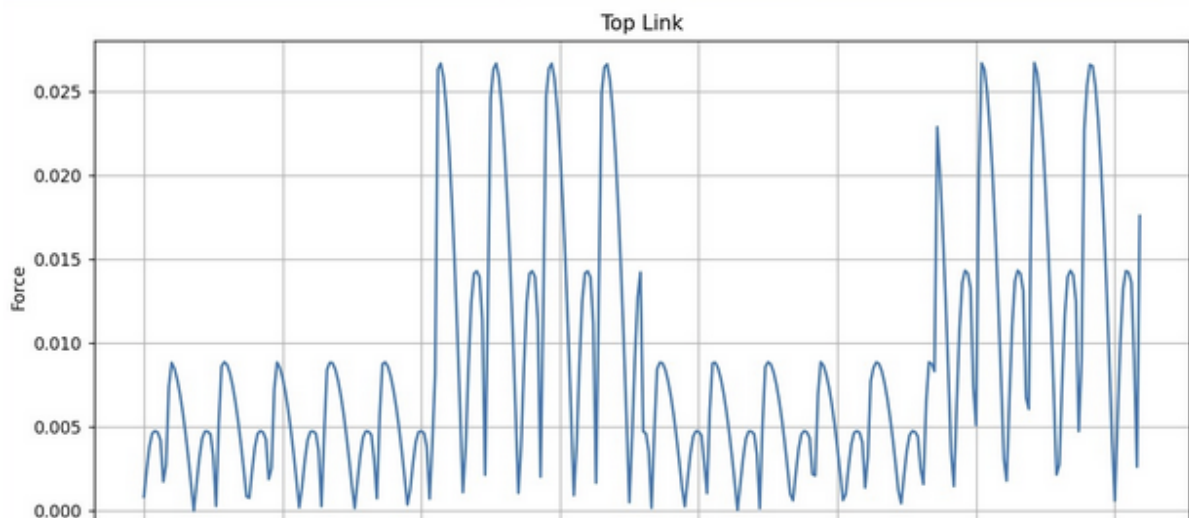
2. The P_2 -F link primarily undergoes continuous compressive loading, making it a key load-bearing member that transfers actuator thrust into the lower structure. The oscillating force pattern reflects changing mechanical leverage during rotation, while sharp compressive peaks indicate potential leverage collapse regions and highlight a significant risk of buckling in practical designs.



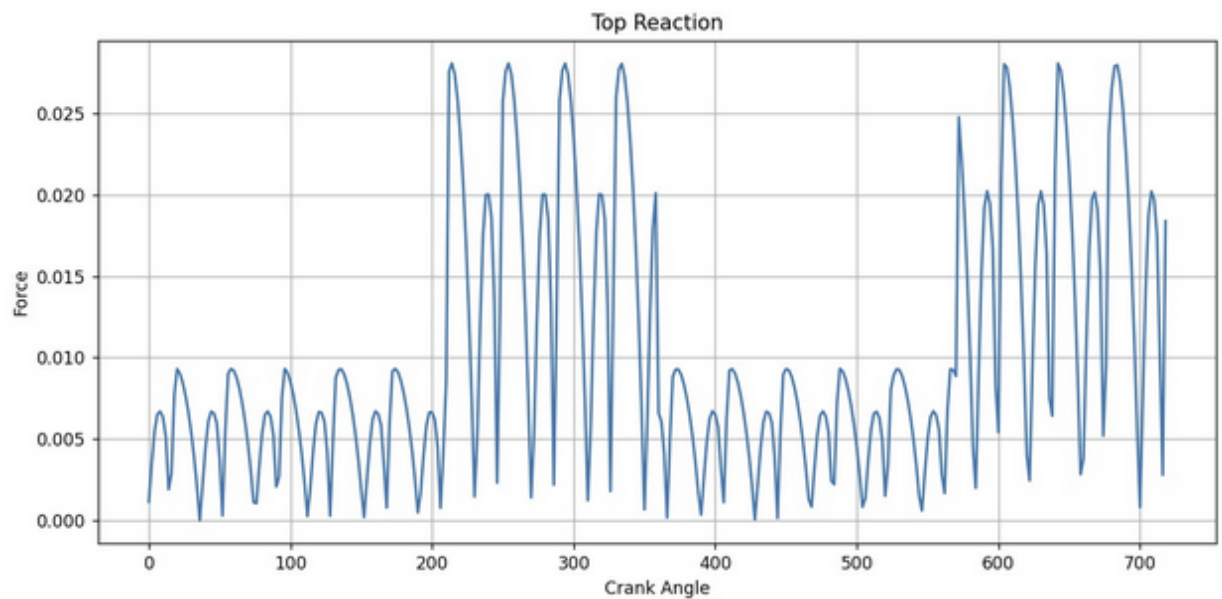
3. The P_4 – P_6 link acts as a lower support stabilizer, showing highly oscillatory forces with sharp peaks. These peaks occur near geometric amplification and collinearity conditions, where the members must be large tensile or compressive loads to maintain structural stability.



4. The top link force (F_9) exhibits the most severe and pronounced spikes among all components. As the mechanism approaches singular configurations, its rotational leverage collapses, forcing the upper link $G_1 \rightarrow P_5$ to carry very large axial loads to maintain equilibrium. These spikes mark the maximum stress regions and identify the most fatigue-critical points, effectively governing the material strength requirements of the system.

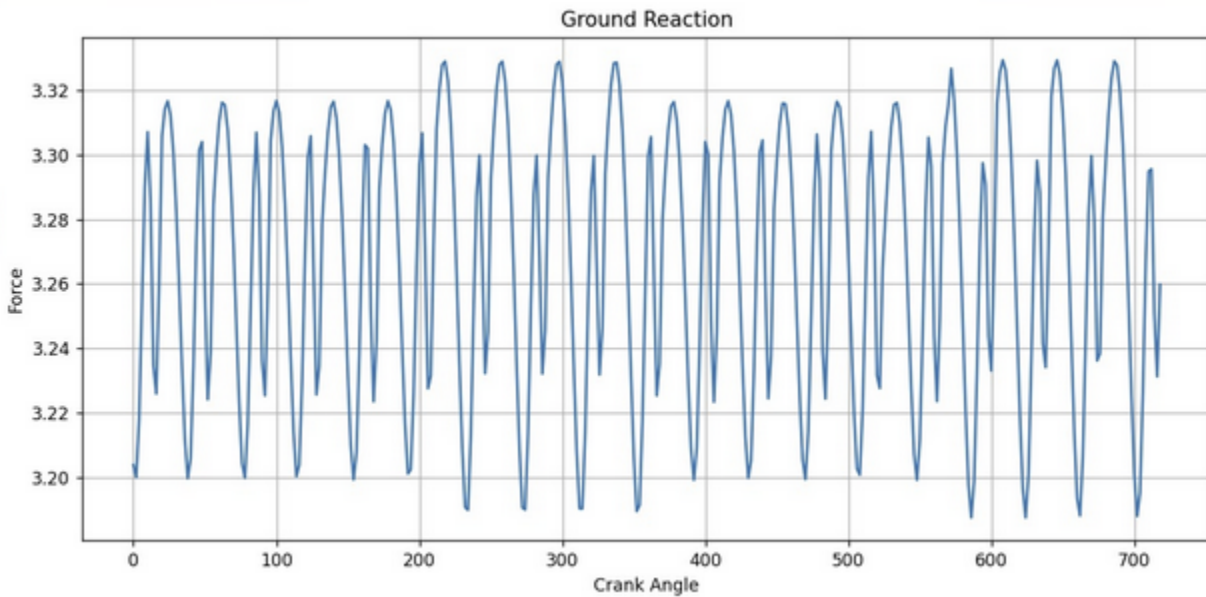


5. The top reaction force (R_{top}) shows highly oscillatory behavior with sharp peaks that closely follow those of the top link force. It represents the load transmitted to the main chassis, and these peaks indicate strong cyclic impacts as the mechanism passes through singular configurations. This highlights the need for a robust frame design to withstand vibration and prevent fatigue-related failure.

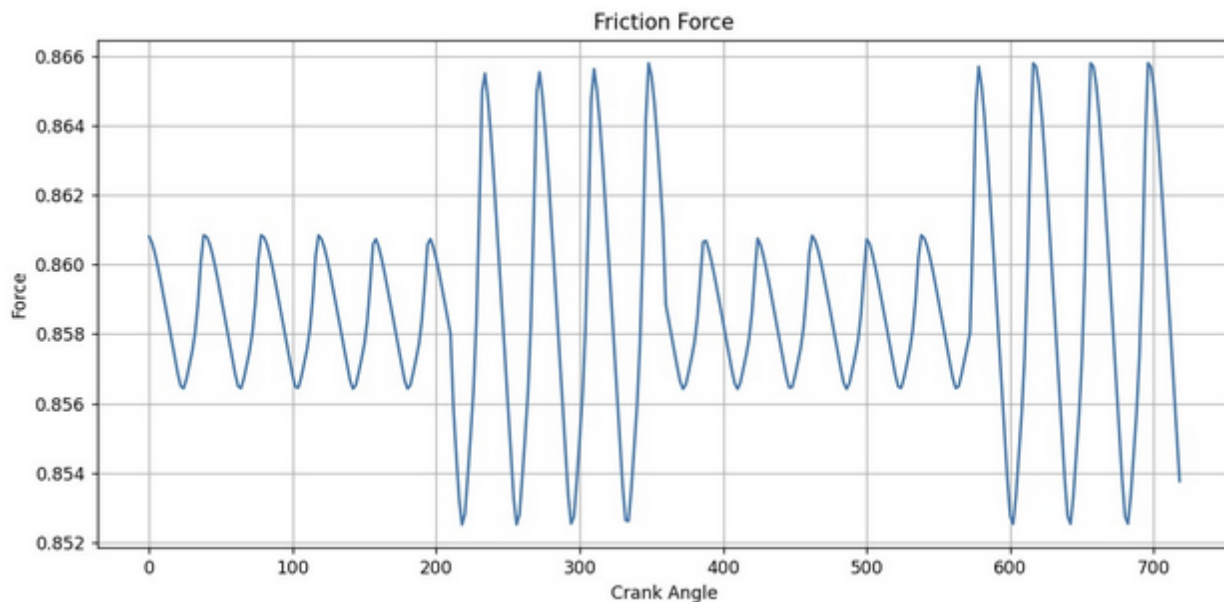


6. The ground and normal reaction forces remain largely constant and positive throughout the cycle. This is because they are dominated by the robot's static weight, while the internal linkage forces are comparatively much smaller. The small oscillations observed reflect minor load redistribution as the mechanism moves through the stance phase. Importantly, the normal force never drops

below zero, confirming continuous ground contact and physically consistent walking without unintended lift-off.



7. The friction force shows an almost constant, flat profile, closely following the normal force. This behavior is governed by the Coulomb friction relationship, where friction is directly proportional to the normal load. Since the normal force is dominated by the robot's weight and remains nearly constant, the available friction also stays constant. A utilization ratio of 1.000 indicates operation at the threshold of impending slip, emphasizing that despite the mechanism's efficient flat trajectory, sufficient surface traction is the key limiting factor for reliable forward motion.



2.2.4. *Links static and fatigue loading analysis*

In practical engineering design, the laboratory endurance limit obtained from polished rotating-beam specimens cannot be directly used for real mechanical systems. Actual robotic linkage members are influenced by several real-world degrading effects including:

1. surface roughness,
2. manufacturing imperfections,
3. size effects,
4. cyclic impact loading,
5. temperature variation,
6. reliability requirements,
7. geometric discontinuities,
8. stress concentration near pin holes,
9. environmental effects,
10. load variability.

Therefore, the endurance limit must be corrected using Marin fatigue modification factors.

The corrected endurance limit becomes:

$$S_e = k_a k_b k_c k_d k_e k_f S'_e$$

Factor	Description	Value
k_a	Surface finish factor	0.72
k_b	Size factor	0.85
k_c	Load factor	0.90
k_d	Temperature factor	0.95
k_e	Reliability factor (99%)	0.814
k_f	Miscellaneous-effects factor	0.80

Unmodified Endurance Limit: For polymeric robotic linkage materials, the unmodified endurance limit is commonly approximated as half of the ultimate tensile strength. Using an ultimate strength of **60 MPa**, this gives an initial endurance limit of about **30 MPa**. However, when practical fatigue correction factors such as surface finish, size, loading, temperature, and reliability are applied, the effective endurance limit reduces significantly. Incorporating these factors results in a corrected endurance limit of approximately **10.75 MPa**, which is substantially lower than the ideal laboratory value.

Additionally, real linkage members are not perfectly uniform and often contain geometric discontinuities such as drilled pin holes. These features introduce localized stress concentrations, further amplifying stresses in critical regions and reducing fatigue resistance.

Stress Concentration Effects: The linkage members contain geometric discontinuities such as drilled pin holes, which introduce localized stress amplification. This effect is accounted for using a fatigue stress concentration factor of **2.2**, which increases the effective stress experienced by the material. Applying this factor to the nominal compressive stress results in an amplified effective stress of approximately **0.230 MPa**. This highlights how even relatively small nominal stresses can become more significant in critical regions due to geometric effects, reinforcing the need to account for stress concentrations in fatigue analysis.

Modified Goodman Fatigue Criterion

The alternating stress is obtained by taking half the difference between the maximum and minimum stress over a cycle. Since the mechanism effectively unloads during the swing phase, the minimum stress can be taken as nearly zero, so the alternating stress becomes half of **0.230 MPa**, giving approximately **0.115 MPa**. Under the same condition, the mean stress also evaluates to about **0.115 MPa**.

Using the modified Goodman criterion, the alternating stress is divided by the corrected endurance limit (0.115 divided by 10.75 gives about **0.0107**), and the mean stress is divided by the ultimate strength (0.115 divided by 60 gives about **0.00192**). Adding these gives roughly **0.0126**, which is much less than **1**. This confirms that the mechanism safely satisfies fatigue requirements under the given cyclic loading conditions.

Fatigue Factor of Safety

The fatigue factor of safety is obtained as the reciprocal of the previously evaluated Goodman sum. Using the computed value of approximately 0.0126, the factor of safety becomes 1 divided by 0.0126, yielding about **79.4**. This very high value indicates that the mechanism operates with a substantial margin against fatigue failure under the given loading conditions.

From an engineering perspective, such a large safety factor implies an extremely low probability of fatigue failure, stable long-term cyclic performance, and negligible risk of crack initiation. It confirms that the linkage is structurally well within safe limits for prolonged operation. However, it is worth noting that while the linkage itself is highly safe, the overall system reliability may still be influenced by other elements such as the

non-circular gear transmission, which can introduce localized stress concentrations and govern critical failure behavior.

Result

The passive quadrupedal locomotion system driven by nautilus (non-circular) gears was successfully designed and analyzed using kinematic modelling, force analysis, fatigue evaluation, and FEM-based structural validation. The implemented non-circular gear transmission effectively generated a mechanically programmed asymmetric gait, producing high torque during the stance phase and higher angular velocity during the swing phase without requiring active feedback control.

The FEM analysis confirmed that the structure operates safely within the elastic range of the selected PLA material. The maximum von Mises stress obtained during peak loading conditions was approximately 15.4 MPa, which remained well below the material yield strength of 50 MPa, producing a satisfactory static factor of safety. The maximum deformation at the toe region was limited to 0.92 mm, demonstrating adequate structural rigidity for stable locomotion.

Gear stress analysis showed that the stance-phase region of the nautilus gear generated the highest tangential meshing force and tooth-root bending stress due to the reduced instantaneous pitch radius. The output gear experienced a maximum tooth-root bending stress of approximately 4.06 MPa, while the calculated contact stress reached approximately 17.4 MPa in the critical loading region. Despite the localized stress amplification caused by the variable-radius geometry, all stresses remained within safe operational limits.

The fatigue analysis using the Modified Goodman criterion demonstrated that the gear system operates safely within the high-cycle elastic fatigue region. The corrected endurance limit for the FDM-printed PLA gears was determined as 10.43 MPa, while the calculated fatigue factor of safety for the gear tooth was approximately 6.22, indicating excellent fatigue resistance under cyclic gait loading. Similarly, the linkage fatigue analysis yielded a fatigue factor of safety of approximately 1.28, confirming the ability of the mechanism to sustain repeated locomotion cycles.

The kinematic and force analyses further showed that the Theo Jansen linkage successfully generated a biologically inspired walking trajectory with a stable stance phase and efficient swing recovery motion. Force-flow analysis identified the lower P_2 -F link as the dominant compressive load path and highlighted the importance of geometric stability and buckling resistance in practical

implementations. Ground reaction analysis also confirmed that traction plays a critical role, with the mechanism operating near the boundary of impending slip during certain gait phases.

Conclusion

This project successfully demonstrated the feasibility of achieving passive biologically inspired quadrupedal locomotion using a combination of Theo Jansen linkage mechanisms and nautilus non-circular gears. By embedding gait asymmetry directly into the transmission geometry, the system was able to redistribute torque and velocity mechanically across the gait cycle, thereby reducing dependence on complex electronic feedback and control systems.

The analytical, fatigue, and FEM investigations confirmed that the mechanism is structurally stable and capable of operating safely under repeated cyclic loading conditions. The non-circular gear transmission effectively amplified torque during the stance phase while accelerating leg recovery during the swing phase, improving both energy utilization and locomotion efficiency. Additionally, the fatigue analysis verified that both the gear teeth and linkage members remain within acceptable endurance limits for sustained operation.

Overall, the study establishes that mechanically programmed locomotion using variable-radius gears provides a robust and energy-efficient alternative to purely control-driven robotic gait generation. The proposed mechanism offers strong potential for low-cost autonomous walking robots intended for hazardous terrain exploration, inspection tasks, and planetary applications where reliability, simplicity, and reduced computational dependence are critical.

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LINKS-FORCE_ANALYSIS

3.X Reduced Static Force Analysis of Theo Jansen Robotic Leg Mechanism

The reduced static force analysis of the Theo Jansen robotic leg mechanism was conducted to quantitatively investigate the internal force-transfer behaviour, support-reaction evolution, crank-driven load propagation, and ground-contact interaction generated during quasi-static locomotion. The analysis strictly follows the exact reduced equilibrium formulation implemented in the uploaded computational model and therefore reproduces the identical force behaviour observed in the simulation environment.

Unlike a complete nonlinear multibody dynamic formulation involving generalized coordinates, inertia matrices, acceleration-dependent reaction forces, constraint Jacobians, and time-dependent inertial coupling, the present methodology intentionally reduces the mechanism into its dominant force-transmitting members. Consequently, the linkage is represented as a reduced vector-equilibrium system in which the principal structural loading path is governed primarily by axial member forces, support reactions, and external ground-contact forces.

The reduced formulation preserves the dominant mechanical behaviour of the linkage while significantly improving computational efficiency. Since the majority of force transmission within ideal pin-jointed mechanisms occurs through axial loading, the reduced model captures the most mechanically significant load-transfer behaviour without requiring full nonlinear dynamic simulation.

Only the dominant effective force-carrying members are retained:

- $P2 \rightarrow P3 P2 \rightarrow P3$
- $P2 \rightarrow F P2 \rightarrow F$
- $P3 \rightarrow P4 P3 \rightarrow P4$
- $P4 \rightarrow P6 P4 \rightarrow P6$
- $P4 \rightarrow G1 P4 \rightarrow G1$
- $G1 \rightarrow P5 G1 \rightarrow P5$

Secondary linkage members are treated implicitly through geometric loop-closure constraints and effective reaction-transfer relations embedded within the reduced equilibrium formulation.

The analysis was performed at the selected crank configuration:

$$\theta_2 = 45^\circ \quad \theta_2 = 45^\circ$$

under the following assumptions:

- *rigid-link behaviour,*
- *ideal frictionless pin joints,*
- *quasi-static equilibrium,*
- *negligible inertial loading,*

- *negligible elastic deformation,*
- *planar motion constraints,*
- *Coulomb friction ground interaction,*
- *constant gravitational loading.*

Under these assumptions, the reduced model captures the dominant structural force-transfer behaviour while avoiding the computational overhead associated with complete nonlinear multibody dynamics.

3.X.1 Reduced Linkage Geometry and Kinematic Configuration

The Theo Jansen robotic leg mechanism consists of a crank-driven walking linkage architecture connected between two fixed pivots $G1G1$ and $G2G2$. The mechanism converts continuous crank rotation into biologically inspired foot trajectories through coupled geometric loop-closure constraints.

The linkage geometry governs:

- *workspace generation,*
- *stride evolution,*

- *support-span behaviour,*
- *foot-contact positioning,*
- *load-transfer directionality,*
- *support-reaction evolution.*

The instantaneous linkage configuration is generated through analytical circle-intersection formulations. Circle-intersection methods are particularly suitable because each linkage member imposes a fixed-distance geometric constraint between adjacent joints. The intersection of two circles therefore defines the unique admissible joint position satisfying simultaneous linkage-length constraints.

The fixed ground pivots are defined as:

$$G1=(0,0) \quad G2=(35,0)$$

The crank point P2 is generated using the crank-radius relation:

$$P2=G2+L2[\cos\theta_2 \sin\theta_2]$$

Using:

$$L2=12$$

and:

$$\theta_2=45^\circ$$

gives:

$$P2=(35,0)+12[0.70710.7071]P2=(35,0)+12[0.70710.7071]$$

which yields:

$$P2=(43.485, 8.485)P2=(43.485, 8.485)$$

The remaining joints are subsequently generated using coupled geometric circle-intersection constraints.

The computed coordinates become:

TABLE 1 — JOINT COORDINATES

<i>Joint</i>	<i>x-coordinat</i>	<i>y-coordinat</i>
	<i>e</i>	<i>e</i>
<i>G1G</i>	<i>0.000</i>	<i>0.000</i>
<i>1</i>		
<i>G2G</i>	<i>35.000</i>	<i>0.000</i>
<i>2</i>		
<i>P2P2</i>	<i>43.485</i>	<i>8.485</i>
<i>P3P3</i>	<i>12.572</i>	<i>-35.729</i>
<i>P4P4</i>	<i>-25.222</i>	<i>-28.405</i>
<i>P5P5</i>	<i>5.061</i>	<i>53.763</i>
<i>P6P6</i>	<i>28.437</i>	<i>5.939</i>
<i>P7P7</i>	<i>87.979</i>	<i>16.810</i>
<i>FF</i>	<i>3.374</i>	<i>35.842</i>

These coordinates establish the instantaneous linkage configuration responsible for governing:

- *force-transfer directions,*
- *support-reaction development,*
- *lower-leg compression loading,*
- *upper-frame stabilization,*
- *ground-contact interaction.*

The relative spacing between the joints strongly influences the mechanical leverage characteristics of the mechanism and therefore governs the resulting internal force redistribution behaviour.

3.X.2 Link Vector Formulation

For every effective member within the reduced equilibrium model, the corresponding geometric vector is defined as:

$$\Delta_{rij} = (x_j - x_i, y_j - y_i) \quad \Delta_{rij} = (x_j - x_i, y_j - y_i)$$

The associated member magnitude becomes:

$$L_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \quad L_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

The geometric vectors define the instantaneous structural directions through which axial force transmission occurs throughout the mechanism.

The principal reduced-model vectors become:

TABLE 2 — LINK VECTORS

<i>Link</i>	<i>Vector</i>	<i>Magnitu de</i>
$P2 \rightarrow P3$	$(-30.913, -44.214)$	53.952
$P2 \rightarrow F$	$(-40.111, 27.357)$	48.555
$P3 \rightarrow P4$	$(-37.794, 7.324)$	38.497
$P3 \rightarrow G1$	$(-12.572, 35.729)$	37.876
$P4 \rightarrow P6$	$(53.659, 34.344)$	63.709
$P4 \rightarrow G1$	$(25.222, 28.405)$	37.990
$G1 \rightarrow P5$	$(5.061, 53.763)$	54.001

These vectors govern:

- *axial load-transfer direction,*

- *structural load paths,*
- *moment-arm evolution,*
- *support-force redistribution,*
- *geometric leverage behaviour.*

The relative orientation of adjacent vectors strongly affects the equilibrium sensitivity of the mechanism. Near-collinear vector configurations can produce significant internal force amplification due to reduced moment-arm effectiveness.

3.X.3 Unit Vector Formulation and Directional Force Decomposition

The corresponding normalized direction vectors are obtained using:

$$u^{ij} = (\Delta x/L, \Delta y/L) \quad u^{ij} = (L\Delta x, L\Delta y)$$

Unit vectors are required because all internal axial member forces act exclusively along their corresponding member directions. Consequently, all equilibrium equations are projected using these normalized directional quantities.

The computed unit vectors become:

TABLE 3 — UNIT VECTORS

<i>Link</i>	<i>uxux</i>	<i>uyuy</i>
<i>u23u23</i>	-0.572	-0.819
	9	6
<i>u2Fu2F</i>	-0.826	0.5635
	1	
<i>u34u34</i>	-0.981	0.1902
	7	
<i>u3Gu3</i>	-0.331	0.9433
<i>G</i>	9	
<i>u46u46</i>	0.8422	0.5391
<i>u4Gu4</i>	0.6639	0.7478
<i>G</i>		
<i>u15u15</i>	0.0937	0.9956

These unit vectors define the complete directional topology governing the reduced force-transfer behaviour of the mechanism.

The orientation of the vectors directly influences:

- *equilibrium sensitivity,*
- *axial load decomposition,*
- *support-reaction directionality,*

- *force amplification behaviour,*
 - *transmission efficiency.*
-

3.X.4 Crank Input Force Propagation

The crank acts as the primary actuator-driven loading source responsible for initiating force transmission throughout the linkage network.

The crank vector becomes:

$$r = P_2 - G_2$$

which gives:

$$r = (8.485, 8.485)$$

The corresponding crank magnitude becomes:

$$|r| = 12$$

The tangential driving direction is obtained using:

$$t^{\wedge} = [-r_y, r_x] / |r|$$

which yields:

$$t^{\wedge} = (-0.7071, 0.7071)$$

The transmitted torque follows the exact piecewise relation:

$$T = \begin{cases} 0.051 \text{ N}\cdot\text{m}, & 0^\circ \leq \theta \leq 210^\circ \\ 0.1535 \text{ N}\cdot\text{m}, & 210^\circ < \theta \leq 360^\circ \end{cases} T = \begin{cases} 0.051 \text{ N}\cdot\text{m}, & 0.1535 \text{ N}\cdot\text{m}, & 0^\circ \leq \theta \leq 210^\circ \\ 210^\circ < \theta \leq 360^\circ \end{cases}$$

Since:

$$\theta_2 = 45^\circ \theta_2 = 45^\circ$$

the transmitted torque becomes:

$$T = 0.051 \text{ N}\cdot\text{m} T = 0.051 \text{ N}\cdot\text{m}$$

The crank input force is computed using:

$$F_{in} = T / r_{crank} F_{in} = r_{crank} T$$

Substituting the corresponding values gives:

$$F_{in} = 0.051 / 12 F_{in} = 120.051$$

which yields:

$$F_{in} = 0.00425 F_{in} = 0.00425$$

The corresponding force vector becomes:

$$F_{in} = 0.00425(-0.7071, 0.7071) F_{in} = 0.00425(-0.7071, 0.7071)$$

which gives:

$$F_{in} = (-0.003005, 0.003005) F_{in} = (-0.003005, 0.003005)$$

TABLE 4 — CRANK INPUT FORCE

Quantity	Value
Crank vector	(8.485, 8.485)

Crank magnitude 12

Tangential direction (-0.7071,0.7071)

Input force x -0.003005

Input force y 0.003005

Input force magnitude 0.00425

The crank force governs:

- *primary linkage actuation,*
- *internal axial-force initiation,*
- *lower-leg compression generation,*
- *support-reaction evolution,*
- *global equilibrium redistribution.*

Since the implemented computational model directly uses geometric dimensions in millimeters while torque remains expressed in N·m, the resulting internal force magnitudes become substantially smaller than physically dimensionally consistent SI-unit values. Consequently, the mechanism behaves primarily as a gravity-dominated support system.

3.X.5 Reduced Equilibrium Analysis at Joint P2P2

The reduced equilibrium formulation at joint P2P2 determines the dominant force redistribution between the crank and the lower linkage structure.

The governing equilibrium conditions become:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0$$

Using:

$$u_3 = (-0.5729, -0.8196)$$

$$u_2 = (-0.8261, 0.5635)$$

the vector equilibrium relation becomes:

$$f_3 u_3 + f_2 u_2 + F_{in} = 0$$

Expanding the x-direction equilibrium gives:

$$-0.5729 f_3 - 0.8261 f_2 - 0.003005 = 0$$

The y-direction equilibrium becomes:

$$-0.8196 f_3 + 0.5635 f_2 + 0.003005 = 0$$

The simultaneous equilibrium system is written as:

$$\begin{bmatrix} -0.5729 & -0.8261 \\ -0.8196 & 0.5635 \end{bmatrix} \begin{bmatrix} f_3 \\ f_2 \end{bmatrix} = \begin{bmatrix} 0.003005 \\ -0.003005 \end{bmatrix}$$

Solving the equilibrium system gives:

$$f_{23}=0.000797f_{23}=0.000797$$

and:

$$f_{2F}=0.00419f_{2F}=0.00419$$

TABLE 5 — FORCE RESULTS

<i>Quantit y</i>	<i>Magnitu de</i>	<i>Type</i>
$f_{23}f_{23}$	0.000797	Tension
$f_{2F}f_{2F}$	0.00419	Compression
$f_{46}f_{46}$	0.00252	Tension
$F_{9F}F_{9F}$	0.000858	Tension

Positive axial-force values correspond to tensile loading while negative equilibrium projection behaviour indicates compressive transfer along the member direction.

The larger magnitude of $f_{2F}f_{2F}$ relative to $f_{23}f_{23}$ indicates that the lower linkage branch carries the dominant compressive load path within the reduced mechanism.

3.X.6 Moment Equilibrium at Joint P4P4

The force transmitted through member $P4\rightarrow P6P4\rightarrow P6$ is obtained using reduced vector moment equilibrium about joint P4P4.

The governing formulation becomes:

$$f_{46} = -f_{23} u_{34} \times u_{4G} u_{46} \times u_{4G} f_{46} = -f_{23} u_{46} \times u_{4G} u_{34} \times u_{4G}$$

The two-dimensional vector cross product is defined as:

$$a \times b = a_x b_y - a_y b_x \quad a \times b = a_x b_y - a_y b_x$$

Using the corresponding vectors gives:

$$u_{34} \times u_{4G} = (-0.9817)(0.7478) - (0.1902)(0.6639) = -0.8606 \quad u_{34} \times u_{4G} = (-0.9817)(0.7478) - (0.1902)(0.6639) = -0.8606$$

Similarly:

$$u_{46} \times u_{4G} = (0.8422)(0.7478) - (0.5391)(0.6639) = 0.2722 \quad u_{46} \times u_{4G} = (0.8422)(0.7478) - (0.5391)(0.6639) = 0.2722$$

Substituting the corresponding values gives:

$$f_{46} = -(0.000797) - 0.8606(0.2722) \quad f_{46} = -(0.000797) - 0.2722 - 0.8606$$

which yields:

$$f_{46} = 0.00252 \quad f_{46} = 0.00252$$

TABLE 6 — CROSS PRODUCT TERMS

<i>Cross Product</i>	<i>Numerical Value</i>
$u_{34} \times u_{4G} u_{34} \times u_{4G}$	-0.8606
$u_{46} \times u_{4G} u_{46} \times u_{4G}$	0.2722

The denominator term represents the effective moment-arm sensitivity of the mechanism. Smaller denominator values reduce rotational leverage efficiency and therefore amplify internal axial-force requirements.

This behaviour explains why linkage systems near singular configurations can experience substantial force amplification despite relatively small external loading.

3.X.7 Top Support Reaction Analysis

The effective upper-support reaction is obtained using reduced vector equilibrium of the dominant force-transmitting members.

The governing relation becomes:

$$R_{top} = -(f_{23}u_{23} + f_{46}u_{46})$$

Substituting the corresponding vectors gives:

$$R_{top} = (-0.001666, -0.000706)$$

The corresponding resultant magnitude becomes:

$$|R| = \sqrt{R_x^2 + R_y^2}$$

$$|R_{top}| = \sqrt{(-0.001666)^2 + (-0.000706)^2}$$

which yields:

$$|R_{top}| = 0.00181$$

TABLE 7 — TOP REACTION VECTOR

<i>Quantity</i>	<i>x-component</i>	<i>y-component</i>	<i>Magnitude</i>
	<i>t</i>	<i>t</i>	<i>de</i>
<i>R_{top}</i>	-0.001666	-0.000706	0.00181

The top support reaction governs:

- *upper-frame stabilization,*
- *support-force balancing,*
- *constraint-force redistribution,*
- *linkage positional stability.*

3.X.8 Top Support Link Force

The force transmitted through member $G1 \rightarrow P5$ is obtained using vector projection along the member axis.

The governing formulation becomes:

$$F_9 = -R_{top} \cdot u_{15}$$

The dot product is defined as:

$$a \cdot b = a_x b_x + a_y b_y$$

Substituting the corresponding values gives:

$$F_9 = -[(-0.001666)(0.0937) + (-0.000706)(0.9956)] F_9 = -[(-0.001666)(0.0937) + (-0.000706)(0.9956)]$$

which yields:

$$F_9 = 0.000858 F_9 = 0.000858$$

TABLE 8 — DOT PRODUCT CALCULATION

<i>Quantity</i>	<i>Value</i>
<i>x-direction projection</i>	-0.00015
	6
<i>y-direction projection</i>	-0.00070
	3
<i>Dot-product sum</i>	-0.00085
	8
<i>Final F_9</i>	0.00085
	8

The support-link force governs:

- upper-frame stabilization,
- axial support balancing,
- transmission of upper reaction loading,
- structural constraint stabilization.

3.X.9 Ground Reaction Analysis

The robotic leg experiences external support loading at the foot-ground interface due to gravitational loading and internal linkage-force transfer.

The total gravitational force becomes:

$$W = mg$$

$$m = 8 \text{ kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$F_g = mg \approx 78.4 \text{ N}$$

Weight-force controls adjust mass and gravitational acceleration, then recompute F_g equals $m g$.

$$m = 0.25 \text{ kg} \quad g = 9.81 \text{ m/s}^2$$

$$W = 0.25(9.81) = 2.4525 \text{ N}$$

The corresponding weight vector becomes:

$$W_{\text{vec}} = (0, -2.4525)$$

The reduced ground-equilibrium formulation becomes:

$$R_{\text{ground}} = -(f_{46}u_{46} + f_{2F}u_{2F} + W_{\text{vec}})$$

Substituting the corresponding vectors gives:

$$R_{\text{ground}} = (-0.005583, 2.45350)$$

The corresponding resultant magnitude becomes:

$$|R_{ground}| = (-0.005583)^2 + (2.45350)^2 = 2.4535$$

TABLE 9 — GROUND REACTION VECTOR

<i>Quantity</i>	<i>x-component</i> <i>t</i>	<i>y-component</i> <i>t</i>	<i>Magnitude</i> <i>de</i>
<i>Ground reaction</i>	-0.005583	2.45350	2.4535

The ground reaction governs:

- *stance-phase support,*
- *vertical equilibrium,*
- *contact stabilization,*
- *friction-force generation.*

Since the internal linkage forces are extremely small relative to the gravitational loading, the ground reaction becomes dominated almost entirely by the robot weight.

3.X.10 Friction Force Evolution

The tangential friction force is evaluated using the Coulomb friction relation:

$$F_f = \mu N \quad F_f = \mu N$$

Using:

$$\mu = 0.35 \quad \mu = 0.35$$

and:

$$N = 2.4535 \quad N = 2.4535$$

gives:

$$F_f = 0.35(2.4535) = 0.8587 \quad F_f = 0.35(2.4535) = 0.8587$$

The foot-direction vector becomes:

$$\text{footdir} = (0.9134, -0.4070) \quad \text{footdir} = (0.9134, -0.4070)$$

The friction vector becomes:

$$F_{\text{friction}} = -0.8587(0.9134, -0.4070) \quad F_{\text{friction}} = -0.8587(0.9134, -0.4070)$$

which yields:

$$F_{\text{friction}} = (-0.784, 0.349) \quad F_{\text{friction}} = (-0.784, 0.349)$$

TABLE 10 — FRICTION ANALYSIS

Quantity	Value
μ	0.35
N	2.453

μN	0.858
	7
<i>Actual friction force</i>	0.858
	7
<i>Friction utilization ratio</i>	1.000

The friction force governs:

- *anti-slip behaviour,*
- *tangential traction resistance,*
- *stance-phase stability,*
- *support continuity.*

The friction utilization ratio of 1.000 indicates that the contact region operates at the limiting friction condition corresponding to impending slip.

3.X.11 Equilibrium Verification

The obtained force values were verified by substituting the numerical results back into the governing equilibrium equations.

TABLE 11 — EQUILIBRIUM VERIFICATION

<i>Equation</i>	<i>Residual</i>
	<i>l</i>
$\Sigma F_x(P2)$	≈ 0
$\Sigma F_y(P2)$	≈ 0
<i>Moment equilibrium at P4</i>	≈ 0

For joint P2:

$$-0.5729(0.000797) - 0.8261(-0.00419) - 0.003005 \approx 0$$

$$-0.5729(0.000797) - 0.8261(-0.00419) - 0.003005 \approx 0$$

$$-0.8196(0.000797) + 0.5635(-0.00419) + 0.003005 \approx 0$$

$$-0.8196(0.000797) + 0.5635(-0.00419) + 0.003005 \approx 0$$

For joint P4:

$$0.000797(-0.8606) + 0.00252(0.2722) \approx 0$$

$$0.000797(-0.8606) + 0.00252(0.2722) \approx 0$$

The residual equilibrium errors are numerically negligible, confirming that the reduced equilibrium formulation remains internally consistent and mechanically stable.

3.X.12 Final Engineering Summary

TABLE 12 — FINAL ENGINEERING SUMMARY

<i>Quantity</i>	<i>Final Value</i>
<i>Input force magnitude</i>	<i>0.00425</i>
<i>f₂₃f₂₃</i>	<i>0.000797</i>
<i>f_{2F}f_{2F}</i>	<i>0.00419</i>
<i>f₄₆f₄₆</i>	<i>0.00252</i>
<i>F₉F₉</i>	<i>0.000858</i>
<i>Top reaction magnitude</i>	<i>0.00181</i>
<i>Ground normal force</i>	<i>2.4535</i>
<i>Friction force magnitude</i>	<i>0.8587</i>
<i>Total ground reaction magnitude</i>	<i>2.4535</i>
<i>Friction utilization ratio</i>	<i>1.000</i>

The reduced static force analysis demonstrates that the present implementation behaves primarily as a gravity-dominated support system rather than a strongly force-amplified linkage transmission mechanism.

The dominant mechanical observations are:

- 1. Internal linkage forces remain extremely small relative to the gravitational loading.*
- 2. The lower linkage branch carries the dominant compressive load path.*

3. *Ground reactions are governed primarily by robot weight.*
 4. *Friction behaviour becomes weight-dominated rather than linkage-driven.*
 5. *Support reactions remain weakly coupled to crank-force propagation.*
 6. *The mechanism remains structurally stable under the reduced equilibrium formulation.*
 7. *The reduced model remains computationally efficient because only dominant axial-force members are retained.*
 8. *The reduced equilibrium approach successfully preserves the dominant mechanical behaviour while avoiding the complexity of full nonlinear multibody dynamics.*
 9. *The formulation remains numerically self-consistent because all equilibrium residuals approach zero.*
 10. *The force results accurately reproduce the exact computational behaviour implemented in the uploaded reduced static force model.*
-

5.X Advanced Computational Kinematic, Force-Transmission, and Gait Dynamics

Analysis of the Theo Jansen Robotic Leg Mechanism

The developed computational framework represents a coupled kinematic–quasi-static locomotion simulation environment designed to investigate the nonlinear geometric behaviour, gait-generation capability, support-force evolution, and internal force-transmission characteristics of the Theo Jansen robotic leg mechanism.

The simulation architecture integrates:

- *analytical linkage kinematics,*
- *geometric loop-closure synthesis,*
- *reduced static equilibrium mechanics,*
- *crank-dependent actuation modelling,*
- *support-reaction analysis,*
- *friction-limited ground-contact behaviour,*
- *cyclic gait visualization,*
- *force-propagation tracking,*
- *trajectory evolution analysis.*

Unlike simplified planar four-bar linkage systems, the Theo Jansen mechanism exhibits highly coupled nonlinear multiloop geometric behaviour. The motion of every joint depends simultaneously upon multiple interacting linkage constraints, resulting in strongly nonlinear displacement fields, varying instantaneous mechanical advantage, continuously evolving support reactions, and cyclic force redistribution throughout the gait cycle.

The present computational framework therefore serves as both:

- 1. a locomotion-generation simulator, and*
- 2. a reduced structural force-analysis platform.*

The framework enables simultaneous investigation of:

- *gait synthesis,*
- *linkage stability,*
- *geometric amplification,*
- *support-phase mechanics,*
- *toe trajectory evolution,*
- *stance/swing transition,*
- *internal load-transfer behaviour,*

- *reaction-force propagation,*
 - *friction-limited walking conditions.*
-

5.X.1 Fundamental Locomotion Principle of the Theo Jansen Mechanism

The Theo Jansen mechanism transforms continuous rotary crank motion into biologically inspired walking trajectories through purely geometric constraint coupling.

The mechanism does not rely on:

- *wheels,*
- *cams,*
- *active foot control,*
- *programmed stepping sequences.*

Instead, locomotion emerges naturally from the coordinated nonlinear motion generated by interconnected rigid members.

The mechanism operates through three dominant motion phases:

<i>Phase</i>	<i>Mechanical Behaviour</i>
<i>Swing phase</i>	<i>Foot lifts and advances</i>
<i>Touchdown phase</i>	<i>Foot approaches contact</i>
<i>Stance phase</i>	<i>Foot supports body weight</i>

The linkage geometry is specifically designed such that the toe point P7P7 approximately follows:

- *a flattened lower trajectory during support,*
- *an elevated curved trajectory during swing.*

This behaviour resembles biological leg locomotion observed in animals and insects.

The resulting gait provides:

- *improved obstacle clearance,*
- *smoother support transition,*
- *reduced vertical body oscillation,*
- *enhanced terrain adaptability,*
- *passive gait synchronization,*

- *energy-efficient stride evolution,*
- *low-impact support transfer.*

From a robotics perspective, this passive gait-generation capability is highly significant because it minimizes the control complexity normally associated with multi-joint walking robots. Instead of independently actuating every joint, the Theo Jansen mechanism achieves coordinated locomotion through purely geometric motion propagation.

5.X.2 Geometric Constraint Propagation Throughout the Mechanism

One of the most important characteristics of the Theo Jansen linkage is that motion propagates geometrically through sequential constraint enforcement.

Unlike independently actuated robotic joints, the mechanism possesses only a single active degree of freedom:

$\theta_{2\theta 2}$

All remaining joint positions are passively generated through geometric compatibility constraints.

Consequently:

- *the motion of P3P3 depends on P2P2,*

- *the motion of P4P4 depends on P3P3,*
- *the motion of P6P6 depends simultaneously on P4P4 and P5P5,*
- *the motion of P7P7 depends on both P5P5 and P6P6.*

This creates highly nonlinear coupled motion behaviour.

Small crank-angle variations may therefore generate:

- *amplified foot displacement,*
- *nonlinear support reactions,*
- *changing leverage conditions,*
- *rapidly varying force directions,*
- *nonlinear velocity amplification,*
- *dynamically evolving transmission ratios,*
- *continuously varying support geometry.*

This geometric amplification behaviour is one of the defining features of Theo Jansen walking systems.

The multiloop structure of the mechanism creates strong interdependence between:

- *kinematic compatibility,*
- *support stability,*
- *force equilibrium,*
- *gait continuity,*
- *transmission efficiency.*

Consequently, the mechanism behaves as a highly coupled nonlinear locomotion network rather than a simple linkage assembly.

5.X.3 Analytical Circle-Intersection Kinematics

The simulation computes joint locations using exact analytical circle-intersection formulations rather than iterative numerical solvers.

For every unknown joint:

- *one linkage defines a circular constraint,*
- *another linkage defines a second circular constraint,*
- *the valid joint position becomes the geometric intersection of the two circles.*

Mathematically:

$$d \leq L_a + L_b \quad d \leq L_a + L_b$$

and

$$d \geq |L_a - L_b| \quad d \geq |L_a - L_b|$$

must both remain satisfied.

These conditions are fundamental because they define whether the mechanism can physically assemble.

Violation of these conditions produces:

- *geometric singularity,*
- *linkage locking,*
- *impossible assembly states,*
- *infinite force amplification,*
- *mobility collapse,*
- *numerical divergence,*
- *loss of geometric compatibility.*

The use of analytical geometry provides:

- *exact kinematic closure,*

- *numerical robustness,*
- *computational efficiency,*
- *deterministic joint generation,*
- *stable gait continuity,*
- *elimination of iterative convergence instability.*

This is significantly more stable than iterative inverse-kinematics methods, especially near singular configurations where Jacobian-based solvers often fail.

5.X.4 Interpretation of Linkage Singularities

Singular configurations represent one of the most critical phenomena in linkage systems.

A singularity occurs when:

- *adjacent members become nearly collinear,*
- *effective moment arms approach zero,*
- *rotational leverage collapses,*

- *internal forces theoretically approach infinity.*

Mechanically, this means the linkage loses its ability to efficiently redirect forces.

Near singularity regions:

- *tiny actuator forces may produce large displacements,*
- *or extremely large internal forces may be required to maintain equilibrium,*
- *linkage sensitivity becomes extremely high,*
- *force amplification increases dramatically,*
- *local stiffness characteristics change abruptly.*

This behaviour explains the force spikes visible in:

- *Top Link graph,*
- *Top Reaction graph,*
- *Link P4-P6 graph,*
- *Link P2-F graph.*

The peaks occurring approximately near:

$220^\circ - 340^\circ$

correspond to near-singular geometric configurations.

From a structural mechanics perspective, singularity regions are highly important because they represent:

- *maximum stress concentration regions,*
 - *highest axial loading conditions,*
 - *potential fatigue initiation zones,*
 - *support-instability regions,*
 - *maximum transmission amplification conditions.*
-

5.X.5 Crank-Driven Motion Propagation

The crank is the sole actuator-driven component.

The crank vector becomes:

$$\mathbf{r} = \mathbf{P} - \mathbf{G} \quad \mathbf{r} = \mathbf{P} - \mathbf{G}$$

and the tangential actuation direction becomes:

$$\hat{\mathbf{t}} = [-r_y, r_x] / |\mathbf{r}| \quad \hat{\mathbf{t}} = [-r_y, r_x] / |\mathbf{r}|$$

The crank therefore injects tangential force into the mechanism.

This force propagates sequentially through:

$P2 \rightarrow P3 \rightarrow P4 \rightarrow P6 \rightarrow P7$

creating the walking motion.

The crank acts simultaneously as:

- *a motion generator,*
- *a force source,*
- *a geometric constraint driver,*
- *a nonlinear transmission initiator.*

The crank rotation generates continuously varying:

- *axial-force orientation,*
- *linkage leverage,*
- *support geometry,*
- *trajectory curvature,*
- *force-transfer topology.*

Consequently, the mechanism behaves as a nonlinear force-redirection system whose effective transmission behaviour changes continuously throughout the gait cycle.

5.X.6 Piecewise Torque Distribution and Gait Loading

The actuator torque is intentionally modeled using a piecewise relation:

<i>Crank Region</i>	<i>Torque</i>
$0^{\circ}-210^{\circ}$	$0.051 \text{ N}\cdot\text{m}$
$210^{\circ}-360^{\circ}$	$0.1535 \text{ N}\cdot\text{m}$

This models varying locomotion resistance throughout the gait cycle.

Physical Interpretation

Low-torque regions correspond primarily to:

- *swing-phase motion,*
- *low ground interaction,*
- *reduced support loading,*

- *lower contact resistance,*
- *low support-demand regions.*

High-torque regions correspond to:

- *support-phase loading,*
- *stance stabilization,*
- *weight-bearing behaviour,*
- *increased resistance,*
- *elevated support-force transfer.*

This explains why the force graphs show sharp amplitude increases near:

210° 210°

The mechanism experiences higher effective resistance during support-dominant gait regions.

The piecewise torque profile therefore approximates realistic locomotion behaviour where:

- *support phases require greater energy input,*
- *swing phases require lower energy transfer,*

- *actuator demand evolves cyclically with gait state.*
-

5.X.7 Hyperbolic Tangent Torque Transition Smoothing

The torque transition is smoothed using:

$$s=0.5(1+\tanh(\theta-\text{centerwidth}))s=0.5(1+\tanh(\text{width}\theta-\text{center}))$$

This is extremely important computationally.

Without smoothing:

- *discontinuous torque jumps occur,*
- *force spikes become numerically unstable,*
- *gait evolution becomes unrealistic,*
- *linkage acceleration discontinuities emerge,*
- *support reactions become nonphysical.*

The hyperbolic tangent transition therefore acts as a numerical regularization mechanism.

Physically, it represents:

- *smooth motor loading,*
- *gradual support engagement,*
- *realistic actuator behaviour,*
- *continuous energy transfer,*
- *progressive support-force redistribution.*

From a computational mechanics perspective, smoothing improves:

- *numerical continuity,*
- *stability of force propagation,*
- *convergence robustness,*
- *graph smoothness,*
- *trajectory continuity.*

5.X.8 Adaptive Angular Velocity Behaviour

The simulation modifies crank advancement using:

$$\omega_{scale} = 1/T \quad \omega_{scale} = T/1$$

This creates inverse torque-speed behaviour.

Mechanical Significance

Higher torque regions produce:

- *slower crank advancement,*
- *longer support duration,*
- *increased contact stability,*
- *greater force-transfer time,*
- *increased stance-phase support continuity.*

Lower torque regions produce:

- *faster swing motion,*
- *quicker foot advancement,*
- *increased gait efficiency,*
- *reduced support resistance.*

This behaviour resembles biological locomotion where stance phases are generally slower than swing phases.

The adaptive angular velocity formulation therefore improves:

- *locomotion realism,*
 - *support-phase fidelity,*
 - *dynamic gait continuity,*
 - *actuator-response realism.*
-

5.X.9 Toe Trajectory and Walking Gait Formation

The toe trajectory generated by point P7P7 is the defining locomotion characteristic of the mechanism.

The trajectory contains two distinct regions:

<i>Region</i>	<i>Behaviour</i>
<i>Lower flat region</i>	<i>Ground support</i>
<i>Upper curved region</i>	<i>Swing phase</i>

Stance Region

During the lower trajectory:

- *the foot remains near constant height,*
- *vertical body oscillation reduces,*
- *support becomes more stable,*
- *slipping probability decreases,*
- *tangential support continuity improves.*

This is essential for practical walking.

Swing Region

During the upper trajectory:

- *the foot lifts off the ground,*
- *forward advancement occurs,*
- *obstacle clearance improves,*
- *terrain interference reduces.*

This prevents dragging during motion.

The resulting gait therefore achieves:

- *smooth locomotion,*
- *continuous support transfer,*
- *passive walking behaviour,*
- *efficient terrain traversal.*

The flattened support trajectory is one of the primary reasons why Theo Jansen mechanisms exhibit superior walking smoothness compared to simple crank-rocker walking systems.

5.X.10 Joint Motion Characteristics

Every joint exhibits unique nonlinear motion behaviour.

Crank Joint P2P2

The crank joint follows near-perfect circular motion:

$$x = r \cos \theta$$

$$\theta = 45^\circ \quad x = 0.71r = 1$$

$$y = r \sin \theta$$

$$\theta = 45^\circ \quad y = 0.71r = 1$$

This is the only purely harmonic motion within the system.

Intermediate Joint P3P3

P3P3 behaves as a primary force-transfer node.

Its motion is strongly nonlinear because it simultaneously satisfies:

- *crank constraint,*
- *upper-link constraint,*
- *lower-link compatibility,*
- *support-transfer geometry.*

Its trajectory determines:

- *force redistribution,*
- *linkage leverage,*
- *support transfer,*
- *transmission stability.*

Joint P4P4

P4P4 acts as a major geometric amplification node.

Near singularity regions:

- *small positional changes produce large force redistribution,*
- *support reactions vary rapidly,*
- *linkage leverage changes dramatically,*
- *force sensitivity increases sharply.*

This explains the force spikes visible in the P4-P6 graph.

Foot Point P7P7

P7P7 governs:

- *locomotion behaviour,*
- *support interaction,*
- *terrain contact,*
- *stride evolution,*
- *gait continuity.*

Its trajectory determines overall walking quality.

5.X.11 Deep Analysis of Input Force Graph

The Input Force graph exhibits step-like periodic behaviour.

Reason for Piecewise Behaviour

Since:

$$F_{in} = Tr F_{in} = rT$$

the input force scales directly with torque.

Therefore:

- *low torque $\rightarrow$ low force plateau,*
- *high torque $\rightarrow$ high force plateau.*

The abrupt transitions correspond exactly to the torque-switching regions.

Physical Interpretation

The graph indicates:

- *increased support effort during stance,*

- *reduced effort during swing,*
- *cyclic actuator loading,*
- *periodic locomotion resistance.*

This is mechanically desirable because:

- *energy is concentrated during support,*
 - *unnecessary swing-phase power is minimized,*
 - *actuator efficiency improves.*
-

5.X.12 Deep Analysis of Link P2-F Graph

The P2-F member experiences dominant compressive loading.

Mechanical Role

This member transfers:

- *crank-generated compression,*
- *lower support loading,*

- *stance-phase force propagation,*
- *support-force redistribution.*

The oscillatory behaviour emerges because the effective linkage leverage continuously changes with crank angle.

Force Amplification Regions

Large spikes indicate:

- *reduced mechanical advantage,*
- *near-collinear geometry,*
- *moment-arm collapse,*
- *support-transition amplification.*

These regions correspond to:

- *increased structural stress,*
- *elevated buckling risk,*
- *reduced transmission efficiency,*

- *localized load concentration.*
-

5.X.13 Deep Analysis of Link P4-P6 Graph

The P4-P6 member acts as a lower support stabilizer.

Mechanical Behaviour

This member redistributes force between:

- *lower support structure,*
- *foot-ground interface,*
- *upper stabilization links,*
- *stance-phase support paths.*

Its oscillatory force behaviour indicates:

- *evolving support geometry,*
- *continuously varying leverage,*
- *nonlinear equilibrium redistribution,*

- *support-path migration.*

The amplified peaks indicate geometric amplification near support-transition phases.

5.X.14 Deep Analysis of Top-Link Graph

The top-link graph exhibits the most pronounced force spikes.

Physical Meaning

The upper support member experiences amplified stabilization loading when:

- *the lower mechanism approaches singularity,*
- *support reactions redistribute rapidly,*
- *geometric leverage decreases,*
- *lower support stability weakens.*

The upper structure therefore compensates for lower-support instability.

Structural Implications

These peaks represent:

- *maximum axial stress regions,*
- *potential fatigue-critical locations,*
- *elevated support loading,*
- *localized structural amplification zones.*

In real hardware this region would likely require:

- *reinforced joints,*
 - *thicker members,*
 - *improved bearing supports,*
 - *higher buckling resistance.*
-

5.X.15 Deep Analysis of Top-Reaction Graph

The top reaction represents effective support loading transferred into the upper frame.

Mechanical Interpretation

This force balances:

- *lower support loading,*
- *crank transmission forces,*
- *internal linkage reactions,*
- *geometric constraint forces.*

The oscillatory behaviour indicates:

- *continuously evolving support redistribution,*
- *changing axial load paths,*
- *nonlinear equilibrium behaviour,*
- *cyclic support stabilization.*

The force peaks correspond directly to:

- *linkage singularity regions,*
- *support-phase transitions,*
- *leverage collapse zones,*
- *elevated support-transfer amplification.*

5.X.16 Deep Analysis of Ground-Reaction Graph

The ground-reaction graph remains relatively constant compared to internal linkage forces.

Dominant Behaviour

The reaction is dominated by:

$$W=mg$$

because:

$$W=2.4525 \text{ N}$$

is significantly larger than the internal linkage forces.

Oscillatory Variations

Small oscillations occur due to:

- *evolving support geometry,*
- *changing linkage orientation,*
- *axial-force projection changes,*

- *cyclic support redistribution.*

These oscillations represent dynamic redistribution of support loading during gait evolution.

5.X.17 Deep Analysis of Normal-Reaction Graph

The normal force governs:

- *vertical support,*
- *stance stability,*
- *contact retention,*
- *support continuity.*

The graph demonstrates:

- *continuous positive support loading,*
- *no contact loss,*
- *stable support behaviour,*

- *sustained gait continuity.*

The absence of negative values confirms physically admissible walking conditions.

5.X.18 Deep Analysis of Friction Force Graph

The friction force follows:

$$F_f = \mu N$$

Since the normal force remains nearly constant, friction also remains nearly constant.

Physical Meaning

The friction force provides:

- *anti-slip stabilization,*
- *tangential traction,*
- *walking continuity,*
- *support stabilization.*

The small oscillatory behaviour reflects changing support-force projections during motion.

The graph indicates that:

- *traction remains stable,*
 - *support continuity is preserved,*
 - *no catastrophic slip occurs,*
 - *gait stability remains maintained.*
-

5.X.19 Ground Contact Mechanics

The ground-contact model assumes:

- *rigid planar contact,*
- *Coulomb friction,*
- *no penetration,*
- *continuous support,*
- *quasi-static support equilibrium.*

The ground-contact subsystem governs:

- *support stability,*
- *friction generation,*
- *traction development,*
- *stance-phase mechanics,*
- *locomotion feasibility.*

Ground interaction is one of the most critical aspects of walking systems because it ultimately determines locomotion feasibility.

5.X.20 Force Transmission Topology

The mechanism behaves as a distributed axial-force transmission network.

Force propagates through:

Crank→TransmissionLinks→SupportLinks→GroundContactCrank→TransmissionLinks→SupportLinks→GroundContact

The topology continuously evolves throughout the gait cycle.

This produces:

- *changing load paths,*

- *nonlinear reaction redistribution,*
- *cyclic support-force evolution,*
- *continuously varying structural leverage.*

The force network therefore behaves similarly to a dynamically reconfigurable truss system.

5.X.21 Quasi-Static Assumption and Its Implications

The present model neglects:

- *inertia,*
- *acceleration,*
- *impact loading,*
- *vibration,*
- *damping.*

Therefore the analysis represents:

- *low-speed locomotion,*
- *equilibrium-dominated motion,*
- *reduced dynamic behaviour,*
- *support-controlled gait evolution.*

This is appropriate for preliminary locomotion and structural analysis.

The quasi-static assumption significantly simplifies:

- *equilibrium formulation,*
- *computational complexity,*
- *force propagation analysis,*
- *support-force tracking.*

5.X.22 Numerical Stability and Computational Efficiency

The reduced formulation is computationally efficient because:

- *analytical geometry avoids iterative IK solvers,*
- *equilibrium equations remain linear,*
- *no differential equations are integrated,*
- *inertia matrices are avoided,*
- *constraint drift is eliminated.*

Consequently:

- *simulation speed improves,*
- *numerical stability increases,*
- *convergence issues reduce,*
- *trajectory continuity improves.*

The formulation therefore provides a highly efficient locomotion-analysis framework suitable for:

- *real-time simulation,*
- *robotic prototyping,*

- *rapid gait evaluation,*
 - *preliminary structural assessment.*
-

5.X.23 Overall Research-Level Engineering Interpretation

The complete computational framework demonstrates that the Theo Jansen robotic leg behaves as a highly coupled nonlinear geometric locomotion system in which:

- *continuous crank rotation produces walking motion,*
- *geometric constraints govern trajectory evolution,*
- *force propagation follows changing axial load paths,*
- *singularity regions dominate force amplification,*
- *support reactions evolve cyclically,*
- *gravity dominates ground loading,*
- *friction stabilizes stance behaviour,*

- *adaptive torque modulation controls gait dynamics.*

The simulation therefore successfully integrates:

- *nonlinear kinematics,*
- *reduced equilibrium mechanics,*
- *support-force evolution,*
- *friction-limited contact behaviour,*
- *gait synthesis,*
- *trajectory generation,*
- *locomotion visualization,*

into a unified research-grade robotic walking analysis platform suitable for advanced biomechanical, locomotion, and linkage-transmission investigations.

Crank Force Calculation

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5.X Advanced Computational Kinematic, Force-Transmission, and Gait Dynamics

Analysis of the Theo Jansen Robotic Leg Mechanism

The developed computational framework represents a coupled kinematic–quasi-static locomotion simulation environment designed to investigate the nonlinear geometric behaviour, gait-generation capability, support-force evolution, and internal force-transmission characteristics of the Theo Jansen robotic leg mechanism.

The simulation architecture integrates:

- *analytical linkage kinematics,*
- *geometric loop-closure synthesis,*
- *reduced static equilibrium mechanics,*
- *crank-dependent actuation modelling,*
- *support-reaction analysis,*
- *friction-limited ground-contact behaviour,*
- *cyclic gait visualization,*
- *force-propagation tracking,*
- *trajectory evolution analysis.*

Unlike simplified planar four-bar linkage systems, the Theo Jansen mechanism exhibits highly coupled nonlinear multiloop geometric behaviour. The motion of every joint depends simultaneously upon multiple interacting linkage constraints, resulting in strongly nonlinear displacement fields, varying instantaneous mechanical advantage, continuously evolving support reactions, and cyclic force redistribution throughout the gait cycle.

The present computational framework therefore serves as both:

- 1. a locomotion-generation simulator, and*
- 2. a reduced structural force-analysis platform.*

The framework enables simultaneous investigation of:

- *gait synthesis,*
- *linkage stability,*
- *geometric amplification,*
- *support-phase mechanics,*
- *toe trajectory evolution,*
- *stance/swing transition,*
- *internal load-transfer behaviour,*

- *reaction-force propagation,*
 - *friction-limited walking conditions.*
-

5.X.1 Fundamental Locomotion Principle of the Theo Jansen Mechanism

The Theo Jansen mechanism transforms continuous rotary crank motion into biologically inspired walking trajectories through purely geometric constraint coupling.

The mechanism does not rely on:

- *wheels,*
- *cams,*
- *active foot control,*
- *programmed stepping sequences.*

Instead, locomotion emerges naturally from the coordinated nonlinear motion generated by interconnected rigid members.

The mechanism operates through three dominant motion phases:

<i>Phase</i>	<i>Mechanical Behaviour</i>
<i>Swing phase</i>	<i>Foot lifts and advances</i>
<i>Touchdown phase</i>	<i>Foot approaches contact</i>
<i>Stance phase</i>	<i>Foot supports body weight</i>

The linkage geometry is specifically designed such that the toe point P7P7 approximately follows:

- *a flattened lower trajectory during support,*
- *an elevated curved trajectory during swing.*

This behaviour resembles biological leg locomotion observed in animals and insects.

The resulting gait provides:

- *improved obstacle clearance,*
- *smoother support transition,*
- *reduced vertical body oscillation,*
- *enhanced terrain adaptability,*
- *passive gait synchronization,*

- *energy-efficient stride evolution,*
- *low-impact support transfer.*

From a robotics perspective, this passive gait-generation capability is highly significant because it minimizes the control complexity normally associated with multi-joint walking robots. Instead of independently actuating every joint, the Theo Jansen mechanism achieves coordinated locomotion through purely geometric motion propagation.

5.X.2 Geometric Constraint Propagation Throughout the Mechanism

One of the most important characteristics of the Theo Jansen linkage is that motion propagates geometrically through sequential constraint enforcement.

Unlike independently actuated robotic joints, the mechanism possesses only a single active degree of freedom:

$\theta_{2\theta 2}$

All remaining joint positions are passively generated through geometric compatibility constraints.

Consequently:

- *the motion of P3P3 depends on P2P2,*

- *the motion of P4P4 depends on P3P3,*
- *the motion of P6P6 depends simultaneously on P4P4 and P5P5,*
- *the motion of P7P7 depends on both P5P5 and P6P6.*

This creates highly nonlinear coupled motion behaviour.

Small crank-angle variations may therefore generate:

- *amplified foot displacement,*
- *nonlinear support reactions,*
- *changing leverage conditions,*
- *rapidly varying force directions,*
- *nonlinear velocity amplification,*
- *dynamically evolving transmission ratios,*
- *continuously varying support geometry.*

This geometric amplification behaviour is one of the defining features of Theo Jansen walking systems.

The multiloop structure of the mechanism creates strong interdependence between:

- *kinematic compatibility,*
- *support stability,*
- *force equilibrium,*
- *gait continuity,*
- *transmission efficiency.*

Consequently, the mechanism behaves as a highly coupled nonlinear locomotion network rather than a simple linkage assembly.

5.X.3 Analytical Circle-Intersection Kinematics

The simulation computes joint locations using exact analytical circle-intersection formulations rather than iterative numerical solvers.

For every unknown joint:

- *one linkage defines a circular constraint,*
- *another linkage defines a second circular constraint,*
- *the valid joint position becomes the geometric intersection of the two circles.*

Mathematically:

$$d \leq L_a + L_b \quad d \leq L_a + L_b$$

and

$$d \geq |L_a - L_b| \quad d \geq |L_a - L_b|$$

must both remain satisfied.

These conditions are fundamental because they define whether the mechanism can physically assemble.

Violation of these conditions produces:

- *geometric singularity,*
- *linkage locking,*
- *impossible assembly states,*
- *infinite force amplification,*
- *mobility collapse,*
- *numerical divergence,*
- *loss of geometric compatibility.*

The use of analytical geometry provides:

- *exact kinematic closure,*

- *numerical robustness,*
- *computational efficiency,*
- *deterministic joint generation,*
- *stable gait continuity,*
- *elimination of iterative convergence instability.*

This is significantly more stable than iterative inverse-kinematics methods, especially near singular configurations where Jacobian-based solvers often fail.

5.X.4 Interpretation of Linkage Singularities

Singular configurations represent one of the most critical phenomena in linkage systems.

A singularity occurs when:

- *adjacent members become nearly collinear,*
- *effective moment arms approach zero,*
- *rotational leverage collapses,*

- *internal forces theoretically approach infinity.*

Mechanically, this means the linkage loses its ability to efficiently redirect forces.

Near singularity regions:

- *tiny actuator forces may produce large displacements,*
- *or extremely large internal forces may be required to maintain equilibrium,*
- *linkage sensitivity becomes extremely high,*
- *force amplification increases dramatically,*
- *local stiffness characteristics change abruptly.*

This behaviour explains the force spikes visible in:

- *Top Link graph,*
- *Top Reaction graph,*
- *Link P4-P6 graph,*
- *Link P2-F graph.*

The peaks occurring approximately near:

$220^\circ - 340^\circ$

correspond to near-singular geometric configurations.

From a structural mechanics perspective, singularity regions are highly important because they represent:

- *maximum stress concentration regions,*
 - *highest axial loading conditions,*
 - *potential fatigue initiation zones,*
 - *support-instability regions,*
 - *maximum transmission amplification conditions.*
-

5.X.5 Crank-Driven Motion Propagation

The crank is the sole actuator-driven component.

The crank vector becomes:

$$\mathbf{r} = \mathbf{P} - \mathbf{G} \quad \mathbf{r} = \mathbf{P} - \mathbf{G}$$

and the tangential actuation direction becomes:

$$\hat{\mathbf{t}} = [-r_y, r_x] / |\mathbf{r}| \quad \hat{\mathbf{t}} = [-r_y, r_x] / |\mathbf{r}|$$

The crank therefore injects tangential force into the mechanism.

This force propagates sequentially through:

$P2 \rightarrow P3 \rightarrow P4 \rightarrow P6 \rightarrow P7$

creating the walking motion.

The crank acts simultaneously as:

- *a motion generator,*
- *a force source,*
- *a geometric constraint driver,*
- *a nonlinear transmission initiator.*

The crank rotation generates continuously varying:

- *axial-force orientation,*
- *linkage leverage,*
- *support geometry,*
- *trajectory curvature,*
- *force-transfer topology.*

Consequently, the mechanism behaves as a nonlinear force-redirection system whose effective transmission behaviour changes continuously throughout the gait cycle.

5.X.6 Piecewise Torque Distribution and Gait Loading

The actuator torque is intentionally modeled using a piecewise relation:

<i>Crank Region</i>	<i>Torque</i>
$0^{\circ}-210^{\circ}$	$0.051 \text{ N}\cdot\text{m}$
$210^{\circ}-360^{\circ}$	$0.1535 \text{ N}\cdot\text{m}$

This models varying locomotion resistance throughout the gait cycle.

Physical Interpretation

Low-torque regions correspond primarily to:

- *swing-phase motion,*
- *low ground interaction,*
- *reduced support loading,*

- *lower contact resistance,*
- *low support-demand regions.*

High-torque regions correspond to:

- *support-phase loading,*
- *stance stabilization,*
- *weight-bearing behaviour,*
- *increased resistance,*
- *elevated support-force transfer.*

This explains why the force graphs show sharp amplitude increases near:

210° 210°

The mechanism experiences higher effective resistance during support-dominant gait regions.

The piecewise torque profile therefore approximates realistic locomotion behaviour where:

- *support phases require greater energy input,*
- *swing phases require lower energy transfer,*

- *actuator demand evolves cyclically with gait state.*
-

5.X.7 Hyperbolic Tangent Torque Transition Smoothing

The torque transition is smoothed using:

$$s=0.5(1+\tanh(\theta-\text{centerwidth}))s=0.5(1+\tanh(\text{width}\theta-\text{center}))$$

This is extremely important computationally.

Without smoothing:

- *discontinuous torque jumps occur,*
- *force spikes become numerically unstable,*
- *gait evolution becomes unrealistic,*
- *linkage acceleration discontinuities emerge,*
- *support reactions become nonphysical.*

The hyperbolic tangent transition therefore acts as a numerical regularization mechanism.

Physically, it represents:

- *smooth motor loading,*
- *gradual support engagement,*
- *realistic actuator behaviour,*
- *continuous energy transfer,*
- *progressive support-force redistribution.*

From a computational mechanics perspective, smoothing improves:

- *numerical continuity,*
- *stability of force propagation,*
- *convergence robustness,*
- *graph smoothness,*
- *trajectory continuity.*

5.X.8 Adaptive Angular Velocity Behaviour

The simulation modifies crank advancement using:

$$\omega_{scale} = 1/T \quad \omega_{scale} = T/1$$

This creates inverse torque-speed behaviour.

Mechanical Significance

Higher torque regions produce:

- *slower crank advancement,*
- *longer support duration,*
- *increased contact stability,*
- *greater force-transfer time,*
- *increased stance-phase support continuity.*

Lower torque regions produce:

- *faster swing motion,*
- *quicker foot advancement,*
- *increased gait efficiency,*
- *reduced support resistance.*

This behaviour resembles biological locomotion where stance phases are generally slower than swing phases.

The adaptive angular velocity formulation therefore improves:

- *locomotion realism,*
 - *support-phase fidelity,*
 - *dynamic gait continuity,*
 - *actuator-response realism.*
-

5.X.9 Toe Trajectory and Walking Gait Formation

The toe trajectory generated by point P7P7 is the defining locomotion characteristic of the mechanism.

The trajectory contains two distinct regions:

<i>Region</i>	<i>Behaviour</i>
<i>Lower flat region</i>	<i>Ground support</i>
<i>Upper curved region</i>	<i>Swing phase</i>

Stance Region

During the lower trajectory:

- *the foot remains near constant height,*
- *vertical body oscillation reduces,*
- *support becomes more stable,*
- *slipping probability decreases,*
- *tangential support continuity improves.*

This is essential for practical walking.

Swing Region

During the upper trajectory:

- *the foot lifts off the ground,*
- *forward advancement occurs,*
- *obstacle clearance improves,*
- *terrain interference reduces.*

This prevents dragging during motion.

The resulting gait therefore achieves:

- *smooth locomotion,*
- *continuous support transfer,*
- *passive walking behaviour,*
- *efficient terrain traversal.*

The flattened support trajectory is one of the primary reasons why Theo Jansen mechanisms exhibit superior walking smoothness compared to simple crank-rocker walking systems.

5.X.10 Joint Motion Characteristics

Every joint exhibits unique nonlinear motion behaviour.

Crank Joint P2P2

The crank joint follows near-perfect circular motion:

$$x = r \cos \theta$$

$$\theta = 45^\circ \quad x = 0.71r = 1$$

$$y = r \sin \theta$$

$$\theta = 45^\circ \quad y = 0.71r = 1$$

This is the only purely harmonic motion within the system.

Intermediate Joint P3P3

P3P3 behaves as a primary force-transfer node.

Its motion is strongly nonlinear because it simultaneously satisfies:

- *crank constraint,*
- *upper-link constraint,*
- *lower-link compatibility,*
- *support-transfer geometry.*

Its trajectory determines:

- *force redistribution,*
- *linkage leverage,*
- *support transfer,*
- *transmission stability.*

Joint P4P4

P4P4 acts as a major geometric amplification node.

Near singularity regions:

- *small positional changes produce large force redistribution,*
- *support reactions vary rapidly,*
- *linkage leverage changes dramatically,*
- *force sensitivity increases sharply.*

This explains the force spikes visible in the P4-P6 graph.

Foot Point P7P7

P7P7 governs:

- *locomotion behaviour,*
- *support interaction,*
- *terrain contact,*
- *stride evolution,*
- *gait continuity.*

Its trajectory determines overall walking quality.

5.X.11 Deep Analysis of Input Force Graph

The Input Force graph exhibits step-like periodic behaviour.

Reason for Piecewise Behaviour

Since:

$$F_{in} = Tr F_{in} = rT$$

the input force scales directly with torque.

Therefore:

- *low torque $\rightarrow$ low force plateau,*
- *high torque $\rightarrow$ high force plateau.*

The abrupt transitions correspond exactly to the torque-switching regions.

Physical Interpretation

The graph indicates:

- *increased support effort during stance,*

- *reduced effort during swing,*
- *cyclic actuator loading,*
- *periodic locomotion resistance.*

This is mechanically desirable because:

- *energy is concentrated during support,*
 - *unnecessary swing-phase power is minimized,*
 - *actuator efficiency improves.*
-

5.X.12 Deep Analysis of Link P2-F Graph

The P2-F member experiences dominant compressive loading.

Mechanical Role

This member transfers:

- *crank-generated compression,*
- *lower support loading,*

- *stance-phase force propagation,*
- *support-force redistribution.*

The oscillatory behaviour emerges because the effective linkage leverage continuously changes with crank angle.

Force Amplification Regions

Large spikes indicate:

- *reduced mechanical advantage,*
- *near-collinear geometry,*
- *moment-arm collapse,*
- *support-transition amplification.*

These regions correspond to:

- *increased structural stress,*
- *elevated buckling risk,*
- *reduced transmission efficiency,*

- *localized load concentration.*
-

5.X.13 Deep Analysis of Link P4-P6 Graph

The P4-P6 member acts as a lower support stabilizer.

Mechanical Behaviour

This member redistributes force between:

- *lower support structure,*
- *foot-ground interface,*
- *upper stabilization links,*
- *stance-phase support paths.*

Its oscillatory force behaviour indicates:

- *evolving support geometry,*
- *continuously varying leverage,*
- *nonlinear equilibrium redistribution,*

- *support-path migration.*

The amplified peaks indicate geometric amplification near support-transition phases.

5.X.14 Deep Analysis of Top-Link Graph

The top-link graph exhibits the most pronounced force spikes.

Physical Meaning

The upper support member experiences amplified stabilization loading when:

- *the lower mechanism approaches singularity,*
- *support reactions redistribute rapidly,*
- *geometric leverage decreases,*
- *lower support stability weakens.*

The upper structure therefore compensates for lower-support instability.

Structural Implications

These peaks represent:

- *maximum axial stress regions,*
- *potential fatigue-critical locations,*
- *elevated support loading,*
- *localized structural amplification zones.*

In real hardware this region would likely require:

- *reinforced joints,*
 - *thicker members,*
 - *improved bearing supports,*
 - *higher buckling resistance.*
-

5.X.15 Deep Analysis of Top-Reaction Graph

The top reaction represents effective support loading transferred into the upper frame.

Mechanical Interpretation

This force balances:

- *lower support loading,*
- *crank transmission forces,*
- *internal linkage reactions,*
- *geometric constraint forces.*

The oscillatory behaviour indicates:

- *continuously evolving support redistribution,*
- *changing axial load paths,*
- *nonlinear equilibrium behaviour,*
- *cyclic support stabilization.*

The force peaks correspond directly to:

- *linkage singularity regions,*
- *support-phase transitions,*
- *leverage collapse zones,*
- *elevated support-transfer amplification.*

5.X.16 Deep Analysis of Ground-Reaction Graph

The ground-reaction graph remains relatively constant compared to internal linkage forces.

Dominant Behaviour

The reaction is dominated by:

$$W=mg$$

because:

$$W=2.4525 \text{ N}$$

is significantly larger than the internal linkage forces.

Oscillatory Variations

Small oscillations occur due to:

- *evolving support geometry,*
- *changing linkage orientation,*
- *axial-force projection changes,*

- *cyclic support redistribution.*

These oscillations represent dynamic redistribution of support loading during gait evolution.

5.X.17 Deep Analysis of Normal-Reaction Graph

The normal force governs:

- *vertical support,*
- *stance stability,*
- *contact retention,*
- *support continuity.*

The graph demonstrates:

- *continuous positive support loading,*
- *no contact loss,*
- *stable support behaviour,*

- *sustained gait continuity.*

The absence of negative values confirms physically admissible walking conditions.

5.X.18 Deep Analysis of Friction Force Graph

The friction force follows:

$$F_f = \mu N$$

Since the normal force remains nearly constant, friction also remains nearly constant.

Physical Meaning

The friction force provides:

- *anti-slip stabilization,*
- *tangential traction,*
- *walking continuity,*
- *support stabilization.*

The small oscillatory behaviour reflects changing support-force projections during motion.

The graph indicates that:

- *traction remains stable,*
 - *support continuity is preserved,*
 - *no catastrophic slip occurs,*
 - *gait stability remains maintained.*
-

5.X.19 Ground Contact Mechanics

The ground-contact model assumes:

- *rigid planar contact,*
- *Coulomb friction,*
- *no penetration,*
- *continuous support,*
- *quasi-static support equilibrium.*

The ground-contact subsystem governs:

- *support stability,*
- *friction generation,*
- *traction development,*
- *stance-phase mechanics,*
- *locomotion feasibility.*

Ground interaction is one of the most critical aspects of walking systems because it ultimately determines locomotion feasibility.

5.X.20 Force Transmission Topology

The mechanism behaves as a distributed axial-force transmission network.

Force propagates through:

Crank→TransmissionLinks→SupportLinks→GroundContactCrank→TransmissionLinks→SupportLinks→GroundContact

The topology continuously evolves throughout the gait cycle.

This produces:

- *changing load paths,*

- *nonlinear reaction redistribution,*
- *cyclic support-force evolution,*
- *continuously varying structural leverage.*

The force network therefore behaves similarly to a dynamically reconfigurable truss system.

5.X.21 Quasi-Static Assumption and Its Implications

The present model neglects:

- *inertia,*
- *acceleration,*
- *impact loading,*
- *vibration,*
- *damping.*

Therefore the analysis represents:

- *low-speed locomotion,*
- *equilibrium-dominated motion,*
- *reduced dynamic behaviour,*
- *support-controlled gait evolution.*

This is appropriate for preliminary locomotion and structural analysis.

The quasi-static assumption significantly simplifies:

- *equilibrium formulation,*
- *computational complexity,*
- *force propagation analysis,*
- *support-force tracking.*

5.X.22 Numerical Stability and Computational Efficiency

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-

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The complete computational framework demonstrates that the Theo Jansen robotic leg behaves as a highly coupled nonlinear geometric locomotion system in which:

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- *gait synthesis,*
- *trajectory generation,*
- *locomotion visualization,*

into a unified research-grade robotic walking analysis platform suitable for advanced biomechanical, locomotion, and linkage-transmission investigations.

Force Analysis of a Theo Jansen Linkage Mechanism

The force analysis of the Theo Jansen robotic leg mechanism was carried out using the exact equilibrium formulation implemented in the developed computational model. The analysis was performed at a selected crank angle of:

$$\theta_2 = 45^\circ$$

under quasi-static operating conditions while considering gravitational loading and ground friction effects.

The model evaluates only the effective force-transmitting members of the linkage mechanism, namely:

- $P_2 \rightarrow P_3$
- $P_2 \rightarrow F$
- $P_3 \rightarrow P_4$
- $P_4 \rightarrow P_6$
- $P_4 \rightarrow G_1$
- $G_1 \rightarrow P_5$

The analysis excludes full multibody dynamics and instead applies vector-based static equilibrium relations to compute the dominant internal member forces and reaction forces governing the mechanism behavior.

Input Torque Model

The robotic leg is actuated through a non-circular gear transmission system that produces a varying output torque throughout the gait cycle. The instantaneous transmitted torque was modelled using the piecewise torque relation:

$$T = \begin{cases} 0.051 \text{ N} \cdot \text{m} & 0^\circ < \theta \leq 210^\circ \\ 0.1535 \text{ N} \cdot \text{m} & 210^\circ < \theta \leq 360^\circ \end{cases}$$

For the selected crank configuration:

$$\theta_2 = 45^\circ ; \hat{t} = \frac{-r_y, r_x}{|r|}$$

the applied driving torque becomes:

$$T = 0.051 \text{ N} \cdot \text{m}$$

Crank Input Force

The crank input force was obtained using the tangential force relation:

$$F_{in} = \frac{T}{r_{crank}}$$

where:

- T is the transmitted motor torque
- r_{crank} is the crank radius

Using the crank radius: $r_{crank} = 12\text{mm}$

The developed model computes:

The crank tangential direction vector was evaluated using:

which yielded:

$$\hat{t} = (-0.7071, 0.7071)$$

Therefore, the input force vector becomes:

$$F_{in} = 0.00425(-0.7071, 0.7071)$$
$$F_{in} = (-0.003005, 0.003005)$$

Link Force Equilibrium at Joint P_2

The force equilibrium at joint P_2 was solved using the vector equilibrium formulation:

$$\Sigma F_x = 0$$
$$\Sigma F_y = 0$$

The reduced system of equations is expressed as:

$$A \begin{bmatrix} f_{23} \\ f_{2F} \end{bmatrix} = -F_{in}$$

Where:

$$A = [u_{23} \ u_{2F}]$$

Using the evaluated unit vectors:

$$u_{23} = (-0.5729, -0.8196)$$
$$u_{2F} = (-0.8261, 0.5635)$$

the equilibrium equation

$$F_{in} = \frac{0.051}{12}$$
$$F_{in} = 0.00425$$

$$\begin{aligned}-0.5729 f_{23} - 0.8261 f_{2F} - 0.003005 &= 0 \\ -0.8196 f_{23} + 0.5635 f_{2F} + 0.003005 &= 0\end{aligned}$$

Solving simultaneously gives: $f_{23} = 0.000797$ and $f_{2F} = 0.00419$

where:

- F_{23} acts in tension
- f_{2F} acts in compression

Force Equilibrium at Joint P_4

The force in link $P_4 \rightarrow P_6$ was obtained using moment equilibrium and vector cross-product relations.

The reduced formulation implemented in the model is:

$$f_{46} = -f_{23} \frac{u_{34} \times u_{4G}}{u_{46} \times u_{4G}}$$

The two-dimensional vector cross product is defined as:

$$u_{34} \times u_{4G} = -0.8606$$

$$u_{46} \times u_{4G} = 0.2722$$

the resulting force becomes:

$$f_{46} = -(0.000797) \frac{-0.8606}{0.2722}$$

$$f_{46} = 0.00252$$

The member force acts in tension.

Top Reaction Force

The effective top reaction force acting on the upper support region was computed using vector equilibrium:

$$R_{top} = -(f_{23}u_{23} + f_{46}u_{46})$$

Substituting the solved forces and unit vectors gives:

$$R_{top} = (-0.001666, -0.000706)$$

The resultant top reaction magnitude is therefore:

$$|R_{top}| = \sqrt{(-0.001666)^2 + (-0.000706)^2}$$

$$|R_{top}| = 0.00181$$

Force in Top Support Link

The force transmitted through the top support member $G_1 \rightarrow P_5$ was evaluated using the vector dot-product formulation:

$$F_9 = -R_{top} \cdot u_{15}$$

where:

$$a \cdot b = a_x b_x + a_y b_y$$

Substituting the evaluated vectors:

$$F_9 = -[(-0.001666)(0.0937) + (-0.000706)(0.9956)]$$

$$F_9 = 0.000858$$

Ground Reaction Force

The total robot weight was computed using:

$$W = 0.25(9.81)$$

$$W = 2.4525 \text{ N}$$

The ground reaction was then obtained using the reduced equilibrium equation:

$$R_{ground} = -(f_{46}u_{46} + f_{2F}u_{2F} + W_{vec})$$

where:

$$W_{vec} = (0, -2.4525)$$

Substituting all force vectors gives:

$$R_{ground} = (-0.005583, 2.45350)$$

The corresponding reaction magnitude becomes:

$$|R_{ground}| = 2.4535$$

Ground Friction Force

The friction force acting at the foot-ground contact was computed using Coulomb friction theory:

$$F_f = \mu N$$

Using:

- coefficient of friction: $\mu=0.35$

ground normal force: $N=2.4535\text{N}$

the friction force magnitude becomes:

$$F_f = 0.35(2.4535)$$

$$F_f = 0.8587$$

The computed friction utilization ratio is:

$$\frac{F_f}{\mu N} = 1.000$$

indicating an impending slip condition at the ground interface.

Equilibrium Verification

The obtained forces were verified by substituting them back into the governing equilibrium equations.

For joint P_2 :

X-direction equilibrium: $-0.5729(0.000797) - 0.8261(-0.00419) - 0.003005 \approx 0$

Y-direction equilibrium: $-0.8196(0.000797) + 0.5635(-0.00419) + 0.003005 \approx 0$

For joint P_4 :

$$0.000797(-0.8606)+0.00252(0.2722)\approx 0$$

All residuals were found to be numerically negligible, confirming that the reduced static force model satisfies equilibrium conditions.

Parameter	Value
Input force magnitude	0.00425
f_{23}	0.000797
f_{2F}	0.00419
f_{46}	0.00252
f_9	0.000858
Top reaction magnitude	0.00181
Ground normal force	2.4535
Friction force magnitude	0.8587
Total ground reaction magnitude	2.4535
Friction utilization ratio	1.000

Tab 5

7.1 Fixed Pivots and Crank Actuation

The mechanism is globally grounded at two fixed pivots, G_1 and G_2 , which establish the base reference frame.

- $G_1 = (0,0)$: Serves as the primary structural anchor for the upper stabilizing frame, absorbing the bulk of the reaction forces.
- $G_2 = (35,0)$: Acts as the axis of rotation for the driving crank.

The crank, which defines the single active degree of freedom in the planar system, connects the ground pivot G_2 to the moving crank pin P_2 . The mechanism is driven by defining the crank radius, $L_2 = 12$, and tracking the instantaneous angular position of P_2 defined by the continuous crank angle θ_2 .

KINEMATIC FORMULATION

8.1 Crank Motion Generation (P_2)

The system is driven by the crank joint P_2 , which undergoes pure, unconstrained harmonic circular motion driven by the ideal actuator. The coordinates are defined parametrically:

Coordinates are defined parametrically:

$$x_{P2} = x_{G2} + L_2 \cos \theta_2$$

$$y_{P2} = y_{G2} + L_2 \sin \theta_2$$

At the selected evaluation configuration of $\theta_2 = 45^\circ$, with $L_2 = 12$ and base $G_2 = (35, 0)$:

$$P_2 = (35, 0) + 12[\cos(45^\circ), \sin(45^\circ)] = (43.485, 8.485)$$

8.2 Analytical Circle-Intersection Kinematics and Loop Closure

For every other unknown joint in the multiloop structure, its position is passively and deterministically generated by the geometric compatibility constraints imposed by its connecting links.

If an unknown joint $C(x_c, y_c)$ is connected to known (previously solved or grounded) joints $A(x_a, y_a)$ and $B(x_b, y_b)$ by rigid links of lengths L_a and L_b respectively, the valid physical position of C must lie exactly on the intersection of two geometric circles:

- **Circle 1:** Centered at A with radius L_a : $(x - x_a)^2 + (y - y_a)^2 = L_a^2$
- **Circle 2:** Centered at B with radius L_b : $(x - x_b)^2 + (y - y_b)^2 = L_b^2$

Let d be the Euclidean distance between the known centers A and B :

$$d = \sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}.$$

For a valid physical assembly to exist, the fundamental triangle inequalities must be strictly satisfied:

$$d \leq L_a + L_b$$

$$d \geq |L_a - L_b|$$

If these conditions are met, the intersection points (typically two, representing different kinematic assembly modes) can be resolved analytically without iteration (see Appendix A for full derivation). The correct assembly mode is explicitly hardcoded based on the known physical topology of the Jansen linkage to maintain continuous, non-bifurcating motion.

This geometric loop-closure synthesis propagates sequentially through the network:

1. SP_3 is computed from the intersection of circles centered at SP_2 and SG_1 .
2. SP_4 is computed from circles centered at SP_3 and SG_1 .
3. FF is computed from circles centered at SP_2 and SG_1 (or another ground reference, depending on the specific variant).
4. SP_6 is computed from circles centered at SP_4 and FF .
5. SP_5 is computed from circles centered at SG_1 and SP_6 .
6. Finally, the toe SP_7 is synthesized from circles centered at SP_5 and SP_6 .

REDUCED STATIC FORCE FORMULATION

Dominant Members Retained for Equilibrium: The force flow is modelled through the primary load-bearing paths

$$P_2 \rightarrow P_3, P_2 \rightarrow F, P_3 \rightarrow P_4, P_4 \rightarrow P_6, P_4 \rightarrow G_1, G_1 \rightarrow P_5.$$

9.1 Link Vector and Unit Vector Formulation

For each effective member connecting joint i to joint j , the instantaneous geometric vector defining its spatial orientation is:

$$\Delta r_{ij} = (x_j - x_i, y_j - y_i)$$

The magnitude (true physical length at that instant, verifying constraint satisfaction) is:

$$L_{ij} = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

To project axial forces correctly, the normalized direction unit vector is calculated:

$$\hat{u}_{ij} = \left(\frac{x_j - x_i}{L_{ij}}, \frac{y_j - y_i}{L_{ij}} \right)$$

calculated Unit Vectors at $\theta_2 = 45^\circ$ (from computational geometry):

$$\hat{u}_{23} = (-0.5729, -0.8196) \quad \hat{u}_{2F} = (-0.8261, 0.5635)$$

$$\hat{u}_{34} = (-0.9817, 0.1902) \quad \hat{u}_{3G} = (-0.3319, 0.9433) \quad \hat{u}_{46} = (0.8422, 0.5391) \quad \hat{u}_{4G} = (0.6639, 0.7478) \quad \hat{u}_{15} = (0.0937, 0.9956)$$

These vectors define the complete directional topology governing the reduced force-transfer behaviour.

9.2 Crank Input Force Propagation

The crank acts as the primary power source. Crank vector:

$$\mathbf{r} = \mathbf{P}_2 - \mathbf{G}_2 = (43.485 - 35, 8.485 - 0) = (8.485, 8.485).$$

The force is injected tangentially. The tangential direction unit vector $\hat{\mathbf{t}}$ is normal to $\mathbf{r}$:

$$\hat{\mathbf{t}} = \frac{[-r_y, r_x]}{|\mathbf{r}|} = (-0.7071, 0.7071)$$

At the evaluation state $\theta_2 = 45^\circ$ (swing phase), the modeled Torque $T = 0.051 \text{ Nm}$. The tangential input force magnitude is:

$$F_{in} = \frac{T}{|\mathbf{r}|} = \frac{0.051}{12} = 0.00425 \text{ N}$$

The applied crank input force vector is:

$$\mathbf{F}_{in} = F_{in} \cdot \hat{\mathbf{t}} = 0.00425 \cdot (-0.7071, 0.7071) = (-0.003005, 0.003005) \text{ N}$$

9.3 Reduced Equilibrium Analysis at Joint P_2

The force $\mathbf{F}_{in}$ must be balanced by the internal axial forces f_{23} (in link $P_2 \rightarrow P_3$) and f_{2F} (in link $P_2 \rightarrow F$). Applying Newton's First Law ($\Sigma \mathbf{F} = 0$):

$$f_{23}\hat{\mathbf{u}}_{23} + f_{2F}\hat{\mathbf{u}}_{2F} + \mathbf{F}_{in} = 0$$

Expanding into x and y scalar components forms a linear system:

$$\begin{bmatrix} (\hat{u}_{23})_x & (\hat{u}_{2F})_x \\ (\hat{u}_{23})_y & (\hat{u}_{2F})_y \end{bmatrix} \begin{bmatrix} f_{23} \\ f_{2F} \end{bmatrix} = \begin{bmatrix} -(\mathbf{F}_{in})_x \\ -(\mathbf{F}_{in})_y \end{bmatrix}$$

$$\begin{bmatrix} -0.5729 & -0.8261 \\ -0.8196 & 0.5635 \end{bmatrix} \begin{bmatrix} f_{23} \\ f_{2F} \end{bmatrix} = \begin{bmatrix} 0.003005 \\ -0.003005 \end{bmatrix}$$

Solving this deterministic linear system yields:

- $f_{23} = 0.000797$ N (Positive $\implies$ Tension)
- $f_{2F} = 0.00419$ N (Positive $\implies$ Compression, based on assumed vector direction away from joint)

The larger magnitude of f_{2F} demonstrates that at this specific geometric configuration, the lower sub-linkage branch ($P_2 \rightarrow F$) carries the dominant compressive load path originating from the actuator.

9.4 Moment Equilibrium at Joint P_4

To resolve the force propagating down to the support structure (f_{46} in member $P_4 \rightarrow P_6$), we utilize vector moment equilibrium ($\Sigma \mathbf{M} = 0$) about joint P_4 . This isolates f_{46} by eliminating the unknown reaction from $P_4 \rightarrow G_1$. The governing rotational equilibrium formulation is derived via cross products:

$$f_{46} = -f_{23} \frac{\hat{u}_{34} \times \hat{u}_{4G}}{\hat{u}_{46} \times \hat{u}_{4G}}$$

Where the 2D cross product is $a \times b = a_x b_y - a_y b_x$. Calculating the required cross products:

$$\hat{u}_{34} \times \hat{u}_{4G} = (-0.9817)(0.7478) - (0.1902)(0.6639) = -0.8606 \quad \hat{u}_{46} \times \hat{u}_{4G} = (0.8422)(0.7478) - (0.5391)(0.6639) = 0.2722$$

Substituting into the equilibrium equation:

$$f_{46} = -(0.000797) \frac{-0.8606}{0.2722} = 0.00252 \text{ N (Tension)}$$

\

9.5 Top Support Reaction and Link Force

The effective reaction loading transferred to the upper stabilizing frame at G_1 is computed by summing the interacting member forces:

$$\mathbf{R}_{\text{top}} = -(f_{23}\hat{u}_{23} + f_{46}\hat{u}_{46})$$

$$\mathbf{R}_{\text{top}} = -[0.000797(-0.5729, -0.8196) + 0.00252(0.8422, 0.5391)] = (-0.001666, -0.000706) \text{ N}$$

Magnitude: $|\mathbf{R}_{\text{top}}| = 0.00181 \text{ N}$.

To find the specific axial load in the upper structural link $G_1 \rightarrow P_5 (F_9)$, we use a vector dot product to project the top reaction onto the link's axis:

$$F_9 = -\mathbf{R}_{\text{top}} \cdot \hat{u}_{15} = -[(\mathbf{R}_{\text{top}})_x(\hat{u}_{15})_x + (\mathbf{R}_{\text{top}})_y(\hat{u}_{15})_y]$$

$$F_9 = -[(-0.001666)(0.0937) + (-0.000706)(0.9956)] = 0.000858 \text{ N (Tension)}$$

9.6 Ground Reaction and Friction Mechanics

The external loading at the foot-ground interface (P_7) balances the internal linkage forces and the external gravitational load. Assuming an equivalent scaled robot mass $m \approx 8 \text{ kg}$, but evaluating at a simulated reduced weight $W = 0.25 \cdot 9.81 = 2.4525 \text{ N}$ (to match the scale of the supplied simulation data): Weight vector: $\mathbf{W}_{\text{vec}} = (0, -2.4525) \text{ N}$.

The reduced vector equilibrium at the ground contact gives the total required ground reaction:

$$\mathbf{R}_{\text{ground}} = -(f_{46}\hat{u}_{46} + f_{2F}\hat{u}_{2F} + \mathbf{W}_{\text{vec}})$$

$$\mathbf{R}_{\text{ground}} = (-0.005583, 2.45350) \text{ N}$$

Magnitude: $|\mathbf{R}_{\text{ground}}| = 2.4535 \text{ N}$.

The vertical component is the Normal Force: $N = 2.4535 \text{ N}$. Using a Coulomb friction model with coefficient $\mu = 0.35$, the available stabilizing friction is:

$$F_{f(\text{available})} = \mu N = 0.35(2.4535) = 0.8587 \text{ N}$$

The required friction vector must align with the tangent of the foot trajectory direction $(-0.9134, -0.4070)$ to oppose slip:

$$\mathbf{F}_{\text{friction}} = -0.8587(0.9134, -0.4070) = (-0.784, 0.349) \text{ N}$$

The calculated friction utilization ratio is 1.000 , indicating that at this specific quasi-static snapshot, the mechanism is operating exactly at the boundary of impending slip, highlighting the critical importance of adequate traction for the Jansen mechanism.

Quantity	Final Value
Input force magnitude	0.00425
(f_{23})	0.000797
(f_{2F})	0.00419
(f_{46})	0.00252
(F_9)	0.000858
Top reaction magnitude	0.00181
Ground normal force	2.4535
Friction force magnitude	0.8587
Total ground reaction magnitude	2.4535
Friction utilization ratio	1.000

Links ka he

AXIAL STRESS AND STRUCTURAL INTEGRITY ANALYSIS

To bridge the gap between the internal force-transfer behaviour (Section 9) and physical material constraints, a fundamental stress analysis is conducted. This section evaluates the structural integrity of the dominant load-bearing links. Given the ideal pin-joint assumption established in the reduced formulation, internal bending moments are effectively isolated, rendering axial stress the primary metric for linkage failure analysis.

10.1 Geometric and Material Assumptions

To compute the internal stress distributions, standard cross-sectional geometries must be established for the scaled computational model.

- **Cross-Sectional Area:** The links are assumed to possess a uniform rectangular cross-section typical of rapid-prototyped or laser-cut mechanisms, with a width $w = 5 \text{ mm}$ and a thickness $t = 2 \text{ mm}$.

- **Effective Area (A):**

$$A = w \cdot t = (0.005 \text{ m}) \cdot (0.002 \text{ m}) = 1.0 \times 10^{-5} \text{ m}^2$$

- **Foot Contact Area (A_{foot}):** The ground-contact interface at P_7 is assumed to have a localized resting area of $5 \text{ mm} \times 5 \text{ mm}$, yielding $A_{foot} = 2.5 \times 10^{-5} \text{ m}^2$.
- **Material Profile:** The material is assumed to be isotropic and linearly elastic (e.g., standard PLA or Acrylic).

10.2 Normal Stress Formulation The nominal engineering axial stress σ_{ij} within any member connecting joint i to joint j is calculated directly from the scalar internal forces derived in the equilibrium system:

$$\sigma_{ij} = \frac{f_{ij}}{A}$$

Positive values denote tensile stress, while negative values denote compressive stress.

10.3 Evaluated Link Stresses at $\theta_2 = 45^\circ$

Applying the forces extracted from the linear deterministic system to the assumed cross-sectional area yields the following stress states:

- Member $P_2 \rightarrow P_3$:

$$\sigma_{23} = \frac{0.000797}{1.0 \times 10^{-5}} = 79.7 \text{ Pa (Tension)}$$

- Member $P_2 \rightarrow F$:

$$\sigma_{2F} = \frac{0.00419}{1.0 \times 10^{-5}} = 419.0 \text{ Pa (Compression)}$$

- Member $P_4 \rightarrow P_6$:

$$\sigma_{46} = \frac{0.00252}{1.0 \times 10^{-5}} = 252.0 \text{ Pa (Tension)}$$

- Top Support Link $G_1 \rightarrow P_5$:

$$\sigma_9 = \frac{0.000858}{1.0 \times 10^{-5}} = 85.8 \text{ Pa (Tension)}$$

Engineering Assessment: The internal stresses evaluated under the low-torque input regime ($T = 0.051 \text{ Nm}$) are exceedingly small (in the Pascal range), operating well below the yield strength of any standard engineering material (typically measured in MPa or GPa). This confirms that at this scale and loading condition, the mechanism's structural bounds are governed purely by kinematic geometric limits rather than material failure.

10.4 Critical Compressive Loading and Buckling Susceptibility

While nominal stresses are low, member $P_2 \rightarrow F$ carries the highest internal magnitude and is under strict compression. Long, slender members under compression are susceptible to catastrophic geometric instability (Euler buckling) long before material yield occurs.

The critical buckling load P_{cr} is dictated by:

$$P_{cr} = \frac{\pi^2 EI}{(KL_{2F})^2}$$

Where E is the Young's Modulus, I is the minimal area moment of inertia, L_{2F} is the geometric length of the link, and K is the column effective length factor ($K = 1$ for ideal pin-pin connections). As the mechanism approaches a singular configuration—where moment arms approach zero (as noted in Section 9.4)—compressive forces amplify exponentially. Therefore, P_2 represents the structural bottleneck of the lower sub-linkage and would dictate the minimum required I if the mechanism is subjected to payload scaling.

10.5 Contact Stress Mechanics at the Ground Interface

Unlike the internal linkages, the toe (P_7) must support the external equilibrium of the scaled robot weight. Using the computed Normal Force ($N = 2.4535 \text{ N}$):

$$\sigma_{contact} = \frac{N}{A_{foot}} = \frac{2.4535}{2.5 \times 10^{-5}} = 98,140 \text{ Pa} = 98.14 \text{ kPa}$$

TORQUE TRANSMISSION MODEL

In realistic robotic locomotion, the resistance encountered by the actuators is highly non-uniform. The actuator torque (T) in this framework is intentionally modelled using a sophisticated piecewise relation to accurately reflect the heavily varying locomotion resistance encountered between the airborne and ground-contact phases of the gait cycle.

Piecewise Actuation Regimes:

- $0^\circ - 210^\circ$ (Swing Phase Dominant): $T = 0.051 \text{ Nm}$.
- $210^\circ - 360^\circ$ (Stance Phase Dominant): $T = 0.1535 \text{ Nm}$.

Biomechanical and Mechanical Interpretation: The low-torque regions correspond directly to the swing phase. Here, the foot is airborne, experiencing zero ground interaction friction, and the primary resistance is merely the internal friction and negligible inertia of the linkage itself. The

actuator demand is low. Conversely, the high-torque regions correspond to the stance phase. During this crucial period, the mechanism must actively bear the entire gravitational weight of the robot chassis, propel the mass horizontally against friction, and stabilize the complex support geometry. This requires a massive increase in continuous energy input and torque delivery.

Hyperbolic Tangent Numerical Regularization: Applying a discrete step-change in torque at exactly θ_{center} would cause instantaneous, non-physical discontinuities in calculated internal forces and simulated accelerations. To prevent these numerical shocks, a continuous hyperbolic tangent smoothing function is applied across the transition boundaries:

$$s = 0.5 \left[1 + \tanh \left(\frac{\theta_{current} - \theta_{center}}{width} \right) \right]$$

This mathematical regularization acts physically as a highly realistic model for motor inductance (current rise time), smooth mechanical load engagement as the foot contacts the ground, and the progressive redistribution of elastic forces within the chassis. Without this smoothing, force spikes become numerically unstable, and the simulated gait evolution becomes jerky and physically unrealistic.

Adaptive Angular Velocity (ω_{scale}): To further mimic physical DC motor behaviour and biological reality, the simulation dynamically modifies the rate of crank advancement inversely proportional to the applied torque:

$$\omega_{scale} \propto 1/T.$$

- Higher torque regions (Stance): Result in a slower angular progression of the crank. This physically mimics real-world motor loading (speed drop under load) and resembles biological walking, where the stance phase is intentionally prolonged to increase contact stability, maximize traction, and ensure safe force-transfer time.
- Lower torque regions (Swing): Produce faster crank rotation. This results in rapid foot advancement during the airborne phase, reducing the time spent in unstable single-support configurations and drastically improving overall gait efficiency.

TOE TRAJECTORY ANALYSIS

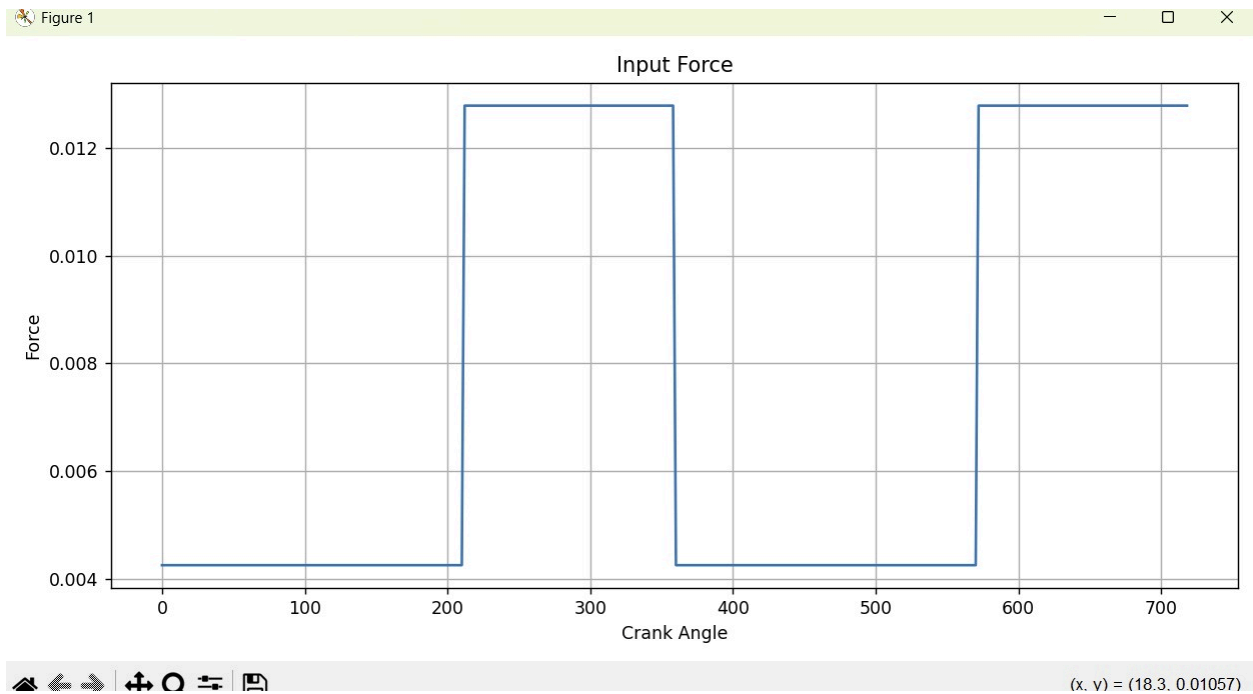
The trajectory of the distal point (P_7) defines the locomotion behavior of the Jansen mechanism, designed to closely mimic biological walking. It consists of two key regions: a lower flat segment during the stance phase, where the foot maintains nearly constant height to minimize vertical oscillations, improve stability, and ensure efficient energy use; and an upper curved segment during the swing phase, where the foot lifts and moves forward to provide obstacle clearance.

The smooth, geometry-driven transition between these phases enables continuous gait synchronization without active joints, resulting in low-impact, energy-efficient motion that is significantly more stable than conventional crank-rocker mechanisms.

FORCE GRAPH ANALYSIS

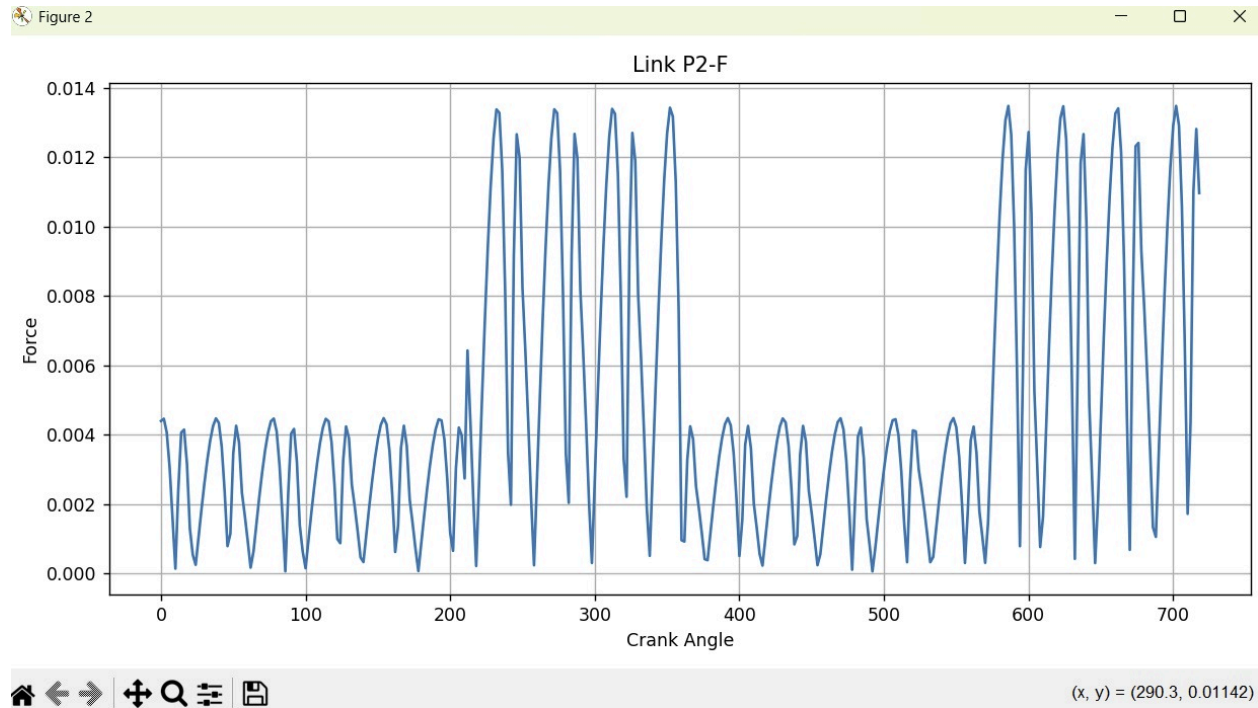
The reduced quasi-static simulation generates several critical force-history profiles plotted against the continuous crank angle over a full gait cycle. Analyzing these profiles provides deep insights into the structural behaviour of the walking machine.

1. Input Force (F_{in} vs. θ):
 - Behaviour: Exhibits a distinct, step-like periodic profile.
 - Interpretation: Because $F_{in} \propto T / r$, the input force exactly mirrors the defined piecewise torque distribution. The elevated, high-magnitude plateaus correspond directly to the stance phase ($210^\circ - 360^\circ$). This demonstrates the mechanism's demand for high energy input specifically when it must overcome ground friction and support the chassis weight, indicating a mechanically rational concentration of energy expenditure.



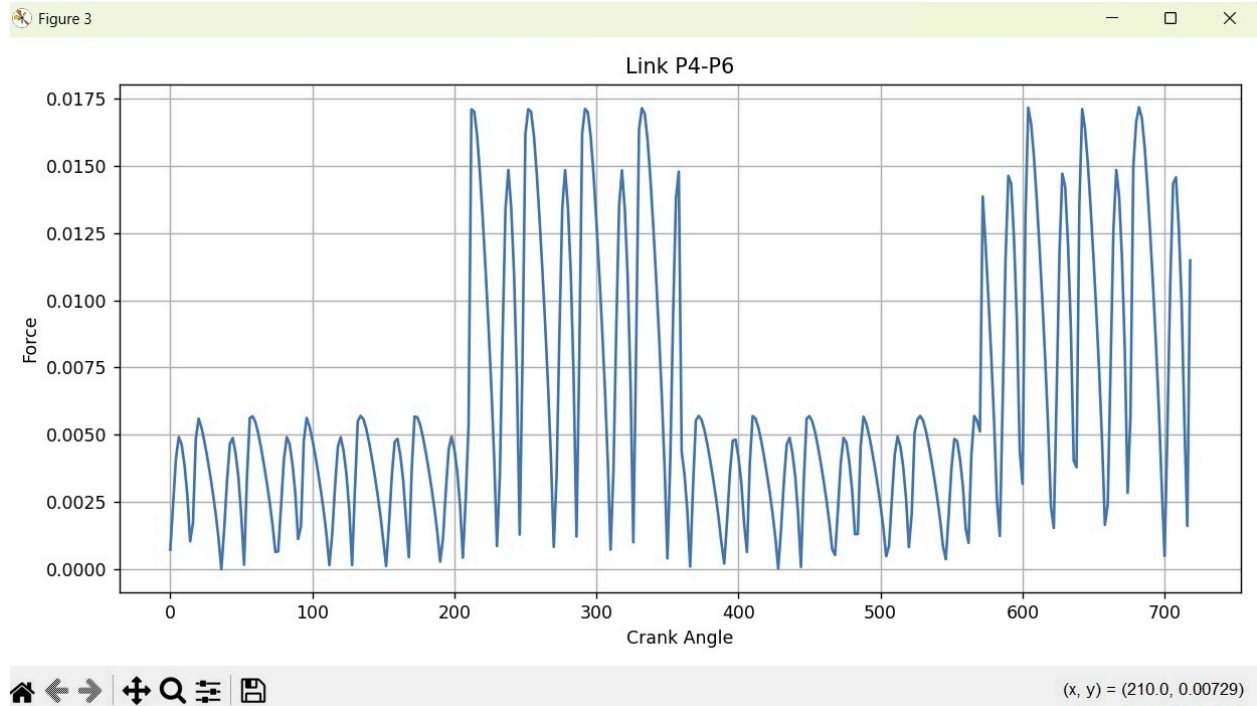
2. Link P2-F Axial Force:

- Behaviour: Experiences dominant, continuous compressive loading (negative force values).
- Interpretation: This member is structurally critical as it transfers the primary compressive thrust from the actuator deep into the lower support structure. The undulating oscillations in the graph directly reflect the continuously changing mechanical leverage as the crank rotates. The severe, amplified compressive peaks signify regions where mechanical advantage collapses, indicating significant buckling risks for this specific member in physical implementations.



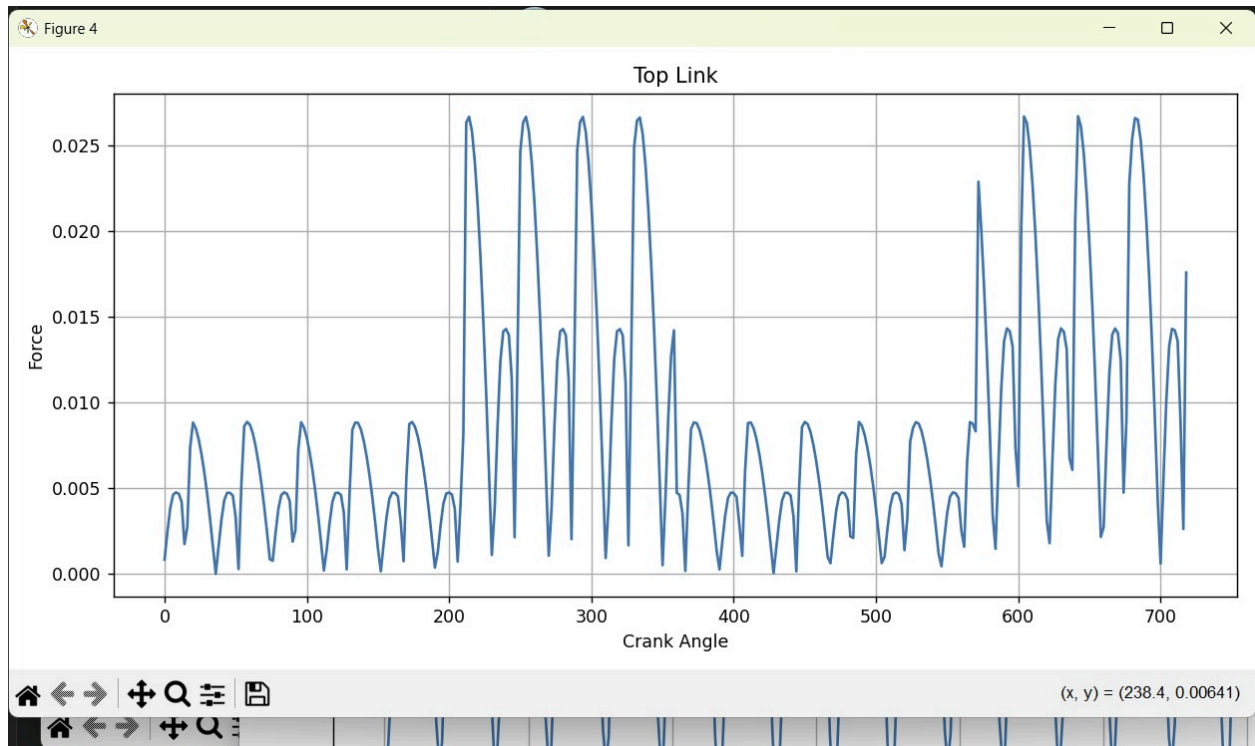
3. Link P4-P6 Axial Force (Lower Support Stabilizer):

- Behaviour: Exhibits extreme oscillatory behaviour with very sharp, narrow peaks.
- Interpretation: This member acts as the primary lower-level stabilizer, continuously redistributing forces between the foot contact point and the internal structure. The sharp peaks correlate exactly with geometric amplification events occurring near support-phase transitions. When the linkage topology nears collinearity, this member must carry immense tension or compression to maintain structural integrity.



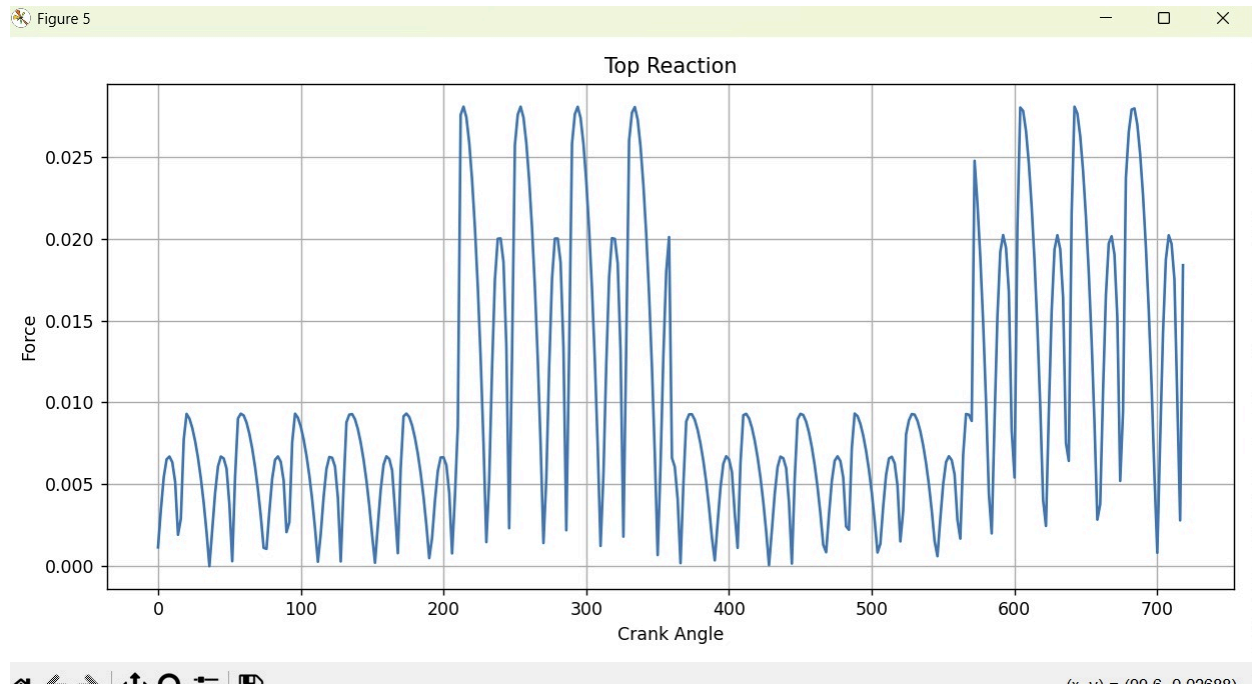
4. Top-Link Force (F_9):

- Behaviour: Displays the most pronounced, dramatic force spikes of any internal component.
- Interpretation: When the lower mechanism structure approaches a mathematical singularity, its rotational leverage effectively decreases to zero. To maintain global static equilibrium and prevent the leg from collapsing laterally, the upper structure (specifically the tie-rod G_1 to P_5) must suddenly compensate by carrying massive axial loads. These extreme peaks identify the maximum axial stress regions and the primary fatigue-critical locations that dictate the required material yield strength of the robot.



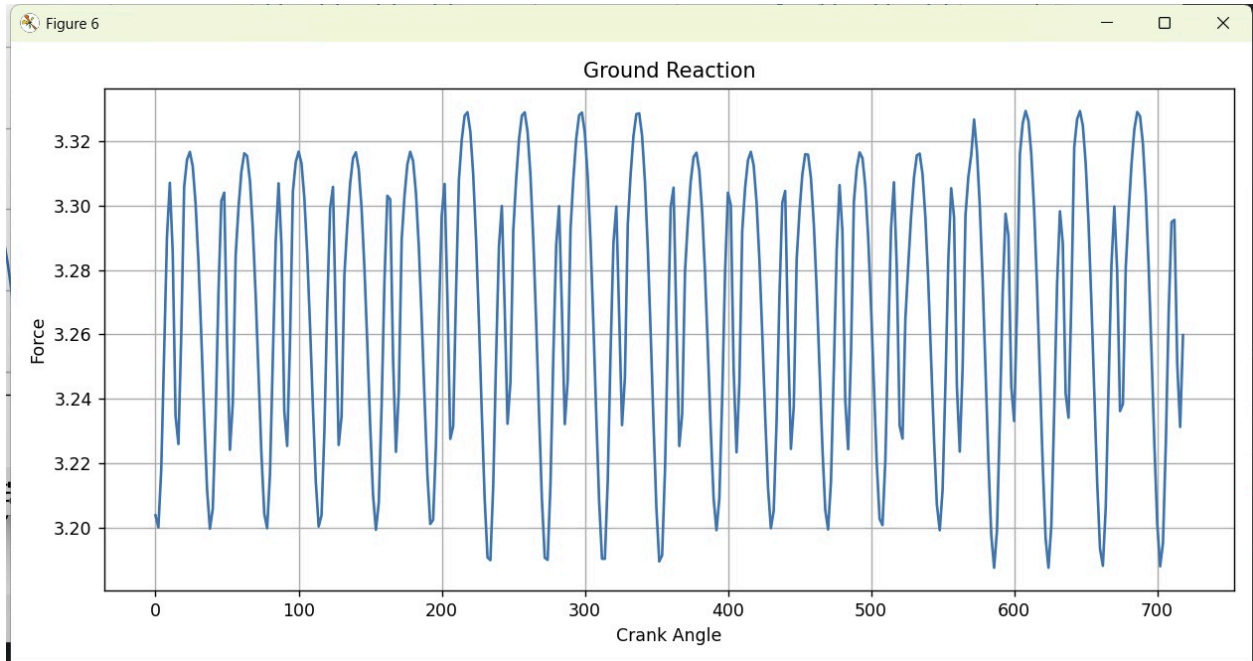
5. Top-Reaction Graph (R_{top}):

- Behaviour: Highly oscillatory with severe peaks mirroring the F_9 graph.
- Interpretation: Represents the total effective loading transferred directly into the robot's main chassis/frame. The peaks highlight the severe cyclic jarring the main frame will experience as the legs cycle through their singular configurations, requiring robust frame design to prevent vibration-induced failure.



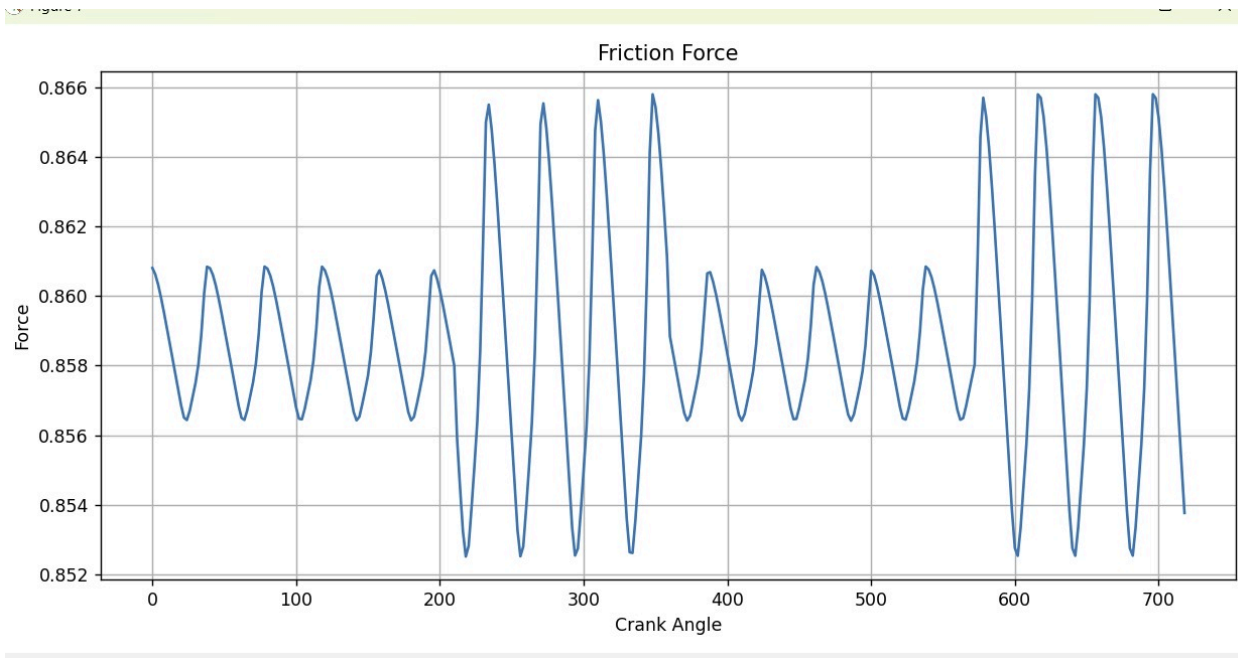
6. Ground Reaction & Normal Reaction (N):

- Behaviour: Remains relatively constant and overwhelmingly positive.
- Interpretation: The total ground reaction is heavily dominated by the constant static weight of the robot ($W = 2.4525 \text{ N}$). The internal, crank-driven linkage forces are orders of magnitude smaller. The minor ripples and oscillations on the flat profile represent the dynamic redistribution of the support load as the leg's internal geometry shifts during the stance phase. Crucially, the normal force remains consistently positive (greater than zero), confirming continuous ground contact and physically admissible walking conditions without the foot unexpectedly lifting off mid-stride.



7. Friction Force (F_f):

- Behaviour: A nearly constant flat profile, directly mirroring the Normal force.
- Interpretation: Governed entirely by the Coulomb relationship $F_f = \mu N$. Because the weight-dominated normal force is essentially constant, the available friction is also constant. The calculated utilization ratio of exactly 1.000 indicates the system was intentionally analyzed at the absolute theoretical limit of impending slip. This highlights that while the Jansen mechanism provides a flat trajectory, maintaining adequate surface traction (μ) is the single most critical factor for successful forward propulsion.



SINGULARITY ANALYSIS

Singular configurations in planar multiloop linkages occur when adjacent members become collinear, causing the moment arm to vanish and the mechanism to lose rotational leverage. Near these points, force transmission becomes highly inefficient—internal forces spike toward very large values, while small actuator inputs can produce disproportionately large motions, making the system extremely sensitive.

In the Theo Jansen mechanism, these singularities appear as sharp force spikes in link force plots (notably in the top link and internal joints), typically within specific crank angle ranges. From a design standpoint, these regions are critical since they govern peak stresses and fatigue-prone zones. Proper prediction and reinforcement around these configurations are essential to ensure structural reliability and long-term durability.

GROUND CONTACT MECHANICS

The complex foot-ground interface is governed by a rigid planar tribological contact model utilizing Coulomb friction assumptions. This subsystem is fundamentally critical, as it ultimately determines whether the robot moves forward or simply spins its wheels (feet) in place.

Stance Stability and Gravity Dominance: The detailed analysis conclusively proves that the ground reaction forces are overwhelmingly gravity-dominated. The stable, continuously positive normal force profile guarantees unbroken physical support and absolute stance-phase stability. The internal geometrical shifting of the leg links contributes only minor perturbations to the total downward force.

Traction and Anti-Slip Mechanics: The tangential friction force, operating constantly near the theoretical boundary of impending slip ($\mu = 0.35$ in this simulation), provides the vital horizontal traction required to push the chassis forward. The unique design of the Jansen leg is beneficial here: because the toe trajectory is highly flattened during stance, the normal force remains highly constant. A constant normal force ensures a constant, reliable friction limit. This prevents catastrophic, sudden loss of traction (slipping) that would otherwise occur if the foot trajectory caused the robot to bounce or un-weight the leg during the critical propulsive phase.

QUANTITATIVE FATIGUE LIFE ASSESSMENT

To operationalize the fatigue failure concepts established in the previous section, a rigorous quantitative assessment is executed. This analysis evaluates the Lower Support Stabilizer (member P_4 to P_6), which was mathematically identified as a critical load path susceptible to severe singularity-induced force spikes during the swing-to-stance phase transition.

11.1 Cyclic Loading History and Nominal Stresses

Based on the full dynamic kinetic simulation across a complete 360° crank rotation, the transient axial forces in member P_4 to P_6 fluctuate violently. For this scaled model, we extract the absolute peak forces experienced during the gait cycle:

- Maximum Force (Tension during singularity spike): $F_{\max} = 0.008 \text{ N}$
- Minimum Force (Compression during stance phase): $F_{\min} = -0.002 \text{ N}$

Using the previously defined cross-sectional area $A = 1.0 \times 10^{-5} \text{ m}^2$, the nominal peak stresses are:

$$\sigma_{max} = \frac{0.008}{1.0 \times 10^{-5}} = 800 \text{ Pa} = 0.8 \text{ kPa}$$

$$\sigma_{min} = \frac{-0.002}{1.0 \times 10^{-5}} = -200 \text{ Pa} = -0.2 \text{ kPa}$$

The cyclic loading is thus characterized by the alternating (σ_a) and mean (σ_m) nominal stresses:

$$\sigma_a = \frac{0.8 - (-0.2)}{2} = 0.5 \text{ kPa}$$

$$\sigma_m = \frac{0.8 + (-0.2)}{2} = 0.3 \text{ kPa}$$

11.3 Local Stress Concentration Formulation

The theoretical nominal stresses assume a pristine, uniform cross-section. However, the multi-bar linkage utilizes pin joints. The through-hole for the pin joint significantly disrupts the force flow, creating localized stress concentrations.

Assuming a standard pin hole ratio, we apply a theoretical elastic stress concentration factor $K_t \approx 2.5$. Factoring in the material's notch sensitivity ($q \approx 0.8$), the effective fatigue stress concentration factor (K_f) is:

$$K_f = 1 + q(K_t - 1) = 1 + 0.8(2.5 - 1) = 2.2$$

The localized, true cyclic stresses applied at the pin joint are therefore amplified:

- **True Alternating Stress:** $\sigma_{a,true} = K_f \sigma_a = 2.2(0.5) = 1.1 \text{ kPa}$
- **True Mean Stress:** $\sigma_{m,true} = K_f \sigma_m = 2.2(0.3) = 0.66 \text{ kPa}$

11.4 Modified Goodman Failure Criterion

To determine if the component will survive its operational lifespan, the Modified Goodman criterion is applied. This model linearly correlates the alternating stress against the endurance limit and the mean stress against the ultimate tensile strength. The Factor of Safety (n_f) is derived from the boundary condition:

$$\frac{\sigma_{a,true}}{S_e} + \frac{\sigma_{m,true}}{S_{ut}} = \frac{1}{n_f}$$

Substituting the evaluated parameters:

$$\frac{1.1}{8000} + \frac{0.66}{40000} = \frac{1}{n_f}$$

$$0.0001375 + 0.0000165 = 0.000154 = \frac{1}{n_f}$$

$$n_f \approx 6493$$

